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A Study on the Optimisation of Extreme Learning Machines for Gallstone Disease Prediction

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Abstract

Gallstone disease is a highly prevalent biliary disorder globally. While conventional diagnostic imaging techniques are inherently reactive, costly, and limited in detecting asymptomatic patients, the transition to early risk prediction using accessible clinical data and machine learning (ML) offers a promising alternative for opportunistic screening during general medical check-ups. The Extreme Learning Machine (ELM) provides rapid prediction capabilities but suffers from performance variance due to random parameter initialisation. To resolve this, this study investigates an Artificial Bee Colony-Regularised Extreme Learning Machine with Internal Cross-Validation (ABC-RELM-CV) following the data refinement of selected features. The ABC metaheuristic algorithm was employed to optimise ELM input weights and hidden biases, while internal cross-validation (CV) was utilised to mitigate overfitting. The model achieved an average accuracy of 0.7939, an F1-score of 0.7739, and a Matthews Correlation Coefficient (MCC) of 0.5947. It demonstrated statistically significant stability improvements over the standard Regularised Extreme Learning Machine (RELM). Conversely, the Logistic Regression Classifier and Support Vector Classifier outperformed ABC-RELM-CV with a posterior probability exceeding 50%. The findings suggest that while metaheuristic optimisation effectively stabilises ELM architectures, sample size constraints restrict the model to the data's linear structure, thereby establishing a complexity ceiling for generalisation.

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I have referenced the open-source repository artificialbeecolony/Java_ABC (https://github.com/artificialbeecolony/Java_ABC.git) for logic understanding and implementation of the ABC algorithm in Python. The underlying methodology is credited to Karaboga and Basturk, and Abobakr et al. as cited in the report.
Furthermore, Python libraries were utilised (NumPy for model implementation, Pandas for data handling, Scikit-learn for splitting/scaling, SciPy, Statsmodels and Baycomp for statistical analysis, and Matplotlib and Seaborn for visualisation).

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Acknowledgements

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Special thanks go to Esen et al. for originally collecting the gallstone disease diagnosis dataset, and to the UC Irvine Machine Learning Repository for making it publicly accessible.

Finally, my deepest and most heartfelt thanks belong to my family for their unconditional love, endless patience, and unwavering moral support.

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1 Introduction

Gallstone disease is a highly prevalent biliary disorder worldwide [1]. While conventional diagnostic imaging techniques, such as ultrasonography and CT scans, provide high sensitivity and specificity, they are inherently reactive, costly, and consequently limited in detecting asymptomatic patients [1–3]. Therefore, transitioning from reactive imaging diagnostics to early risk prediction and prevention is a promising alternative, which can be accomplished through accessible clinical data, such as demographic profiling and laboratory blood tests.

Despite the growing application of machine learning (ML) in gallstone prediction, current research is predominantly focused on image-based models [4,5]. This project investigates the Extreme Learning Machine (ELM) for early prediction. Although ELM avoids computationally expensive gradient descent through random parameter assignment, this inherent randomness induces performance variance. Consequently, this study integrates a metaheuristic algorithm to optimise the ELM's input weights and hidden biases.

1.1 Background

1.1.1 Gallstone Disease

Gallstone disease, also known as cholelithiasis, is a prevalent biliary disorder worldwide which generally develops through the progressive deposition of cholesterol on an initial precipitation nidus [1]. The prevalence differs by region and country, reported to be approximately 10-20% in Western countries and 5-12% in Eastern countries during certain periods [6]. Even though later reports indicated that approximately 6% of the global population has had gallstones since the 21st century, some journals proposed that 50-70% of patients are asymptomatic at diagnosis, while the incidence rate has almost tripled in the past 30 years [2,3,7,8]. However, it is certain that the situation has not improved, as the mortality rate and social burden caused by cholelithiasis are increasing [8–10].

Currently, the diagnosis of cholelithiasis is primarily conducted through systematic transcutaneous sonography (abdominal ultrasound), which achieves 95% sensitivity and practically 100% specificity, providing high confidence among medical staff [11]. Following diagnosis, treatment is initiated, with the currently preferred method being cholecystectomy. The ultrasound image and the clinical process are illustrated in Figure 1 and Figure 2. Although this treatment process is standard and effective, it also comes with disadvantages, such as patients experiencing issues with fat digestion due to the removal of the gallbladder along with the gallstones [1]. Furthermore, there are other challenges, including the lack of more suitable treatments, high treatment costs, the failure to detect asymptomatic patients due to a lack of proactive medical check-ups, and various complications or postoperative sequelae [12–14]. To address these challenges, artificial intelligence (AI) has been introduced to the field, enhancing medical processes in various areas such as the early detection of the disease, course prognosis, and risk estimation [15].

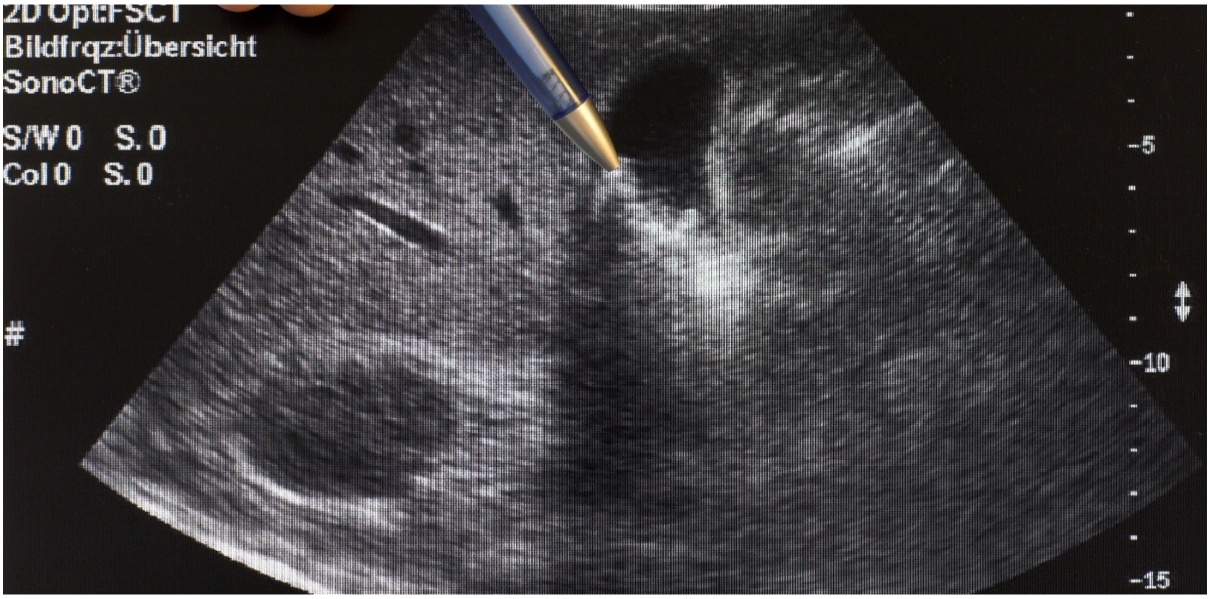

Figure 1 Clinical Gallbladder Ultrasound [16]

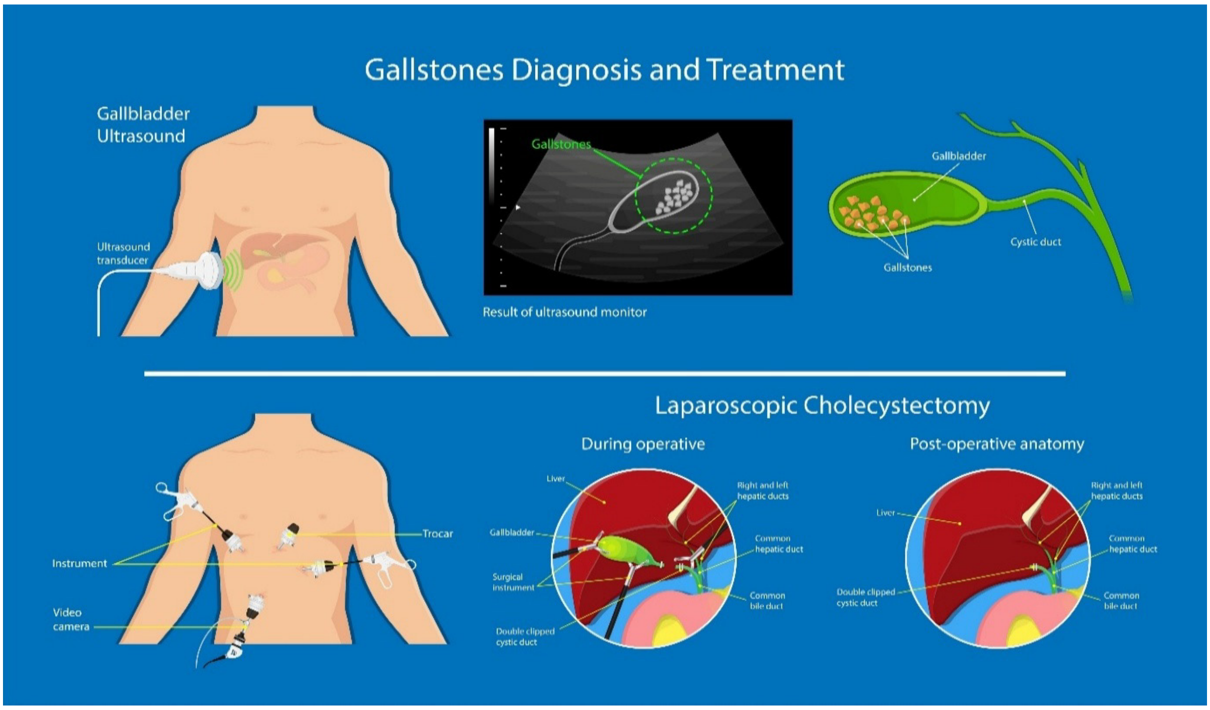

Figure 2 Gallstone Diagnosis and Treatment [17]

1.1.2 Machine Learning

ML is a subfield of AI, with the concept of building a system that can improve its performance on certain tasks through experience [18]. ML is built on three key elements: data (the 'experience' for learning), algorithms (the core calculations performed to obtain results), and models (the combination and formation of a 'black box' that describes the generalised relationship between input and output variables) [19]. It aims to autonomously detect and learn the underlying patterns of data through limited training data and learning algorithms, and then perform tasks on previously unseen data, addressing specific problems clearly defined by humans [20]. By applying ML, robust support is provided for complex scenarios, such as tasks requiring tremendous tuning, tasks performed in fluctuating environments, complex problems lacking established solutions, or the extraction of insights from enormous datasets [21].

1.2 Literature Review

1.2.1 Machine Learning in Healthcare

ML is widely utilised in the healthcare field today to support medical professionals in the diagnosis and treatment processes, resulting in enhanced diagnostic precision, simplified administrative procedures, and individualised treatment regimens [22]. Notable results include the conversion of large-scale clinical data into actionable insights, tumour detection and classification in oncology, and treatment guidance and antimicrobial resistance monitoring in clinical microbiology [23,24].

The hepatobiliary department is no exception; there is an increasing number of studies on the incorporation of ML or AI in hepatobiliary-related diseases. For example, Srivastava et al. [25] evaluated three ML models based on electronic health record datasets for the prediction and classification of liver diseases. Taking a different approach, Akabane et al. [26] proposed a model for predicting the likelihood of requiring Endoscopic Retrograde Cholangiopancreatography (ERCP) following emergency laparoscopic cholecystectomy, using both medical record data and CT/ultrasound features.

1.2.2 Machine Learning in Gallstone Disease

As one of the main hepatobiliary diseases, the diagnosis of cholelithiasis is highly reliant on medical records and medical images currently; therefore, attempts of improvement through ML techniques are clearly appropriate. Ahmed et al. [4] conducted a systematic review of gallstone diagnosis using ML-based image analysis, the results indicated that most models demonstrated good accuracy and speed. As hepatobiliary disease diagnosis primarily relies on clinical images, Convolutional Neural Networks (CNNs) have emerged as the most common choice for deployment in computer-aided diagnosis systems and model development [4,5]. For example, Pang et al. [27] built a novel YOLOv3-based neural network architecture for cholelithiasis identification and gallstone classification while Kumar et al. [5] developed a custom CNN architecture for gallstone classification and prediction.

A variety of ML models have been explored and proposed over time, with cross-validation area under the receiver operating characteristic curve (AUROC) and accuracy as the most common evaluation methods. However, the computational demands and robustness of models across different settings remain a concern due to certain limitations [4]. Researchers have attempted various methods to overcome these limitations. For instance, an AI diagnostic system for cholelithiasis recognition capable of running on Android mobile devices was developed by Pang et al. [28]. A model that takes point-of-care ultrasound (POCUS) as input for cholelithiasis, and cholecystitis detection was proposed by Yu et al. [29] to assist emergency physicians even in remote areas.

Although various attempts have been made to overcome the limitations, some challenges such as cost and reactive diagnostics were inherited from the ultrasound and CT techniques themselves. Consequently, some methods such as the use of clinical laboratory data have been employed for early prediction. For example, Salem et al. [30] investigated ML approaches for cholelithiasis prediction during their research on the identification of metabolites and clinical features associated with cholelithiasis, while Blum et al. [31] demonstrated an ML model that predicts choledocholithiasis by using patients' digital medical records. In 2024, Esen et al. [32] curated and constructed a new dataset of bioimpedance and laboratory data for early prediction of cholelithiasis, followed by an evaluation of nine baseline models. The dataset was uploaded to the UCI Machine Learning Repository. There were several follow-up studies of gallstone prediction based on this, such as metaheuristic optimisation by Sarker et al. [33] and ensemble model optimisation by Indirani et al. [34], which are summarised in Table 1.

Despite the progress in this field, laboratory-data-driven studies remain insufficient compared to image-based research for cholelithiasis prediction. Although multiple models and algorithms have been attempted, there are still many models that have yet to be utilised. In addition, there is still a gap between the evaluation results of laboratory data-based models and images-based models, with image-based models generally performing 5-10% better than data-based models.

This is likely due to the difference between reactive diagnosis, which only focuses on a localised region and early prediction using laboratory data, which is affected by systemic physiological status. Therefore, there is still a need to find new ways to refine and optimise ML models that rely on clinical and laboratory data to narrow this gap.

1.2.3 Extreme Learning Machine and Disease Prediction

To address this need, many ML models and deep learning techniques (such as neural networks) have been explored. In the healthcare domain, the study and application of ELM in disease prediction has been increasing recently due to the elimination of the requirement for backpropagation or gradient descent. Research has demonstrated that ELM combined with optimisation or dimensionality reduction can improve the speed and accuracy of disease prediction [35]. These studies have included not only common chronic diseases such as diabetes and heart disease, but also other diseases such as anaemia and Parkinson's disease [35–40]. However, to the best of the author's knowledge, ELM has not yet been used for cholelithiasis prediction. Therefore, this study explores the application of ELM for the early prediction of cholelithiasis utilising clinical laboratory data. To establish a robust predictive model, a metaheuristic algorithm, specifically the Artificial Bee Colony (ABC) algorithm, is utilised to optimise its architecture.

Table 1 Summary of previous works on the UCI Gallstone dataset

Reference	Data Processing	Feature Selection	Models Used	Optimisation Method	Results
Esen et al., 2024 [32]	70% Train (10-fold CV) / 30% Test split	Statistical ANOVA F-score (reduced from 32 to 9 core features)	Gradient Boosting (GB), KNN, MLP, SVM, DT, RF, AdaBoost, Naive Bayes, LR	Grid Search	Accuracy: 85.42%
Sarker et al., 2025 [33]	10-fold CV, Internal fixed 70% Train / 30% Test split per fold	-	Random Forest (RF)	Metaheuristic: Sand Cat Swarm Optimisation (SCSO) for simultaneous dimensionality reduction (38 to 13) and hyperparameter tuning, XAI: SHAP and DiCE for counterfactual explanations	Accuracy: 78.32%, F1-Score: 77.44%, Max Single-fold AUC: 88.70%, Precision: 80.36%, Recall: 75.01%
Cansu et al., 2025 [41]	Unmentioned	Recursive Feature Elimination (RFE) for dynamic joint optimisation of retained features	SVM, MLP, LR, Gaussian NB, RF	Metaheuristic: Slime Mould Algorithm (SMA) for hyperparameter tuning	Accuracy: 80.21%, F1-Score: 80.00%, AUC: 79.90%

Akben and Yumrutas, 2026 [42]	Global 10-fold CV	-	Meta Learner: 3-layer Deep Neural Network (DNN) DT, SVM, KNN, LDA	Ensemble: Stacking Ensemble (Base predictions one-hot encoded into an 8D meta-matrix), Tuning: Bayesian Optimisation for DNN hyperparameters	Accuracy: 88.40% (±2.4%), F1-Score: 88.39%, AUC: 95.30%, Precision: 88.54%, Recall: 88.37%
Indirani et al., 2026 [34]	Nested Stratified CV (Strict isolation of feature selection and model calibration to prevent data leakage)	Top 25 features selected via SHAP	Voting Based Ensemble: RF, XGBoost, LR, GB, Extra Trees, LightGBM, SVM	Ensemble: Soft Voting probability fusion, Post-hoc Tuning: Decision threshold calibration (Threshold = 0.42)	Baseline Ensemble Acc: 79.69%, Best Calibrated Acc: 87.50%
Rodriguez et al., 2026 [43]	70% Train (5-fold Stratified CV) / 30% Test (Stratified Split)	Comparison of 4 strategies (e.g., Heatmap correlation exclusion, ANOVA top 25%)	SVM, RF, GB, LR, AdaBoost, DT, Bagging	XAI: SHAP (Global/Local) and LIME (Individual distribution)	Accuracy: 83.33%, F1-Score: 86.37%, AUC: 88.80%, Precision: 82.00%, Recall: 85.42%
Shrestha et al., 2026 [44]	10-fold CV, Internal fixed 70% Train / 30% Test split per fold	-	Rotational Forest (RoF)	Metaheuristic: Bald Eagle Search (BES) algorithm for optimising retained features (17 features) and estimator count, XAI: SHAP for feature weight attribution	Accuracy: 75.78%, F1-Score: 75.41%, AUC: 86.00%

Note: The results reported in some of these studies warrant careful interpretation. Several documentation inconsistencies were identified during the review; a detailed record of these observed variations is provided in Table A2 (Appendix B).

1.3 Problem Statement

Current models for early gallstone prediction relying on clinical laboratory data consistently trail traditional image-based diagnostics by 5% to 10%. This performance gap stems primarily from the noisy nature of clinical data, which is sensitive to a patient's overall physiological status rather than focused on a localised disease region. While various conventional ML algorithms have been tested to improve these predictions, the ELM has not yet been explored for gallstone disease prediction. Furthermore, due to the inherent initialisation variance of the ELM architecture, metaheuristic optimisation is required to establish a robust and stable predictive model.

1.4 Project Objectives

The primary aim of this study is to design an optimised ML predictive model for gallstone disease. To achieve this, the specific objectives are formulated as follows:

1. To refine selected features utilising physiological consistency to ensure data integrity.
2. To develop a baseline ML model utilising the ELM architecture for predicting the risk of gallstone disease.
3. To optimise the ELM predictive model by integrating the ABC metaheuristic algorithm and to compare its performance metrics against the pre-optimised baseline.

1.5 Project Scope

This study focused on developing an ELM predictive model for gallstone disease utilising publicly available clinical datasets. The methodological scope was bounded to analysing clinical data, executing physiological data consistency corrections, and integrating the ABC algorithm to optimise the ELM architecture. To define its boundaries clearly, this research specifically excluded clinical trials, direct patient data collection, and the design of new diagnostic hardware. The evaluated outcomes are classification performance metrics and a comparative analysis between the baseline and optimised models.

2 Data Preparation

2.1 Data Analysis

The gallstone disease diagnosis dataset utilised in this study was sourced from the UC Irvine (UCI) Machine Learning Repository, originally collected and provided by Esen et al. [32]. The dataset comprises a total of 319 instances and 39 features. It consists of a target variable, 'Gallstone Status', and 38 predictive features categorised into four distinct groups: demographics, comorbidities, bioimpedance analysis (BIA), and laboratory blood tests. These data are detailed in Table 2. The dataset was preprocessed by its creators, resulting in a complete data source that required no missing value imputation. Additionally, the target class is largely balanced, comprising 161 negative (50.5%) and 158 positive (49.5%) cases. All features are numerically encoded: continuous variables as numeric values, and binary categorical variables as 0 or 1, with the exception of 'Comorbidity' and 'Hepatic Fat Accumulation' (HFA), which are encoded as categorical integers. Notably, feature names have been retained in their original spelling to maintain consistency.

Despite its overall completeness, several notable documentation discrepancies between the original paper and the published dataset require special attention. Firstly, while the paper claims a total of 38 features, it describes only 37; the dataset contains 39 columns, including an undocumented feature, 'Extracellular Fluid/Total Body Water' (ECF/TBW). The primary feature list also omitted 'Total Body Fat Ratio' (TBFR), though it sporadically appears later in the text. Furthermore, the original paper lists hypertension as one of the cardiovascular comorbidities, which is a variable entirely absent from the dataset and has been replaced by Coronary Artery Disease (CAD). Moreover, the comorbidity feature is described as binary, yet appears in the dataset with categorical values ranging from 0 to 3. Additionally, a variation in gender statistics is observed, the paper reporting 187 females while the dataset records 157.

The most significant discrepancy concerns the encoding of the target variable, 'Gallstone Status'. The paper explicitly defines 1 as a positive diagnosis representing 161 cases and 0 as a negative one representing 158 cases, whereas the actual dataset records 158 instances of '1' and 161 instances of '0' [32]. The official UCI repository describes the data using an unconventional inverse encoding where 0 denotes a positive status and 1 denotes a negative one [45]. While this inverse encoding mathematically aligns with the authors' reported count of 161 positive cases, it conflicts with the paper's description and is inconsistent with the other binary features in the dataset, which conventionally use 1 to denote a positive status. Faced with this ambiguity, this study elected not to apply any modifications or reverse mappings to the raw target column. Instead, the methodology aligns with the original paper's clinical definition by retaining 1 as the positive value and 0 as the negative value. Consequently, this study proceeds with the analysis of 158 positive cases and 161 negative cases, operating under the assumption that the case frequencies reported in the original manuscript may reflect documentation variations.

Table 3 provides a statistical summary for the categorical features, with corresponding histograms and demographic distributions presented in Figures 3–6. Similar to gallstone status, the gender distribution is largely balanced. Additionally, the distributions of Age, Height, and Weight follow a relatively balanced normal distribution. Notably, while only nearly 32% of patients have a positive comorbidity score, nearly 60% present with HFA, indicating that HFA was excluded from the comorbidity scoring criteria. Among specific comorbidities, CAD, Hypothyroidism and Hyperlipidemia all remain below 5%, whereas Diabetes Mellitus (DM) is more prevalent in this dataset, reaching 13.5%.

Furthermore, Table 4 provides the statistical summary for the quantitative features, while frequency distributions and boxplots of BIA and laboratory blood tests are visualised in Figures 7–10. Some features exhibit a standard deviation higher than their mean, namely obesity, Alanin Aminotransferaz (ALT) and C-Reactive Protein (CRP). While the latter two are highly sensitive to underlying health conditions, making a direct judgment impossible without professional background, the standard deviation for obesity, which is more than three times its mean, clearly indicates an anomaly. Upon further inspection, there was a biologically implausible value of 1954% with some outliers more than 100% to be verified.

Besides the extreme variance, some physiological inconsistencies data points appear in BIA such as instances where Total Body Water is less than Intracellular Water (ICW), and Muscle Mass (MM) value being lower than Visceral Muscle Area (VMA) (both in kg). This suggests that the data requires additional processing before utilisation. Most of the BIA features exhibit a normal distribution except for obesity, which is extremely right-skewed due to the 1954% outlier. In contrast, nearly half of the laboratory data exhibit severe right-skewness, and the remaining are nearly normal while only Glomerular Filtration Rate (GFR) shows slight left-skewness. As noted earlier, because blood test parameters are highly sensitive to individual physiological states, the presence of erroneous inputs cannot be definitively determined based solely on statistical distributions without professional clinical evaluation.

Table 2 Categorisation and description of the UCI Gallstone dataset features

Feature	Type	Description	Value / Unit
Target			
Gallstone Status	Cat.	Presence of Gallstone	0: No, 1: Yes
Demographics			
Age	Cont.	Age of the person	Years
Gender	Cat.	Gender of the person	0: Male, 1: Female
Height	Cont.	Barefoot height	cm
Weight	Cont.	Body weight with thin clothing	kg
Comorbidities			
Comorbidity	Cat.	Concomitant diseases	0,1,2,3
Coronary Artery Disease (CAD)	Cat.	Cardiovascular disease	0: No, 1: Yes
Hypothyroidism	Cat.	Underactive thyroid gland	0: No, 1: Yes
Hyperlipidemia	Cat.	High levels of fat in the blood	0: No, 1: Yes
Diabetes Mellitus (DM)	Cat.	High blood sugar	0: No, 1: Yes
Hepatic Fat Accumulation (HFA)	Cat.	Fatty liver stage	0,1,2,3,4
Bioimpedance			
Body Mass Index (BMI)	Cont.	Weight / Height^2	kg/m^2
Total Body Water (TBW)	Cont.	Total water in the body	kg
Extracellular Water (ECW)	Cont.	Total Extracellular water	kg
Intracellular Water (ICW)	Cont.	Total Intracellular water	kg
Extracellular Fluid/Total Body Water (ECF/TBW)	Cont.	Edema Index	
Total Body Fat Ratio (TBFR)	Cont.	Total fat content	%
Lean Mass (LM)	Cont.	Lean body mass	%
Body Protein Content (Protein)	Cont.	Protein content of the body	%
Visceral Fat Rating (VFR)	Cont.	Visceral organ fat level	
Bone Mass (BM)	Cont.	Bone weight	kg
Muscle Mass (MM)	Cont.	Amount of muscle	kg
Obesity	Cont.	Degree of obesity	%
Total Fat Content (TFC)	Cont.	Total fat amount	kg
Visceral Fat Area (VFA)	Cont.	Inner adipose tissue area	kg
Visceral Muscle Area (VMA)	Cont.	Inner muscle area	kg
Laboratory Blood Tests			
Glucose	Cont.	Blood glucose level	mg/dL
Total Cholesterol (TC)	Cont.	TC level	mg/dL

Low Density Lipoprotein (LDL)	Cont.	LDL level	mg/dL
High Density Lipoprotein (HDL)	Cont.	HDL level	mg/dL
Triglyceride	Cont.	Triglyceride level	mg/dL
Aspartat Aminotransferaz (AST)	Cont.	Liver enzyme level	U/L
Alanin Aminotransferaz (ALT)	Cont.	Liver related enzyme level	U/L
Alkaline Phosphatase (ALP)	Cont.	Liver and bone enzyme level	U/L
Creatinine	Cont.	Kidney function indicator	mg/dL
Glomerular Filtration Rate (GFR)	Cont.	Kidney filtration rate	ml/seconds
C-Reactive Protein (CRP)	Cont.	Inflammation indicator	mg/L
Hemoglobin (HGB)	Cont.	Hemoglobin concentration	g/dL

Note: Cat. = Categorical; Cont. = Continuous.

Table 3 Summary of categorical and demographic features

Category	***n*** **(Percentage)**	
Gallstone Status		
No	161	(50.5%)
Yes	158	(49.5%)
Gender		
Male	162	(50.8%)
Female	157	(49.2%)
Comorbidity Score		
0	217	(68.0%)
1	99	(31.0%)
2	1	(0.3%)
3	2	(0.6%)
Coronary Artery Disease (CAD)		
No	307	(96.2%)
Yes	12	(3.8%)
Hypothyroidism		
No	310	(97.2%)
Yes	9	(2.8%)
Hyperlipidemia		
No	311	(97.5%)
Yes	8	(2.5%)
Diabetes Mellitus (DM)		
No	276	(86.5%)
Yes	43	(13.5%)
Hepatic Fat Accumulation (HFA)		
No Fat Accumulation	129	(40.4%)
Mild	41	(12.9%)
Moderate	122	(38.2%)
Severe	26	(8.2%)
Very Severe	1	(0.3%)

Note: Percentages may not total 100% due to rounding.

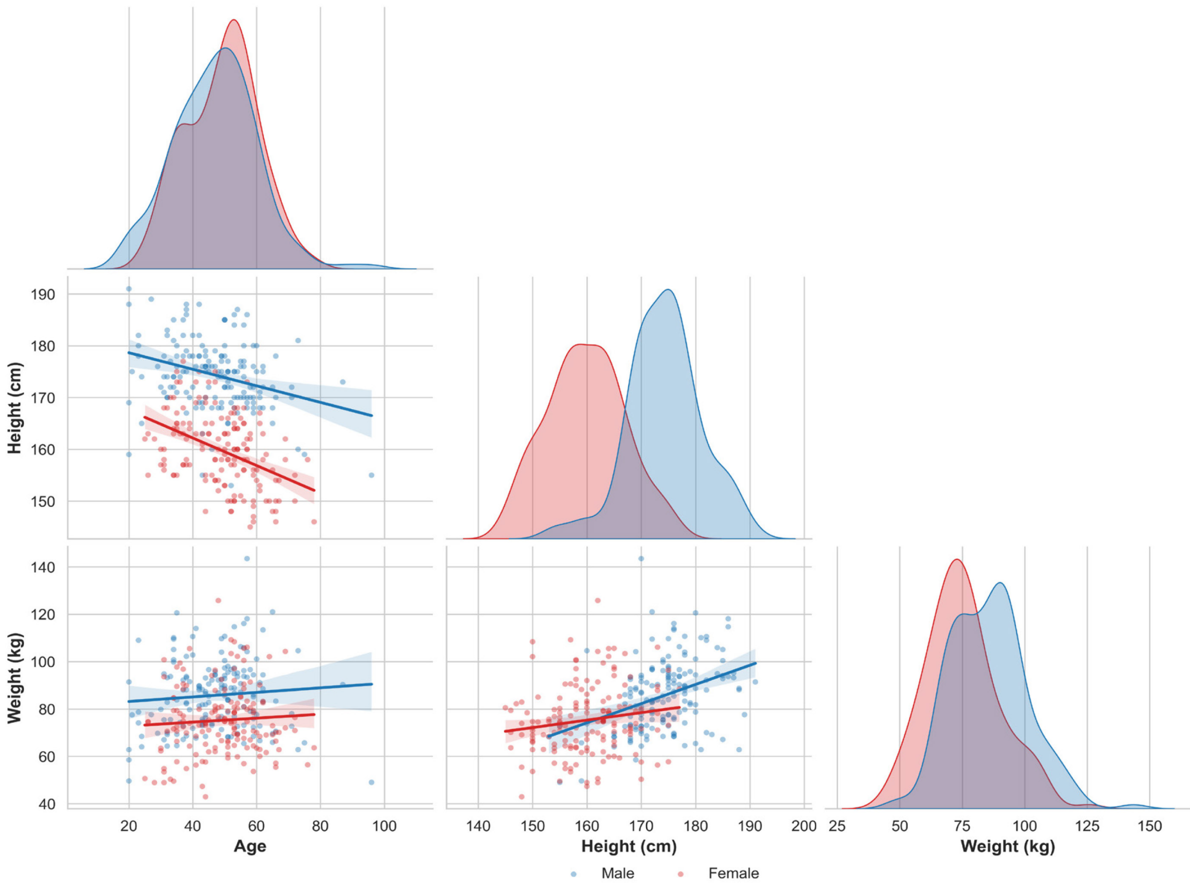

Figure 3 Demographic pair plots and density distributions by gender

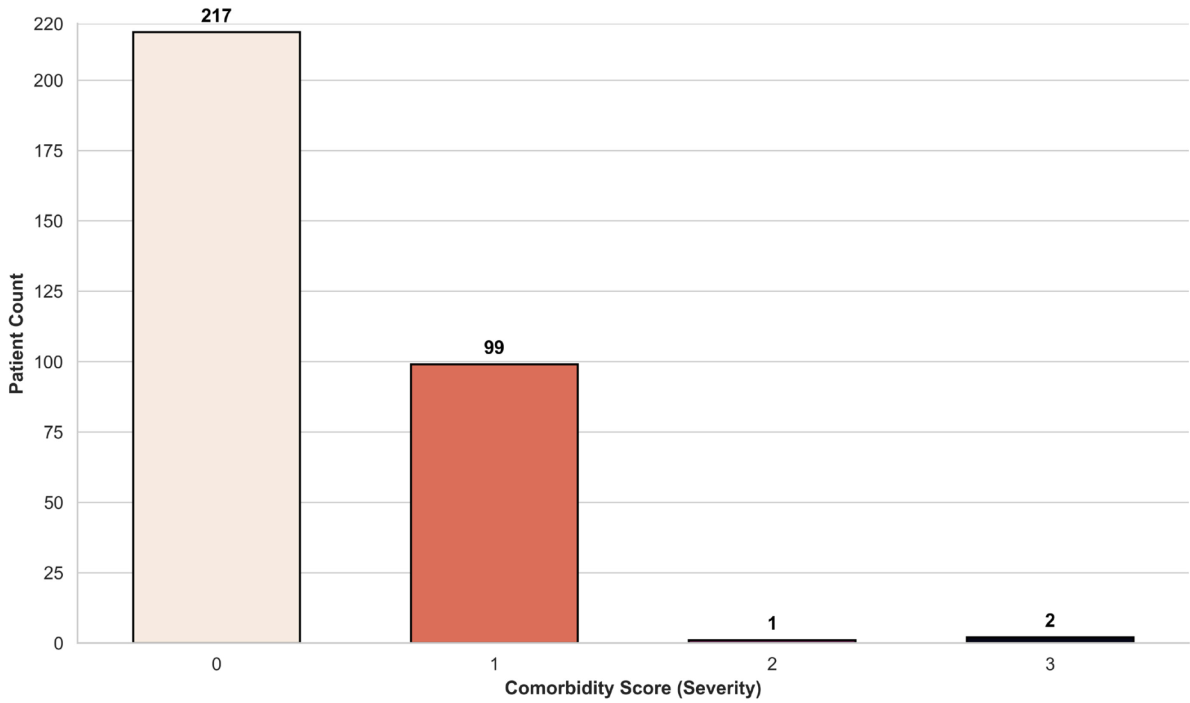

Figure 4 Distribution of comorbidity severity

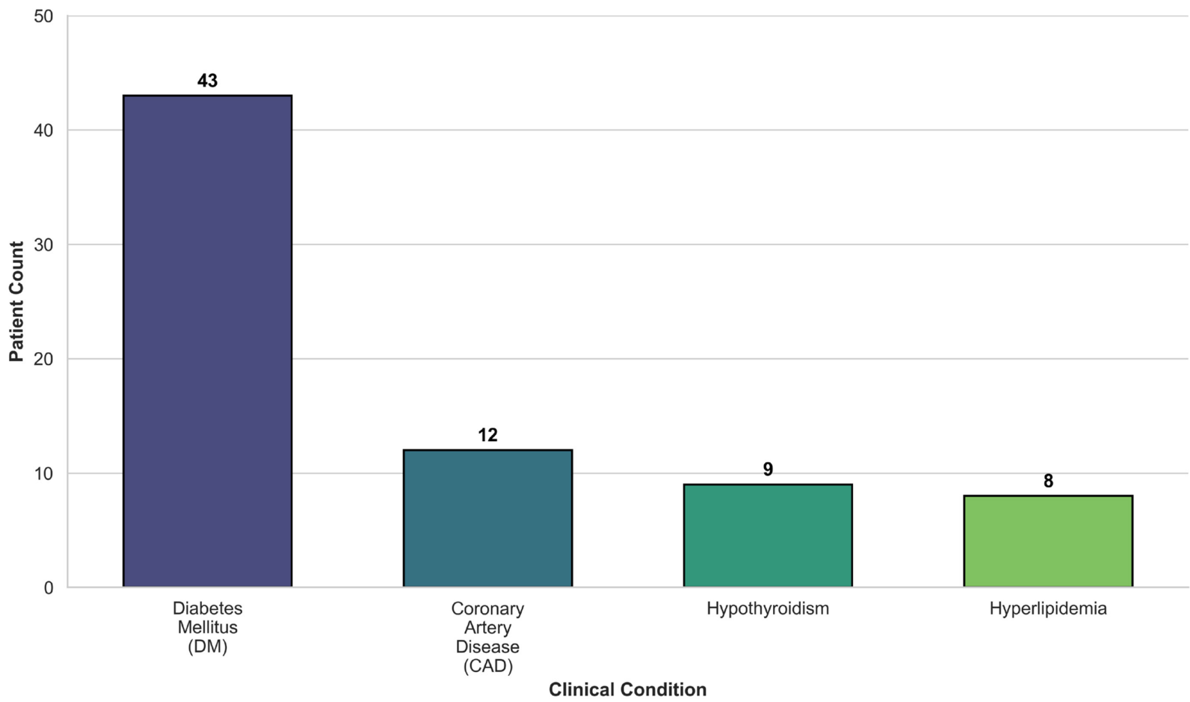

Figure 5 Prevalence of specific comorbidities among patients with gallstone disease

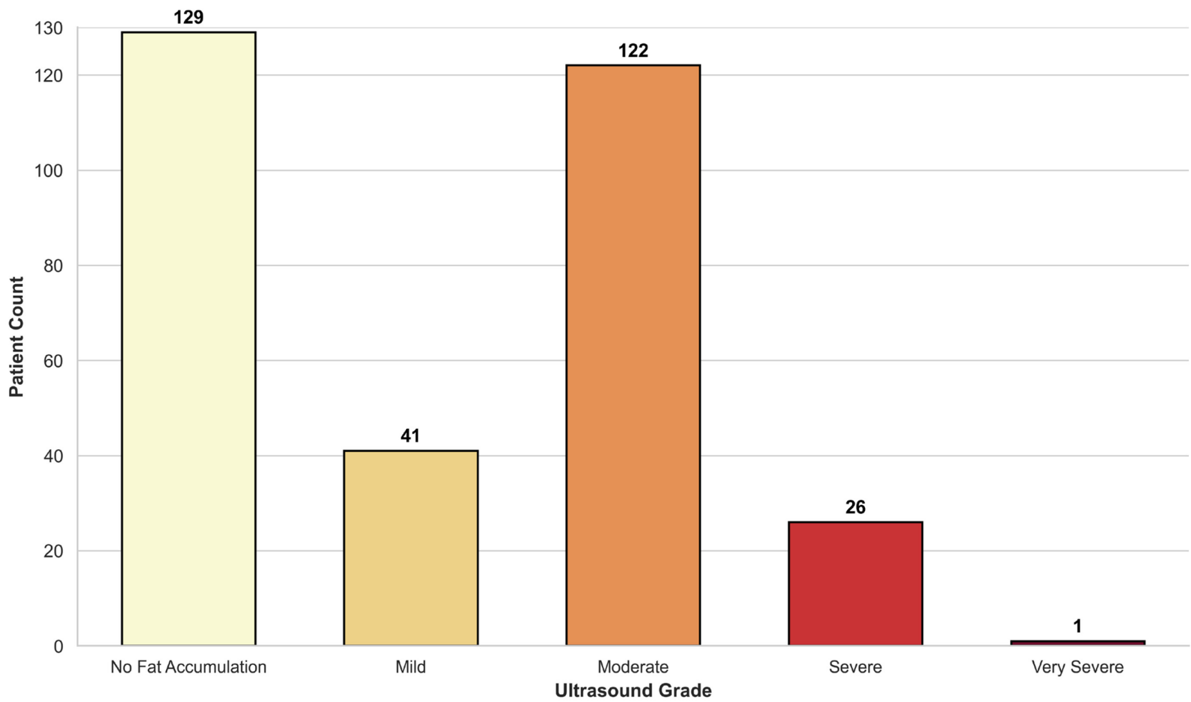

Figure 6 Ultrasound grading for hepatic fat accumulation

Table 4 Statistical summary of quantitative features in the dataset

Feature	*N*	Mean	*SD*	Min	Q_1	Median	Q_3	Max
Age	319	48.1	12.1	20.0	38.5	49.0	56.0	96.0
Height	319	167.2	10.1	145.0	159.5	168.0	175.0	191.0
Weight	319	80.6	15.7	42.9	69.6	78.8	91.2	143.5
Body Mass Index (BMI)	319	28.9	5.3	17.4	25.2	28.3	31.8	49.7
Total Body Water (TBW)	319	40.6	7.9	13.0	34.2	39.8	47.0	66.2
Extracellular Water (ECW)	319	17.1	3.2	9.0	14.8	17.1	19.4	27.8
Intracellular Water (ICW)	319	23.6	5.3	13.8	19.3	23.0	27.6	57.1
Extracellular Fluid /Total Body Water (ECF/TBW)	319	42.2	3.2	29.2	40.1	42.0	44.0	52.0
Total Body Fat Ratio (TBFR) (%)	319	28.3	8.4	6.3	22.0	27.8	34.8	50.9
Lean Mass (LM) (%)	319	71.6	8.4	49.0	65.2	72.1	77.9	93.7
Body Protein Content (Protein) (%)	319	15.9	2.3	5.6	14.5	15.9	17.4	24.8
Visceral Fat Rating (VFR)	319	9.1	4.3	1.0	6.0	9.0	12.0	31.0
Bone Mass (BM)	319	2.8	0.5	1.4	2.4	2.8	3.2	4.0
Muscle Mass (MM)	319	54.3	10.6	4.7	45.8	53.9	62.6	78.8
Obesity (%)	319	35.9	109.8	0.4	13.9	25.6	41.8	1954.0
Total Fat Content (TFC)	319	23.5	9.6	3.1	17.0	22.6	28.6	62.5
Visceral Fat Area (VFA)	319	12.2	5.3	0.9	8.6	11.6	15.1	41.0
Visceral Muscle Area (VMA) (Kg)	319	30.4	4.5	18.9	27.2	30.4	33.8	41.1
Glucose	319	108.7	44.8	69.0	92.0	98.0	109.0	575.0
Total Cholesterol (TC)	319	203.5	45.8	60.0	172.0	198.0	233.0	360.0
Low Density Lipoprotein (LDL)	319	126.7	38.5	11.0	100.5	122.0	151.0	293.0

High Density Lipoprotein (HDL)	319	49.5	17.7	25.0	40.0	46.5	56.0	273.0
Triglyceride	319	144.5	97.9	1.4	83.0	119.0	172.0	838.0
Aspartat Aminotransferaz (AST)	319	21.7	16.7	8.0	15.0	18.0	23.0	195.0
Alanin Aminotransferaz (ALT)	319	26.9	27.9	3.0	14.2	19.0	30.0	372.0
Alkaline Phosphatase (ALP)	319	73.1	24.2	7.0	58.0	71.0	86.0	197.0
Creatinine	319	0.8	0.2	0.5	0.7	0.8	0.9	1.5
Glomerular Filtration Rate (GFR)	319	100.8	17.0	10.6	94.2	104.0	110.7	132.0
C-Reactive Protein (CRP)	319	1.9	5.0	0.0	0.0	0.2	1.6	43.4
Hemoglobin (HGB)	319	14.4	1.8	8.5	13.3	14.4	15.7	18.8
Vitamin D	319	21.4	10.0	3.5	13.3	22.0	28.1	53.1

Note: N = sample size; SD = standard deviation; Q_1 = first quartile; Q_3 = third quartile.

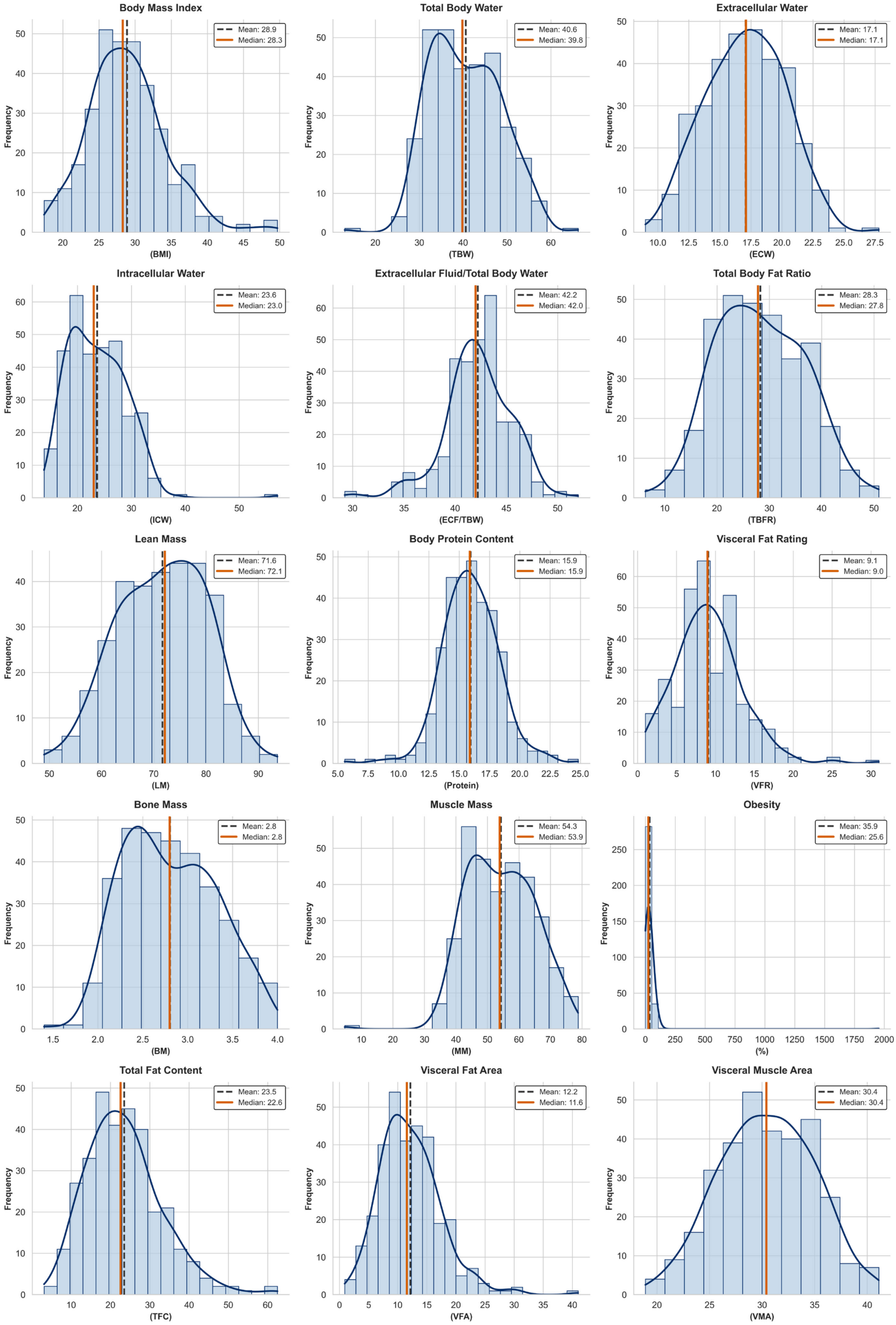

Figure 7 Frequency distributions of BIA features (histograms)

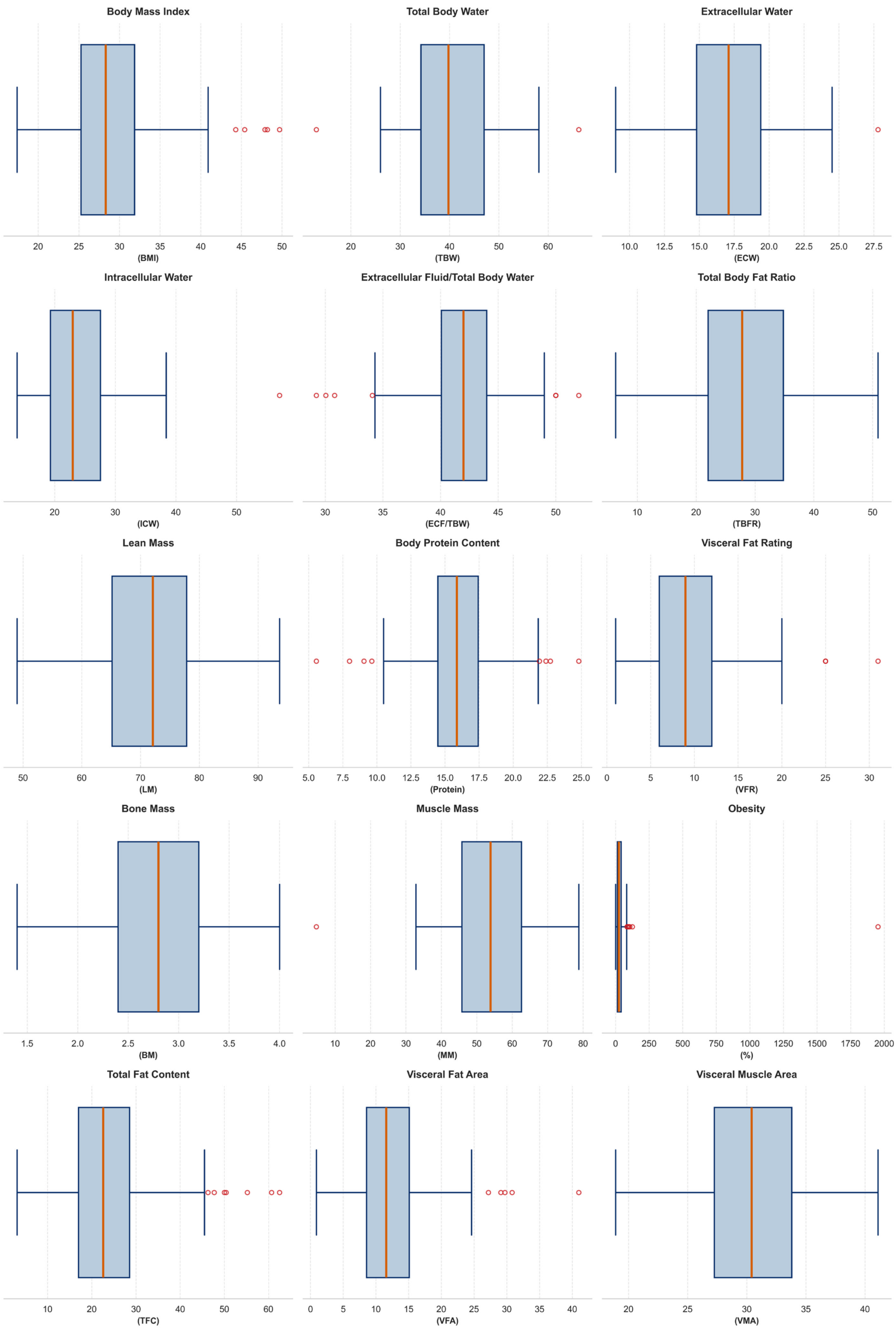

Figure 8 Distribution profiles and outliers of BIA features (boxplots)

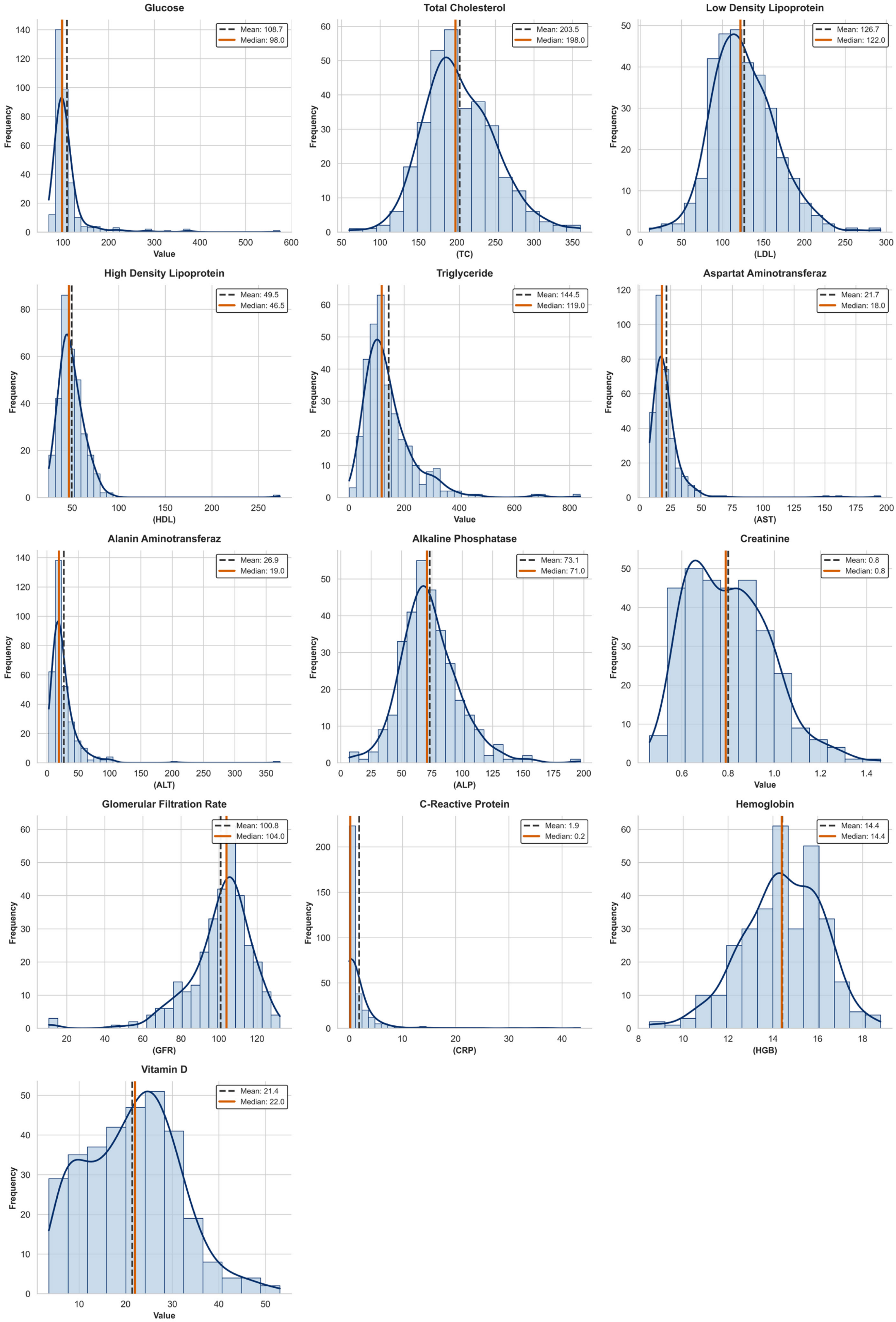

Figure 9 Frequency distributions of laboratory blood test features (histograms)

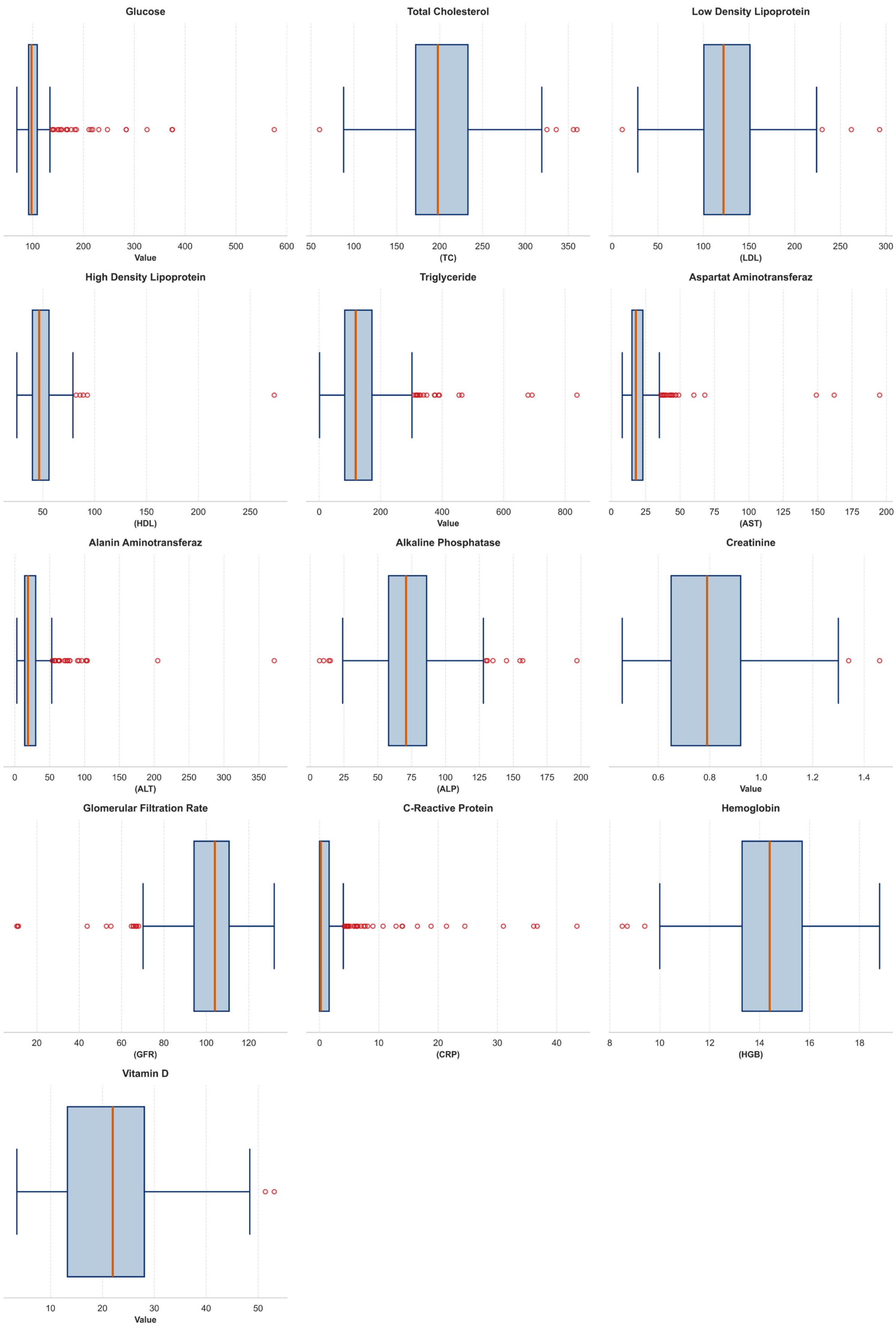

Figure 10 Distribution profiles and outliers of laboratory blood test features (boxplots)

A Pearson correlation analysis was conducted to illustrate the linear relationship between each of the 38 predictive features and the target variable (Gallstone Status). This analysis evaluates the initial linear association of these features with the target variable, rather than their absolute predictive performance. The results are presented in Table 5 and Figure 11.

Among the positively correlated features, C-Reactive Protein (CRP) exhibits the strongest association, with a coefficient of 0.2820, suggesting it is a primary marker linked to the disease within this dataset. This is closely followed by lipid and body fat metrics, such as Total Body Fat Ratio (TBFR) and Total Fat Content (TFC), yielding coefficients of 0.2255 and 0.1702, respectively. Furthermore, among the comorbidity variables, Hyperlipidemia exhibits the strongest positive correlation with gallstone status. This suggests an apparent positive association between lipid-related metabolism and gallstone disease.

Conversely, Vitamin D shows the most pronounced negative correlation, with a coefficient of -0.3549, suggesting a potential inverse relationship with disease prevalence. Unsurprisingly, the physiological counterpart of TBFR, Lean Mass (LM), demonstrates a coefficient of similar magnitude but in the opposite direction. Nevertheless, Bone Mass (BM), which is an absolute measure rather than a percentage, ranks as the third-strongest negative correlate with a coefficient of -0.2166. This indicates that the association between body fat metrics and gallstone risk extends beyond mere lipid metabolism; it likely involves a multifactorial relationship driven by latent variables such as overall body composition and lifestyle habits.

Interestingly, important metabolic and kidney function indicators, such as Glomerular Filtration Rate (GFR), Glucose, and Total Cholesterol (TC), exhibit negligible linear correlations. This implies that their direct linear contribution to gallstone prediction within this specific cohort is limited, although potential non-linear relationships cannot be ruled out.

Notably, the results show that Vitamin D exhibits the most negative coefficient and CRP the most positive coefficient in relation to label '1'. These findings support the target variable encoding where '1' represents gallstone patients and '0' represents healthy individuals, as the original paper identifies Vitamin D deficiency and elevated CRP levels as primary indicators of gallstone formation [32]. Further literature supports that elevated CRP directly indicates gallbladder disease severity [46]. This alignment suggests that the case counts reported in the original manuscript were likely typographical inconsistencies. Since this study adopted a full-feature approach to capture the overall physiological status, all features were retained for the subsequent optimisation process without being filtered or excluded.

Table 5 Pearson Correlation between Predictive Features and Gallstone Status

Feature	**Pearson Correlation Coefficient (*r*)**
Vitamin D	-0.3549
Lean Mass (LM) (%)	-0.2257
Bone Mass (BM)	-0.2166
Hemoglobin (HGB)	-0.1969
Extracellular Water (ECW)	-0.1784
Extracellular Fluid/Total Body Water (ECF/TBW)	-0.1698
Aspartat Aminotransferaz (AST)	-0.1349
Creatinine	-0.1323
Total Body Water (TBW)	-0.1112
Height	-0.1079
Coronary Artery Disease (CAD)	-0.0970
Muscle Mass (MM)	-0.0933
Body Protein Content (Protein) (%)	-0.0864
Visceral Muscle Area (VMA) (Kg)	-0.0730
Low Density Lipoprotein (LDL)	-0.0565
Hypothyroidism	-0.0552
Triglyceride	-0.0508
Comorbidity	-0.0485
Alanin Aminotransferaz (ALT)	-0.0419
Intracellular Water (ICW)	-0.0353
Glomerular Filtration Rate (GFR)	-0.0283
Glucose	-0.0115
Total Cholesterol (TC)	0.0141
Visceral Fat Rating (VFR)	0.0183
Age	0.0363
Weight	0.0487
Obesity (%)	0.0539
Hepatic Fat Accumulation (HFA)	0.0903
Diabetes Mellitus (DM)	0.1047
Alkaline Phosphatase (ALP)	0.1099
Body Mass Index (BMI)	0.1215
Visceral Fat Area (VFA)	0.1404
Gender	0.1535
High Density Lipoprotein (HDL)	0.1586

Hyperlipidemia	0.1619
Total Fat Content (TFC)	0.1702
Total Body Fat Ratio (TBFR) (%)	0.2255
C-Reactive Protein (CRP)	0.2820

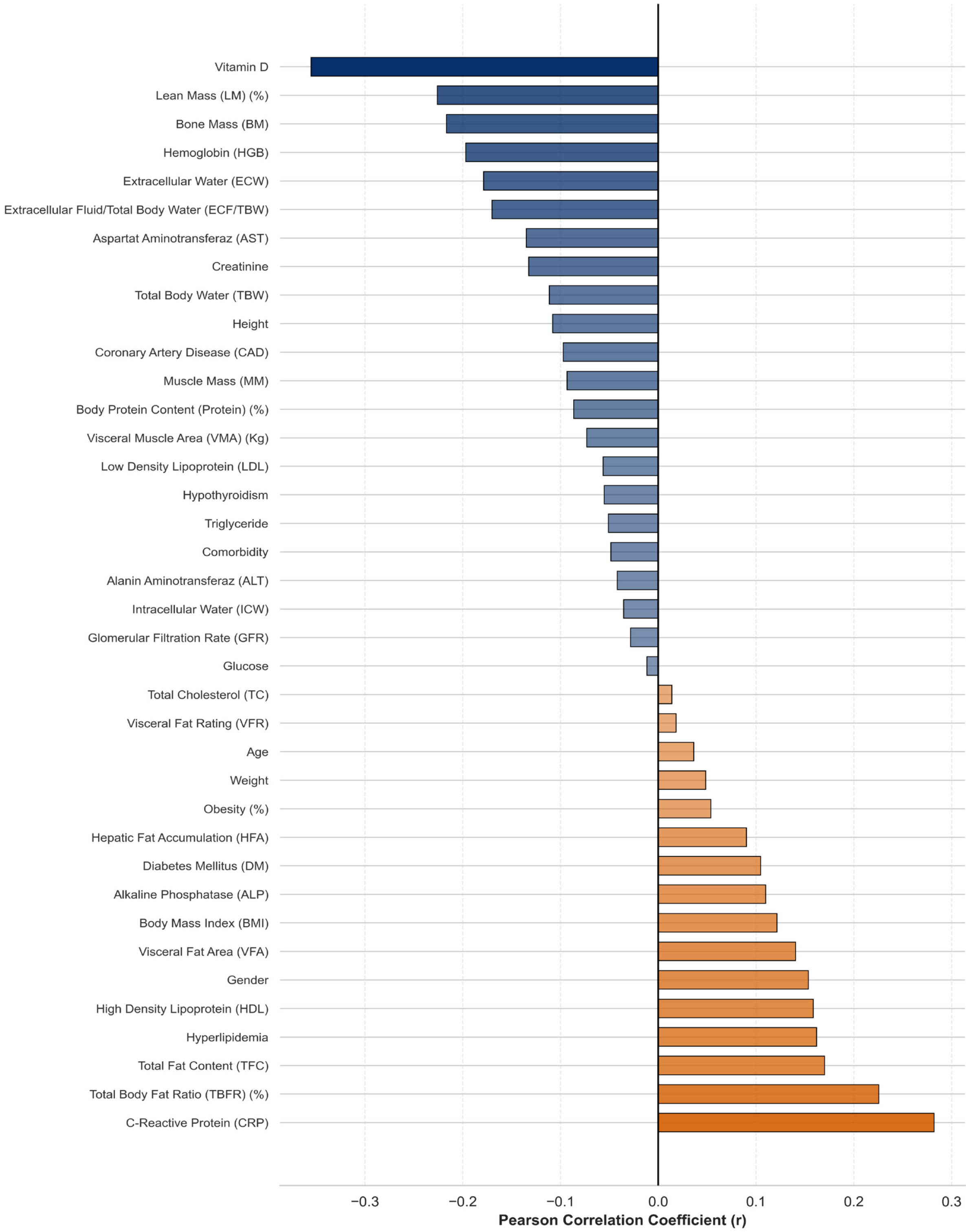

Figure 11 Pearson Correlation between Predictive Features and Gallstone Status

2.2 Data Validation and Correction

During the data validation and correction phase, demographic data and categorical comorbidity data were considered accurate ground truth and retained without further modification. Although the laboratory blood test data exhibited extreme distributions, these features were also retained without alteration; as previously discussed, due to their inherent sensitivity to pathological conditions, classifying the extreme values as errors without external clinical validation is unfeasible.

In contrast, the Bioimpedance Analysis (BIA) data were processed using logical flow constraints. Body Mass Index (BMI), Total Body Water (TBW), Total Fat Content (TFC), Body Protein Content (Protein), and their interdependent variables formed the core of this consistency evaluation. For a given feature $\boldsymbol{X}_a$ of sample j, consistency was evaluated by determining whether its observed value $x_j^{(a)}$ aligned with its theoretically computed value $\hat{x}_j^{(a)}$ derived from established physiological relationships. The discrepancy $\varepsilon_j^{(a)}$ between the observed and computed values was defined as Equation (1).

$$\varepsilon_j^{(a)} = x_j^{(a)} - \hat{x}_j^{(a)} \tag{1}$$

BMI served as a deterministic anchor feature, mathematically reliant on Weight and Height. Outliers in the observed BMI were identified and corrected using a modified Z-score threshold. If the discrepancy calculated from Equation (1) exceeded the predefined threshold, the observed value was replaced by the computed value as defined in Equation (2).

$$x_j^{(\mathrm{BMI})} = \begin{cases} \dfrac{100^2 \cdot x_j^{(\mathrm{Weight})}}{{x_j^{(\mathrm{Height})}}^2}, & \dfrac{0.6745^2 \times \left(\left|\varepsilon_j^{(\mathrm{BMI})} - \mathrm{median}_j\left(\varepsilon_j^{(\mathrm{BMI})}\right)\right|\right)}{\mathrm{median}_j\left(\left|\varepsilon_j^{(\mathrm{BMI})} - \mathrm{median}_j\left(\varepsilon_j^{(\mathrm{BMI})}\right)\right|\right)} > 3.5 \\ x_j^{(\mathrm{BMI})} & ,\mathrm{otherwise} \end{cases} \tag{2}$$

Figure 12 presents a Bland-Altman analysis of BMI consistency before and after the correction [47,48]. Regarding obesity, the analysis indicated the outlier of 1954% violated the normal relationship between BMI and obesity. This outlier was identified as a transcriptional error, while other outliers did not exhibit any violations. It was corrected via decadic shifting to align with the population's linear trend, as shown in Figure 13.

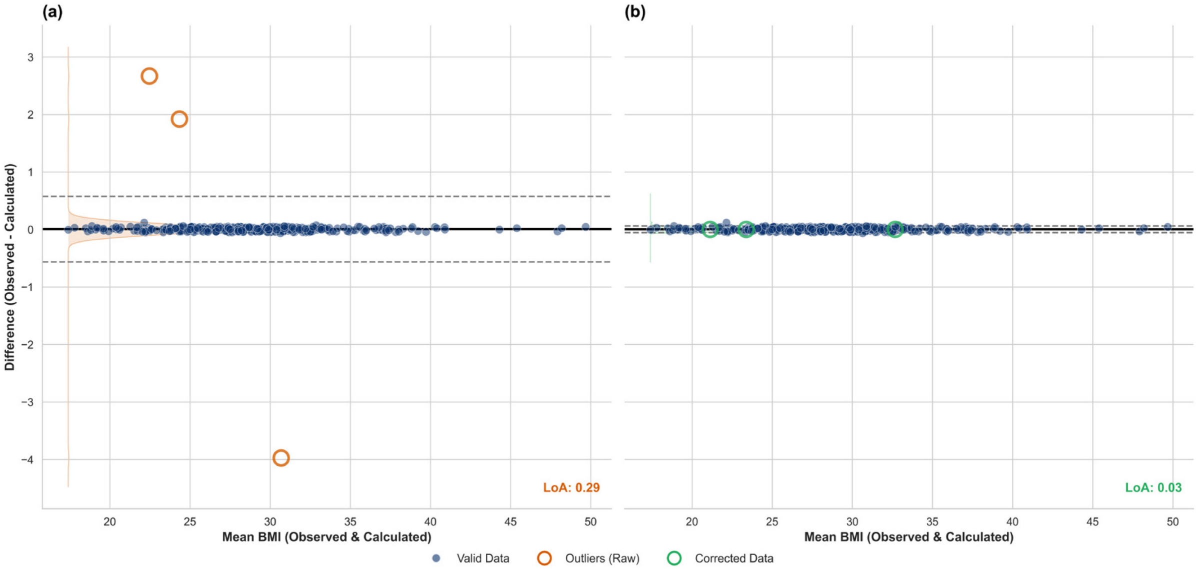

Figure 12 Bland-Altman analysis of BMI consistency
(a) Before correction (b) After correction

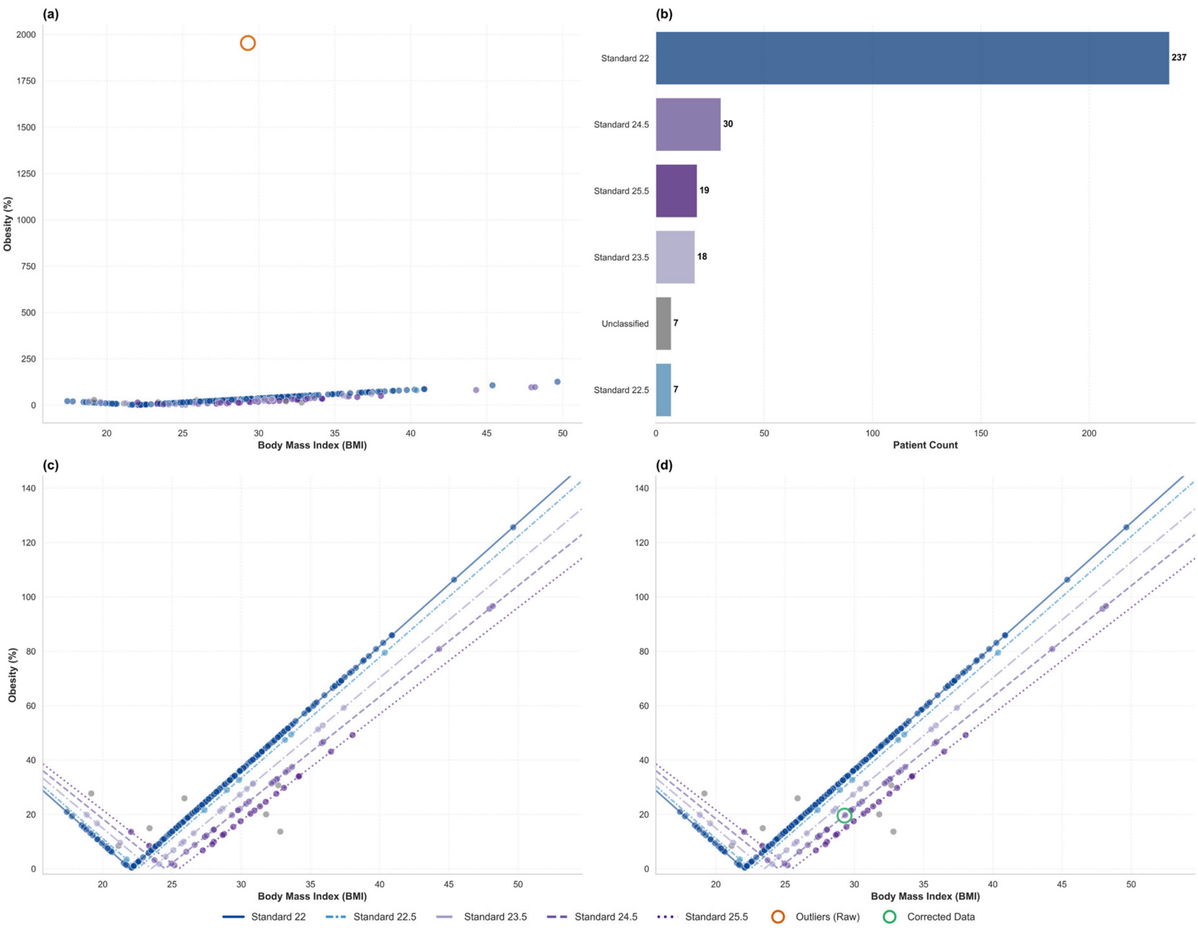

Figure 13 Outlier correction and standard-relative trends of obesity
(a) Raw data displayed on a global scale (b) Population profile distribution (c) Raw data excluded outlier (d) Data displayed with corrected outlier

Among the features not mathematically reliant on the accurate ground truth, the integrity of each feature $\boldsymbol{X}_a$ was evaluated using a robust statistical threshold, τ_a , which was derived from its associated reference variables Ref_a (TBW or TFC). The threshold was calculated based on the Median Absolute Deviation (MAD), as shown in Equation (3a), to ensure resilience against extreme outliers, resulting in Equation (3b).

$$\mathrm{MAD}_a^{\mathrm{normal}} = 1.4826 \cdot \mathrm{median}_j \left(\left| \varepsilon_j^{(a)} - \mathrm{median}_j \left(\varepsilon_j^{(a)} \right) \right| \right) \tag{3a}$$

$$\tau_a^{stat} = \alpha_{\mathrm{MAD}} \cdot \mathrm{MAD}_a^{\mathrm{normal}} \tag{3b}$$

where $\alpha_{\mathrm{MAD}} = 3.5$.

To prevent overly stringent limits in cases of near-zero variance, a dynamic floor threshold, τ_a^{floor}, was established based on the availability of reference variables as in Equation (3c).

$$\tau_a^{\mathrm{floor}} = \begin{cases} 0.015 \cdot \left| \mathrm{median}_j \left(x_j^{(a)} \right) \right| & , \mathrm{Ref}_a \neq \emptyset \\ 0.5 & , \mathrm{Ref}_a = \emptyset \end{cases} \tag{3c}$$

The final robust limit was then defined as the maximum of Equation (3b) and Equation (3c), denoted as Equation (3d).

$$\tau_a = \max\left(\tau_a^{\mathrm{stat}}, \tau_a^{\mathrm{floor}} \right) \tag{3d}$$

During the evaluation phase, if any inconsistencies within a set of interdependent relational constraints exceeded this robust limit (τ_a), yet one specific constraint remained strictly within bounds, that bounded constraint was deemed reliable. Consequently, the single aberrant variable isolated by this valid constraint was identified as the source of the residual error and was subsequently corrected by satisfying the established relationship. Conversely, samples exhibiting inconsistencies across all relational constraints were removed from the dataset, as no reliable physiological reference remained to anchor a valid correction.

The subsequent process details the correction of body water-related features. To cross-validate their interdependent physiological relationships, TBW and its associated parameters were evaluated across four distinct computational tracks. This generated specific discrepancy metrics involving TBW, Intracellular Water (ICW), Extracellular Water (ECW), and the Extracellular Fluid to Total Body Water ratio (ECF/TBW), the tracks are defined as follows in Equations (4) to (7).

$$\varepsilon_j^{(\mathrm{TBW}_{\mathrm{Track\,A}})} = x_j^{(\mathrm{TBW})} - \left(x_j^{(\mathrm{ICW})} + x_j^{(\mathrm{ECW})}\right) \tag{4}$$

$$\varepsilon_j^{(\mathrm{TBW}_{\mathrm{Track\,B}})} = x_j^{(\mathrm{TBW})} - \left(\frac{100 \cdot x_j^{(\mathrm{ECW})}}{x_j^{(\mathrm{ECF/TBW})}}\right) \tag{5}$$

$$\varepsilon_j^{(\mathrm{RatioSum}_{\mathrm{Track\,C}})} = 100 - \left(x_j^{(\mathrm{ECF/TBW})} + \left(\frac{100 \cdot x_j^{(\mathrm{ICW})}}{x_j^{(\mathrm{TBW})}}\right)\right) \tag{6}$$

$$\varepsilon_j^{(\mathrm{TBW}_{\mathrm{Track\,D}})} = \left(x_j^{(\mathrm{ICW})} + x_j^{(\mathrm{ECW})}\right) - \left(\frac{100 \cdot x_j^{(\mathrm{ECW})}}{x_j^{(\mathrm{ECF/TBW})}}\right) \tag{7}$$

The related isolated aberrant features were then corrected to their computed theoretical values, as shown in Equations (8) to (11).

$$x_j^{(\mathrm{ECF/TBW})} = \begin{cases} \dfrac{100 \cdot x_j^{(\mathrm{ECW})}}{x_j^{(\mathrm{TBW})}}, & \left|\varepsilon_j^{(\mathrm{TBW}_{\mathrm{Track\,B}})}\right| > \tau_{\mathrm{TBW}_{\mathrm{Track\,B}}} \\ & \text{and } \left|\varepsilon_j^{(\mathrm{TBW}_{\mathrm{Track\,A}})}\right| < \tau_{\mathrm{TBW}_{\mathrm{Track\,A}}} \\ x_j^{(\mathrm{ECF/TBW})}, & \text{otherwise} \end{cases} \tag{8}$$

$$x_j^{(\mathrm{ICW})} = \begin{cases} x_j^{(\mathrm{TBW})} - x_j^{(\mathrm{ECW})}, & \left|\varepsilon_j^{(\mathrm{TBW}_{\mathrm{Track\,A}})}\right| > \tau_{\mathrm{TBW}_{\mathrm{Track\,A}}} \\ & \text{and } \left|\varepsilon_j^{(\mathrm{TBW}_{\mathrm{Track\,B}})}\right| < \tau_{\mathrm{TBW}_{\mathrm{Track\,B}}} \\ x_j^{(\mathrm{ICW})}, & \text{otherwise} \end{cases} \tag{9}$$

$$x_j^{(\mathrm{ECW})} = \begin{cases} x_j^{(\mathrm{TBW})} - x_j^{(\mathrm{ICW})}, & \left|\varepsilon_j^{(\mathrm{TBW}_{\mathrm{Track\,A}})}\right| > \tau_{\mathrm{TBW}_{\mathrm{Track\,A}}} \\ & \text{and } \left|\varepsilon_j^{(\mathrm{RatioSum}_{\mathrm{Track\,C}})}\right| < \tau_{\mathrm{RatioSum}_{\mathrm{Track\,C}}} \\ x_j^{(\mathrm{ECW})}, & \text{otherwise} \end{cases} \tag{10}$$

$$x_j^{(\mathrm{TBW})} = \begin{cases} x_j^{(\mathrm{ICW})} + x_j^{(\mathrm{ECW})}, & \left|\varepsilon_j^{(\mathrm{TBW}_{\mathrm{Track\,A}})}\right| > \tau_{\mathrm{TBW}_{\mathrm{Track\,A}}} \\ & \text{and } \left|\varepsilon_j^{(\mathrm{TBW}_{\mathrm{Track\,D}})}\right| < \tau_{\mathrm{TBW}_{\mathrm{Track\,D}}} \\ x_j^{(\mathrm{TBW})}, & \text{otherwise} \end{cases} \tag{11}$$

A comparison of the consistency results across these multiple body water correction tracks is illustrated in Figure 14.

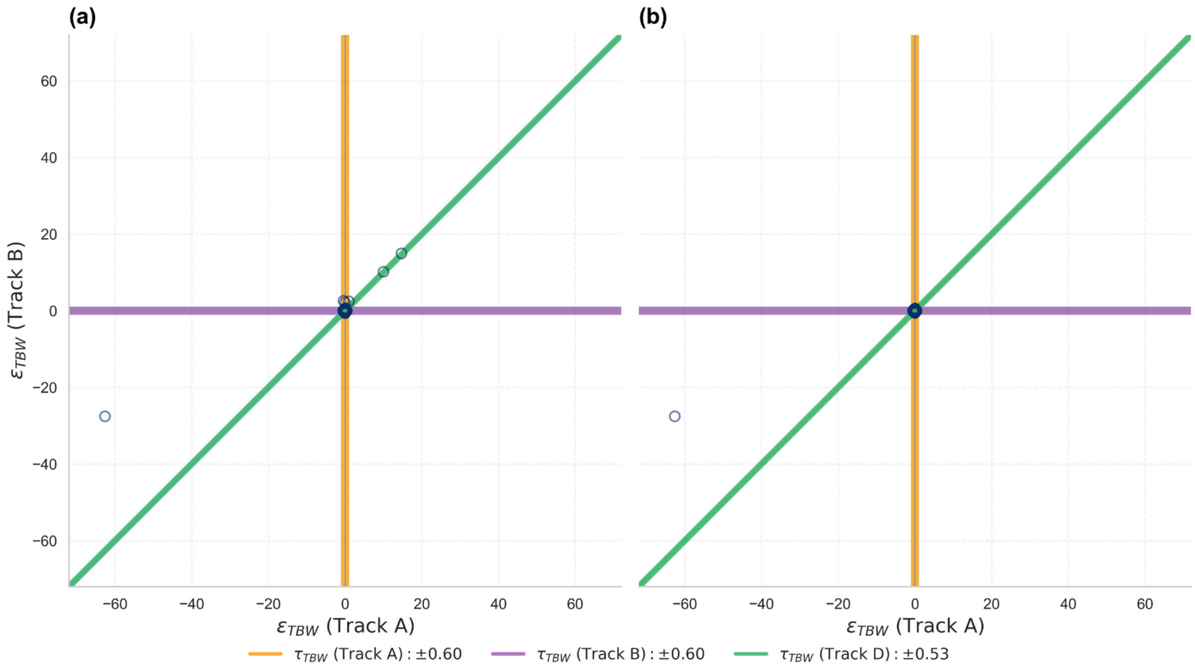

Figure 14 Multi-track consistency evaluation of TBW discrepancies (a) Before correction (b) After correction

Following this, the correction process of body fat-related features was detailed. Similar to the body water evaluation, TFC and its associated parameters, including Total Body Fat Ratio (TBFR) and Lean Mass (LM), were cross-validated across three distinct computational tracks. This generated specific discrepancy metrics, with Equations (12) to (14) establishing the relationships between these variables and the patient's Weight.

$$\varepsilon_j^{(\mathrm{TFC_{Track\ E}})} = x_j^{(\mathrm{TFC})} - \left(x_j^{(\mathrm{Weight})} - \frac{x_j^{(\mathrm{Weight})} \cdot x_j^{(\mathrm{LM})}}{100} \right) \tag{12}$$

$$\varepsilon_j^{(\mathrm{TFC_{Track\ F}})} = x_j^{(\mathrm{TFC})} - \left(\frac{x_j^{(\mathrm{Weight})} \cdot x_j^{(\mathrm{TBFR})}}{100} \right) \tag{13}$$

$$\varepsilon_j^{(\mathrm{RatioSum_{Track\ G}})} = 100 - \left(x_j^{(\mathrm{LM})} + x_j^{(\mathrm{TBFR})} \right) \tag{14}$$

Applying the same logical constraints, the related isolated aberrant features were corrected as shown in Equations (15) to (17).

$$x_j^{(\mathrm{TBFR})} = \begin{cases} \dfrac{100 \cdot x_j^{(\mathrm{TFC})}}{x_j^{(\mathrm{Weight})}}, & \left|\varepsilon_j^{(\mathrm{TFC_{Track\,F}})}\right| > \tau_{\mathrm{TFC_{Track\,F}}} \\ & \text{and} \left|\varepsilon_j^{(\mathrm{TFC_{Track\,E}})}\right| < \tau_{\mathrm{TFC_{Track\,E}}} \\ x_j^{(\mathrm{TBFR})}, & \text{otherwise} \end{cases} \tag{15}$$

$$x_j^{(\mathrm{LM})} = \begin{cases} 100 - x_j^{(\mathrm{TBFR})}, & \left|\varepsilon_j^{(\mathrm{TFC_{Track\,E}})}\right| > \tau_{\mathrm{TFC_{Track\,E}}} \\ & \text{and} \left|\varepsilon_j^{(\mathrm{TFC_{Track\,F}})}\right| < \tau_{\mathrm{TFC_{Track\,F}}} \\ x_j^{(LM)}, & \text{otherwise} \end{cases} \tag{16}$$

$$x_j^{(\mathrm{TFC})} = \begin{cases} \dfrac{x_j^{(\mathrm{Weight})} \cdot x_j^{(\mathrm{TBFR})}}{100}, & \left(\forall a \in \{\mathrm{TFC_{Track\,E}}, \mathrm{TFC_{Track\,F}}\}, \left|\varepsilon_j^{(a)}\right| > \tau_a\right) \\ & \text{and} \left|\varepsilon_j^{(\mathrm{RatioSum_{Track\,G}})}\right| < \tau_{\mathrm{RatioSum_{Track\,G}}} \\ x_j^{(\mathrm{TFC})}, & \text{otherwise} \end{cases} \tag{17}$$

A comparison of the consistency results across these multiple body fat correction tracks is illustrated in Figure 15.

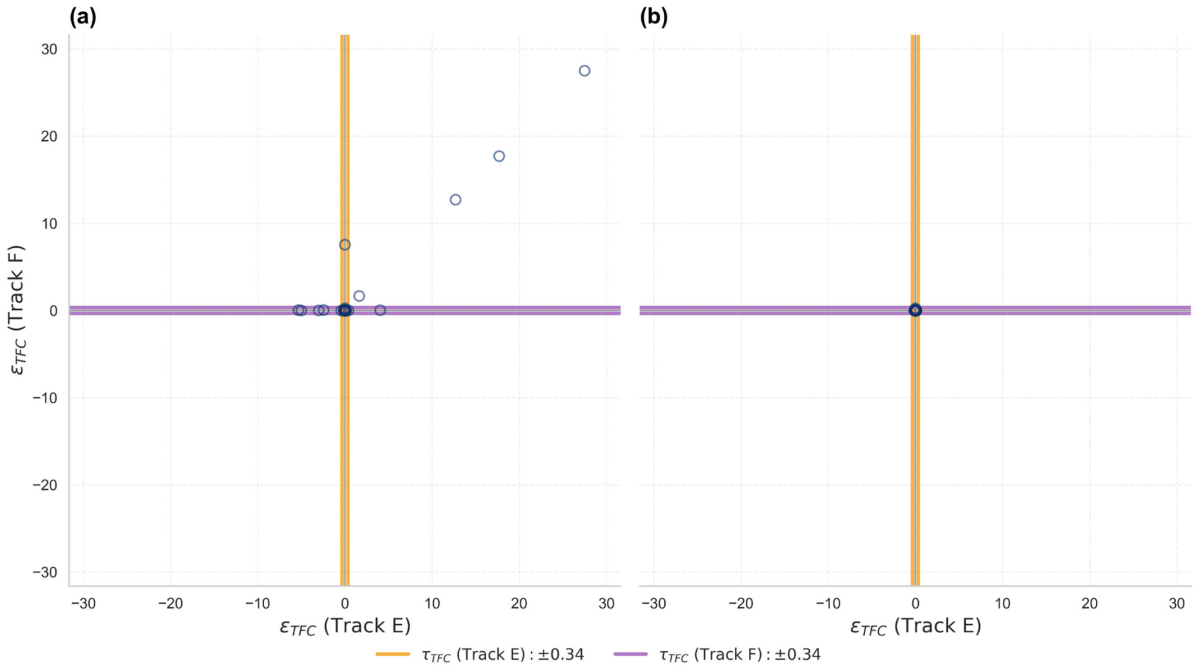

Figure 15 Multi-track consistency evaluation of TFC discrepancies (a) Before correction (b) After correction

The correction protocol for protein and muscle-related features, including Muscle Mass (MM), Visceral Muscle Area (VMA), and Protein, followed a distinct methodology. Given the specific physiological distributions of these variables, anomalies were flagged using a robust 95% interquantile range. A binary error indicator, shown in Equation (18), was generated for each evaluated parameter, assigning 0 if the value fell within the acceptable boundaries and 1 if it constituted an error.

$$\varepsilon_j^{(a)} = \begin{cases} 0, Q_{0.025}(\boldsymbol{X}_a) \leq x_j^{(a)} \leq Q_{0.975}(\boldsymbol{X}_a) \\ 1, \text{otherwise} \end{cases} \tag{18}$$

where $a \in A = \left\{\text{MM}, \text{VMA}, \text{Protein}, \frac{\text{VMA}}{\text{MM}}, \frac{\text{Weight}\cdot\text{Protein}}{100\cdot\text{MM}}, \frac{\text{Weight}\cdot\text{Protein}}{100\cdot\text{VMA}}\right\}$, the latter three features were derivations from the first three initial features for temporary calculations.

The anomaly features were corrected as shown in Equations (19) to (21).

$$x_j^{(\text{MM})} = \begin{cases} \hat{x}_j^{(\text{MM})}, & \left(\prod_{a\in A} \varepsilon_j^{(a)}\right) - \varepsilon_j^{\left(\frac{\text{Weight}\cdot\text{Protein}}{100\cdot\text{VMA}}\right)} = 1 \\ & \text{where } A = \left\{\text{MM}, \frac{\text{VMA}}{\text{MM}}, \frac{\text{Weight}\cdot\text{Protein}}{100\cdot\text{MM}}\right\} \\ x_j^{(\text{MM})}, & \text{otherwise} \end{cases} \tag{19}$$

$$x_j^{(\text{VMA})} = \begin{cases} \hat{x}_j^{(\text{VMA})}, & \left(\prod_{a\in A} \varepsilon_j^{(a)}\right) - \varepsilon_j^{\left(\frac{\text{Weight}\cdot\text{Protein}}{100\cdot\text{MM}}\right)} = 1 \\ & \text{where } A = \left\{\text{VMA}, \frac{\text{VMA}}{\text{MM}}, \frac{\text{Weight}\cdot\text{Protein}}{100\cdot\text{VMA}}\right\} \\ x_j^{(\text{VMA})}, & \text{otherwise} \end{cases} \tag{20}$$

$$x_j^{(\text{Protein})} = \begin{cases} \hat{x}_j^{(\text{Protein})}, & \left(\prod_{a\in A} \varepsilon_j^{(a)}\right) - \varepsilon_j^{\left(\frac{\text{VMA}}{\text{MM}}\right)} = 1 \\ & \text{where } A = \left\{\text{Protein}, \frac{\text{Weight}\cdot\text{Protein}}{100\cdot\text{MM}}, \frac{\text{Weight}\cdot\text{Protein}}{100\cdot\text{VMA}}\right\} \\ x_j^{(\text{Protein})}, & \text{otherwise} \end{cases} \tag{21}$$

where the theoretically computed values were derived as follows in Equations (22) to (24).

$$\hat{x}_j^{(\text{MM})} = \frac{1}{2}\left(\frac{x_j^{(\text{VMA})}}{\text{median}_j\left(x_j^{\left(\frac{\text{VMA}}{\text{MM}}\right)}\right)} + \frac{x_j^{\left(\frac{\text{Weight}\cdot\text{Protein}}{100}\right)}}{\text{median}_j\left(x_j^{\left(\frac{\text{Weight}\cdot\text{Protein}}{100\cdot\text{MM}}\right)}\right)}\right) \tag{22}$$

$$\hat{x}_j^{(\text{VMA})} = \frac{1}{2}\left(x_j^{(\text{MM})}\cdot\text{median}_j\left(x_j^{\left(\frac{\text{VMA}}{\text{MM}}\right)}\right) + \frac{x_j^{\left(\frac{\text{Weight}\cdot\text{Protein}}{100}\right)}}{\text{median}_j\left(x_j^{\left(\frac{\text{Weight}\cdot\text{Protein}}{100\cdot\text{VMA}}\right)}\right)}\right) \tag{23}$$

$$\hat{x}_j^{(\text{Protein})} = \frac{100 \cdot \left(x_j^{(\text{MM})} \cdot \text{median}_j \left(x_j^{\left(\frac{\text{Weight}\cdot\text{Protein}}{100\cdot\text{MM}}\right)} \right) + x_j^{(\text{VMA})} \cdot \text{median}_j \left(x_j^{\left(\frac{\text{Weight}\cdot\text{Protein}}{100\cdot\text{VMA}}\right)} \right) \right)}{2 \cdot x_j^{(\text{Weight})}} \tag{24}$$

A comparison of the consistency results for protein and muscle-related features, illustrating the empirical distributions before and after correction, is presented in Figure 16.

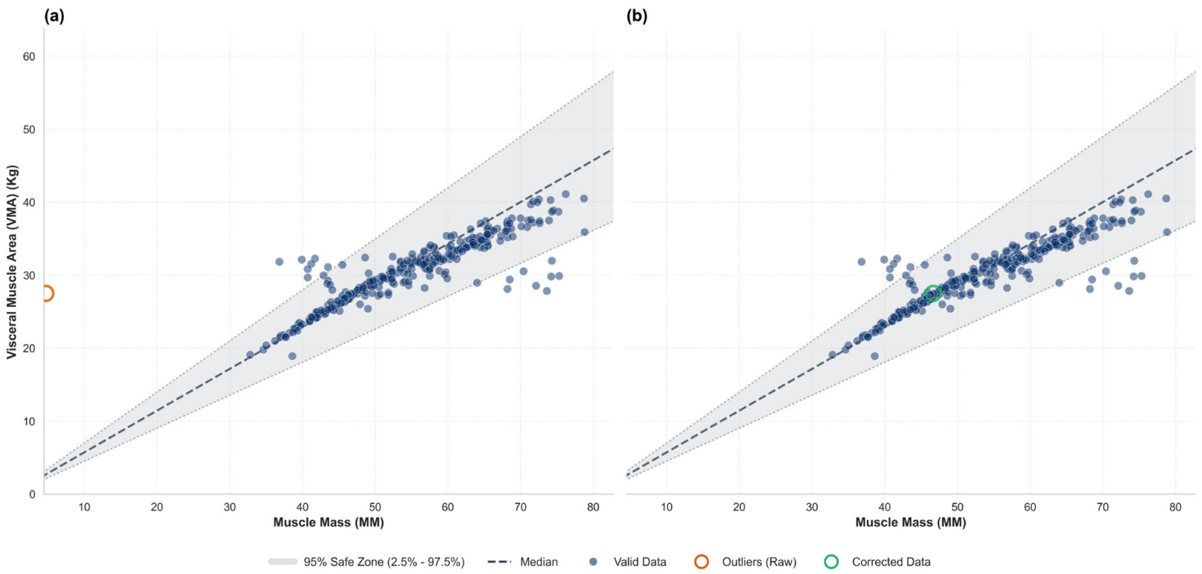

Figure 16 Quantile-based consistency evaluation of VMA–MM discrepancies
(a) Before correction (b) After correction

The corrected data samples for TBW, TFC, and Protein were subsequently validated against the patient's total weight using a four-compartment (4C) model constraint (body weight = total body water + fat mass + protein mass + bone mineral content), the value calculated for validation was calculated as shown in Equation (25).

$$\hat{x}_j^{(\text{Weight})} = x_j^{(\text{TBW})} + x_j^{(\text{TFC})} + x_j^{\left(\frac{\text{Weight}\cdot\text{Protein}}{100}\right)} + \varepsilon_j^{4\text{C}} \tag{25}$$

where $\varepsilon_j^{4\text{C}} = \alpha_{4\text{C}} \times x_j^{(\text{Weight})}$ represents the bone mineral content not in dataset. The parameter $\alpha_{4\text{C}}$ denotes the proportional coefficient of the trace element relative to total body mass, calculated as shown in Equation (26).

$$\alpha_{4\text{C}} = \text{median}_j \left(1 - \frac{x_j^{(\text{TBW})} + x_j^{(\text{TFC})} + x_j^{\left(\frac{\text{Weight}\cdot\text{Protein}}{100}\right)}}{x_j^{(\text{Weight})}} \right) \tag{26}$$

Figure 17 presents a Bland-Altman analysis of the 4C model mass balance consistency before and after the application of these BIA corrections [47–49].

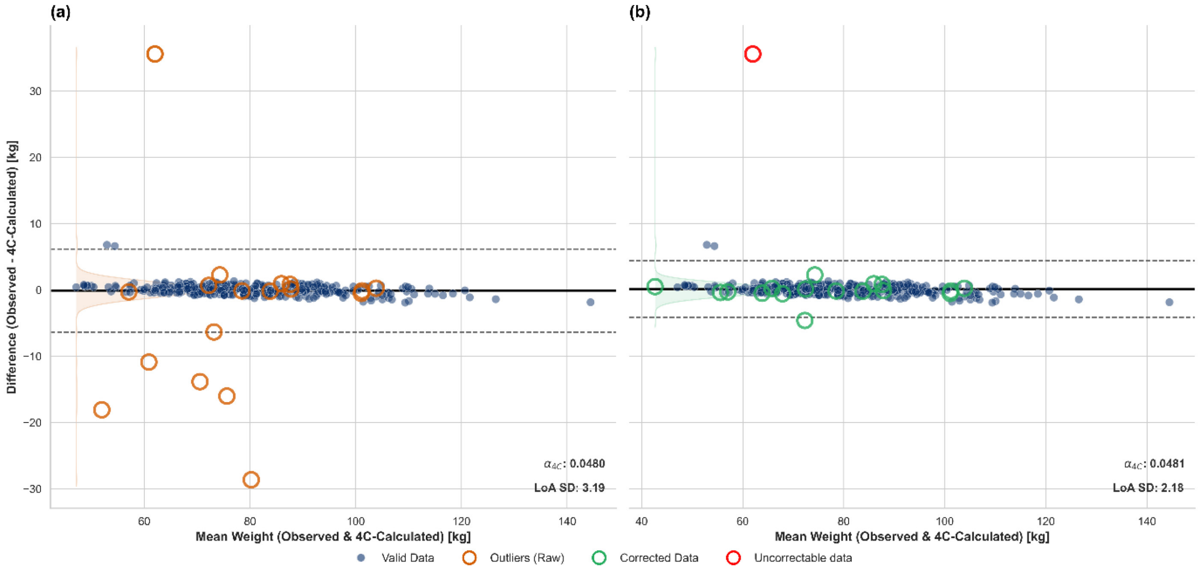

Figure 17 Bland-Altman analysis of 4C model mass balance consistency
(a) Before correction (b) After correction

The red data point represents a sample that exhibited inconsistencies across all relational tracks for a specific parameter, either TBW, TFC, or Protein. Since no reliable physiological reference remained within that parameter to anchor a valid correction, this single sample was subsequently excluded.

An additional post-correction Pearson analysis was conducted to confirm that the BIA adjustments maintained the underlying feature correlations while preserving data integrity. These results are detailed in Table A3 (Appendix C), confirming that the established dataset was not severely distorted prior to the subsequent predictive modelling phase.

3 Methodology

Figure 18 outlines the overall methodological workflow for the subsequent process based on the previously validated dataset.

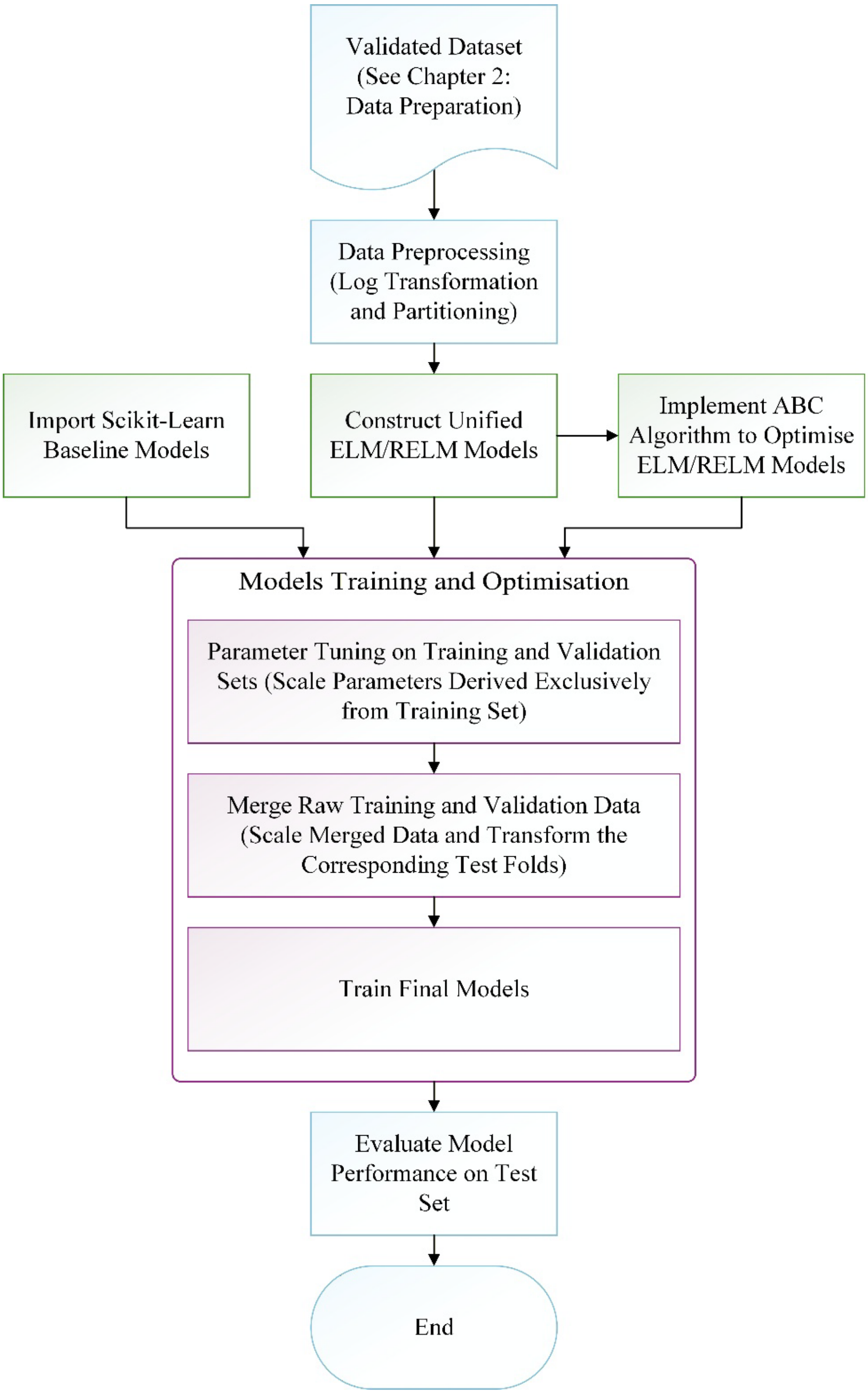

Figure 18 Overall methodological workflow of the training and evaluation process

3.1 Data Splitting and Preprocessing

As approximately half of the laboratory blood test features exhibited extreme right-skewness, a $\log(1 + x)$ transformation was implemented as the initial preprocessing step. To maintain methodological objectivity, this was applied uniformly to all laboratory features, thereby avoiding arbitrary, feature-specific adjustments.

To robustly evaluate the generalisation ability of the models, the dataset was partitioned using a 5-fold cross-validation (CV) approach incorporating a dedicated validation set. Within each fold, 20% of the data was isolated as an unseen testing set, while the remaining 80% was further divided at an 80:20 ratio into a 64% training set and a 16% validation set. As different models exhibit varying sensitivities to data scaling, the scaling strategies were tailored to accommodate each specific model. While raw data was retained for scale-invariant models, a Min-Max Scaler was applied to the remaining models to explicitly constrain the data to the range $[0, 1]$, as defined in Equation (27):

$$x_{j,\text{scaled}}^{(a)} = \frac{x_j^{(a)} - x_{\text{min}}^{(a)}}{x_{\text{max}}^{(a)} - x_{\text{min}}^{(a)}} \tag{27}$$

Subsequently, data normalisation was performed dynamically to prevent data leakage. The scaling parameters were computed solely from the active training set and subsequently applied to the corresponding validation or testing sets, including within any internal K-fold CV subsets. During the final evaluation phase, the overall performance of the models was evaluated across the five outer folds. Specifically, the models were trained on the combined 80% dataset (comprising the merged training and validation sets) and evaluated on the 20% unseen testing set; correspondingly, the scaler was fitted exclusively on this 80% dataset before being used to transform the testing set.

3.2 Risk-Averse Evaluation

In ML studies, multiple experimental factors are often interdependent. To mitigate the evaluation noise induced by these interdependent factors and to obtain robust performance estimates during model tuning, the specific model architectures are evaluated using a risk-averse strategy. Specifically, a hierarchical risk-averse scheme is employed, evaluating a specific target factor by conditioning it on predefined blocking factors. For instance, the overall performance of a model initialised with a specific random seed can be robustly evaluated by treating the K-fold CV performances as the blocking factor.

To evaluate the first-level factor, the risk-adjusted score $\mathrm{LCB}_F^{(1)}$ is applied to assess individual stability based exclusively on its variance. Given a target metric t with W observations under a specific blocking factor F, the risk-adjusted score $\mathrm{LCB}_F^{(1)}$ after isolating factor F is calculated as follows in Equation (28):

$$\begin{cases} \mu_F = \dfrac{1}{W}\sum_{z=1}^{W} t_z^{(F)} \\ \sigma_F = \sqrt{\dfrac{\sum_{z\&=1}^{W}\left(t_z^{(F)} - \mu_F\right)^2}{W-1}} \\ \mathrm{LCB}_F^{(1)} = \mu_F - (\alpha_1 \cdot \sigma_F) \end{cases} \tag{28}$$

During further aggregation, the trustworthiness of the distribution at the previous level is evaluated based on the derived LCB values. Assuming an $(s-1)^{th}$ order of recursion, where a total of G samples of $\mathrm{LCB}_F^{(s-1)}$ have been obtained, the s^{th} order score $\mathrm{LCB}_F^{(s)}$ is calculated based on the sample set $\left\{\mathrm{LCB}_{Fg}^{(s-1)}\right\}_{g=1}^{G}$, the formula is shown in Equation (29).

$$\begin{cases} \mu_F^{(s-1)} = \dfrac{1}{G}\sum_{g=1}^{G} \mathrm{LCB}_{Fg}^{(s-1)} \\ \sigma_F^{(s-1)} = \sqrt{\dfrac{\sum_{g=1}^{G}\left(\mathrm{LCB}_{Fg}^{(s-1)} - \mu_F^{(s-1)}\right)^2}{G-1}} \\ \mathrm{LCB}_F^{(s)} = \mu_F^{(s-1)} - \alpha_s \cdot \dfrac{\sigma_F^{(s-1)}}{\sqrt{G}} \end{cases} \tag{29}$$

3.3 Extreme Learning Machine

Extreme Learning Machine (ELM) is a Single-Layer Feedforward Neural Network (SLFN) proposed by Huang et al. [50] in 2006. ELM is generally considered faster because it avoids the complicated iterative parameter tuning required by traditional gradient descent-based feedforward neural networks. According to ELM theory, the hidden layer does not need to be tuned, where input weights and hidden layer biases can be randomly assigned and fixed. Once these parameters are fixed and applied to an infinitely differentiable (or merely piecewise continuous) activation function, the hidden layer outputs form a constant matrix. Consequently, the relationship between the hidden layer outputs and the target values can be modelled as a linear system, allowing the output weights (β) to be analytically determined using the Moore-Penrose pseudo-inverse, as illustrated in Figure 19.

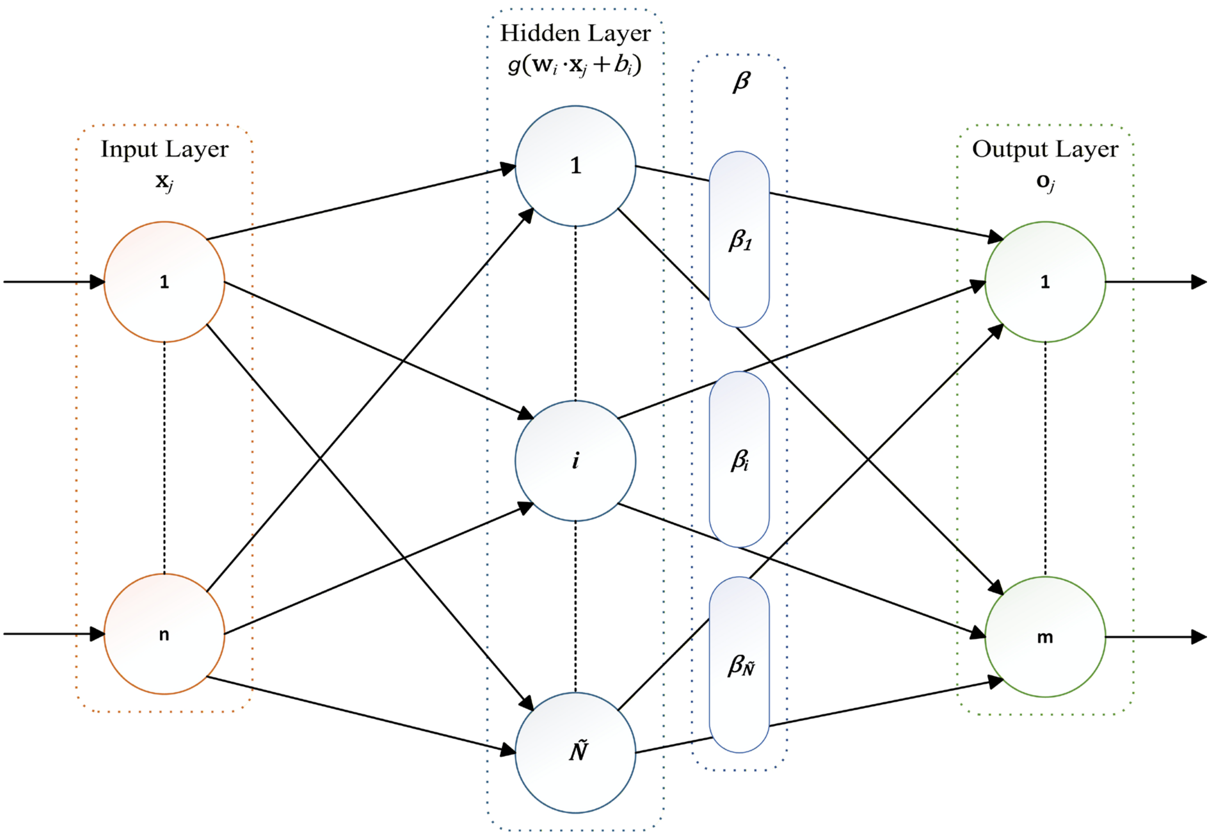

Figure 19 Schematic structure of the ELM

For a standard SLFN, given N arbitrary distinct training samples $(\mathbf{x}_j, \mathbf{t}_j)$, where $\mathbf{x}_j = [x_{j1}, x_{j2}, \dots, x_{jn}]^T \in R^n$ and $\mathbf{t}_j = [t_{j1}, t_{j2}, \dots, t_{jm}]^T \in R^m$, with $\tilde{N}$ hidden nodes and an activation function $g(x)$, it can be mathematically modelled as Equation (30):

$$\sum_{i=1}^{\tilde{N}} \beta_i g_i(\mathbf{x}_j) = \sum_{i=1}^{\tilde{N}} \beta_i g(\mathbf{w}_i \cdot \mathbf{x}_j + b_i) = o_j, \qquad j = 1, \dots, N \tag{30}$$

where $\mathbf{w}_i = [w_{i1}, w_{i2}, \dots, w_{in}]^T$ and $\beta_i = [\beta_{i1}, \beta_{i2}, \dots, \beta_{im}]^T$. If an SLFN achieves zero training error, it satisfies the following condition: $\sum_{j=1}^{N} \|\mathbf{o}_j - \mathbf{t}_j\| = 0$, therefore, there exist β_i , $\mathbf{w}_i$ and b_i such that Equation (30) can lead to Equation (31).

$$\sum_{i=1}^{\tilde{N}} \beta_i g(\mathbf{w}_i \cdot \mathbf{x}_j + b_i) = \mathbf{t}_j, \qquad j = 1, \dots, N \tag{31}$$

Hence, it can be compactly expressed as: $\mathbf{H}\beta = \mathbf{T}$ **,** where **H** denotes the hidden layer output matrix, as defined in Equation (32a), while β and **T** represent the output weights and target values, respectively, as shown in Equation (32b).

$$\mathbf{H}(\mathbf{w}_1, \dots, \mathbf{w}_{\tilde{N}}, b_1, \dots, b_{\tilde{N}}, \mathbf{x}_1, \dots, \mathbf{x}_N) = \begin{bmatrix} g(\mathbf{w}_1 \cdot \mathbf{x}_1 + b_1) & \cdots & g(\mathbf{w}_{\tilde{N}} \cdot \mathbf{x}_1 + b_{\tilde{N}}) \\ \vdots & \cdots & \vdots \\ g(\mathbf{w}_1 \cdot \mathbf{x}_N + b_1) & \cdots & g(\mathbf{w}_{\tilde{N}} \cdot \mathbf{x}_N + b_{\tilde{N}}) \end{bmatrix}_{N \times \tilde{N}}, \tag{32a}$$

$$\beta = \begin{bmatrix} \beta_1^T \\ \vdots \\ \beta_{\tilde{N}}^T \end{bmatrix}_{\tilde{N} \times m} \text{ and } \mathbf{T} = \begin{bmatrix} \mathbf{t}_1^T \\ \vdots \\ \mathbf{t}_N^T \end{bmatrix}_{N \times m}. \tag{32b}$$

According to Huang et al. [50] , training an SLFN is simply equivalent to finding the least-squares solution of the linear system $\mathbf{H}\beta = \mathbf{T}$. As the objective is to minimise the training error, the process can be defined as finding $\hat{\beta} = arg \min_{\beta} \|\mathbf{H}\beta - \mathbf{T}\|$. This produces a unique minimum norm least-squares solution for $\mathbf{H}\beta = \mathbf{T}$, which is $\hat{\beta} = \mathbf{H}^{\dagger}\mathbf{T}$ where $\mathbf{H}^{\dagger}$ is the Moore–Penrose generalised inverse of **H**.

Based on ELM, Wanyu Deng et al. [51] proposed the Regularised Extreme Learning Machine (RELM) in 2009 to address the overfitting issue. The core idea is to minimise the weighted sum of the empirical risk $\|\varepsilon\|^2$ and the structural risk $\|\beta\|^2$ with a weight factor γ. Additionally, the error variable ε_j can be weighted with factors v_i to obtain a robust estimate by mitigating outlier interference, extending $\|\varepsilon\|^2$ to $\|D\varepsilon\|^2$. The optimisation objective and its corresponding constraints are expressed in Equations (33a) and (33b), respectively.

$$\min \frac{1}{2}\|\beta\|^2 + \frac{1}{2}\gamma\|D\varepsilon\|^2 \tag{33a}$$

$$s.t. \sum_{i=1}^{\widetilde{N}} \beta_i g\left(\mathbf{w}_i \cdot \mathbf{x}_j + b_i\right) - \mathbf{t}_j = \varepsilon_j, \qquad j = 1, \dots, N \tag{33b}$$

where $D = \mathrm{diag}(v_1, v_2, \dots, v_N)$ and $\varepsilon = [\varepsilon_1, \varepsilon_2, \dots, \varepsilon_N]$. The best generalisation performance is achieved by balancing the two risks using a Lagrangian approach. The solution for RELM, optimal $\hat{\beta}$ is derived as shown in Equation (34a):

$$\hat{\beta} = \left(\frac{\boldsymbol{I}}{\boldsymbol{\gamma}} + \mathbf{H}^{\mathrm{T}} D^2 \mathbf{H}\right)^{\dagger} \mathbf{H}^{\mathrm{T}} D^2 \mathbf{T} \tag{34a}$$

When D is a unit matrix $\boldsymbol{I}$ (non-weighted), $\hat{\beta}$ simplifies to Equation (34b):

$$\hat{\beta} = \left(\frac{\boldsymbol{I}}{\boldsymbol{\gamma}} + \mathbf{H}^{\mathrm{T}} \mathbf{H}\right)^{\dagger} \mathbf{H}^{\mathrm{T}} \mathbf{T} \tag{34b}$$

For the case where $\widetilde{N} \gg N$, based on the push-through property, $\hat{\beta}$ can be calculated by Equation (34c):

$$\hat{\beta} = \mathbf{H}^{\mathrm{T}} \left(\mathbf{H}\mathbf{H}^{\mathrm{T}} + \frac{\boldsymbol{I}}{\boldsymbol{\gamma}}\right)^{\dagger} \mathbf{T} \tag{34c}$$

Notably, only a unified programmatic model was constructed in this study, designed to dynamically implement the standard ELM and RELM architectures based on the input regularisation parameter, $\lambda \left(\text{where } \lambda = \frac{1}{\gamma}\right)$ value. For clarity, the methodological derivations below utilise the ELM framework but are equally valid for RELM, as the two architectures differ only in the inclusion of the regularisation term. The activation function utilised for both architectures in this study was the standard sigmoid function, as shown in Equation (35):

$$g(x) = \frac{1}{1 + \mathrm{e}^{-x}} \tag{35}$$

The corresponding workflow of this unified model is illustrated in Figure 20.

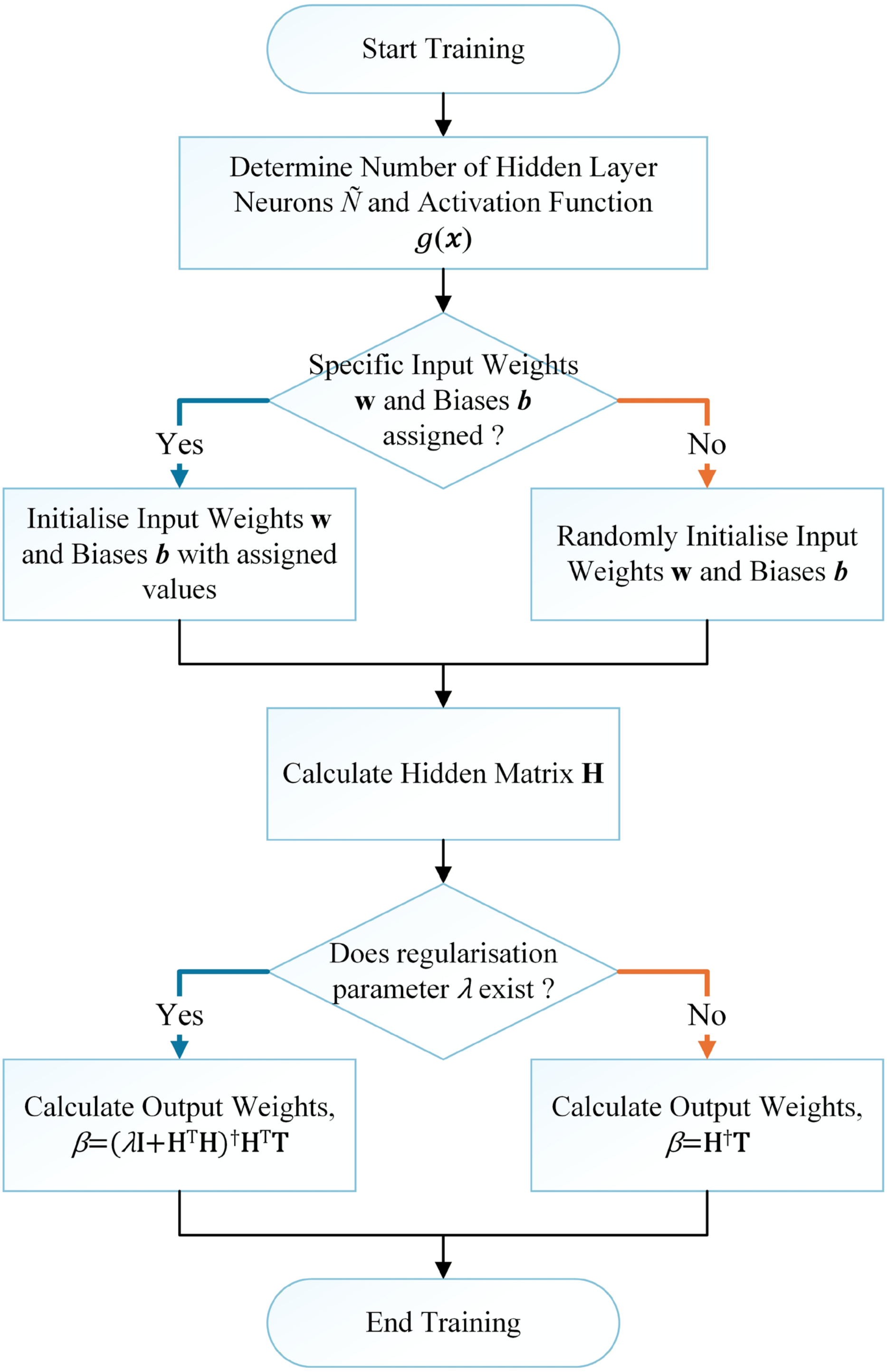

Figure 20 Flowchart of the unified training algorithm for ELM and RELM

3.4 Artificial Bee Colony Algorithm

The Artificial Bee Colony (ABC) algorithm is a swarm intelligence-based metaheuristic that simulates the foraging behaviour of honeybees. The concept of utilising bee swarms was initially introduced by Karaboga [52] in 2005, while the formal algorithm was proposed by Karaboga and Basturk [53] in 2007. Conceptually, the ABC model is composed of three components: food sources, employed foragers, and unemployed foragers. A food source represents a nectar source, while employed foragers are the bees currently exploiting or associating with a respective food source and sharing this information with the hive. The unemployed foragers consist of two types of bees: onlooker bees, which receive information from employed bees and evaluate the food sources, and scout bees which randomly explore the surroundings for new food sources.

Based on this concept, the ABC algorithm was designed around three groups of bees and the food sources. The number of employed bees is equal to both the number of onlooker bees and the number of food sources, while the number of scout bees is initially zero. A food source represents a candidate solution in the search space; an employed bee represents the evaluation and potential modification of the solution; an onlooker bee represents the evaluation and exploitation of a solution based on the information shared by employed bees; a scout bee is converted from an employed bee or scout bee if its associated solution cannot be improved further, thus randomly exploring a new solution. The quality of the solutions is evaluated by a fitness function.

In the ABC algorithm, the basic parameters are defined as follows: Population size (SN), which is the number of food sources or solutions; Trial limit (L), which is the mining round, or the maximum number of neighbourhood exploitation attempts for a solution before abandonment; and the termination criteria, normally defined by the maximum number of iterations (MI), or a non-improvement iteration count for an early stopping mechanism. The main steps of the ABC algorithm, as illustrated in Figure 21, are as follows:

1. Initialise the algorithm by randomly generating candidate solutions
2. Evaluate the fitness of the initial solutions
3. Repeat the iteration process:
    i. Employed Bee Phase: Employed bees explore new solutions in the neighbourhood search space of existing solutions, evaluate their fitness and compare them with their respectively existing solutions. An existing solution

will be replaced if the new solutions are better. The solution information is then shared with onlooker bees.

ii. Onlooker Bee Phase: Onlooker bees selectively explore new solutions or exploit the neighbour search space based on the solution information received from employed bees. The higher the fitness of a solution, the higher its probability of being selected for exploitation. Similar to the employed bees, a new solution with better fitness will replace the existing solution.

iii. Scout Bee Phase: A scout bee will only be triggered when employed bees and onlooker bees are unable to improve the fitness of the solution after exceeding the predefined trial limit (L). The solution is abandoned and replaced with a randomly generated new solution. The scout bee then reverts to an employed bee to evaluate this new solution and share the information instantly.

4. If the termination criteria are met, end the process.

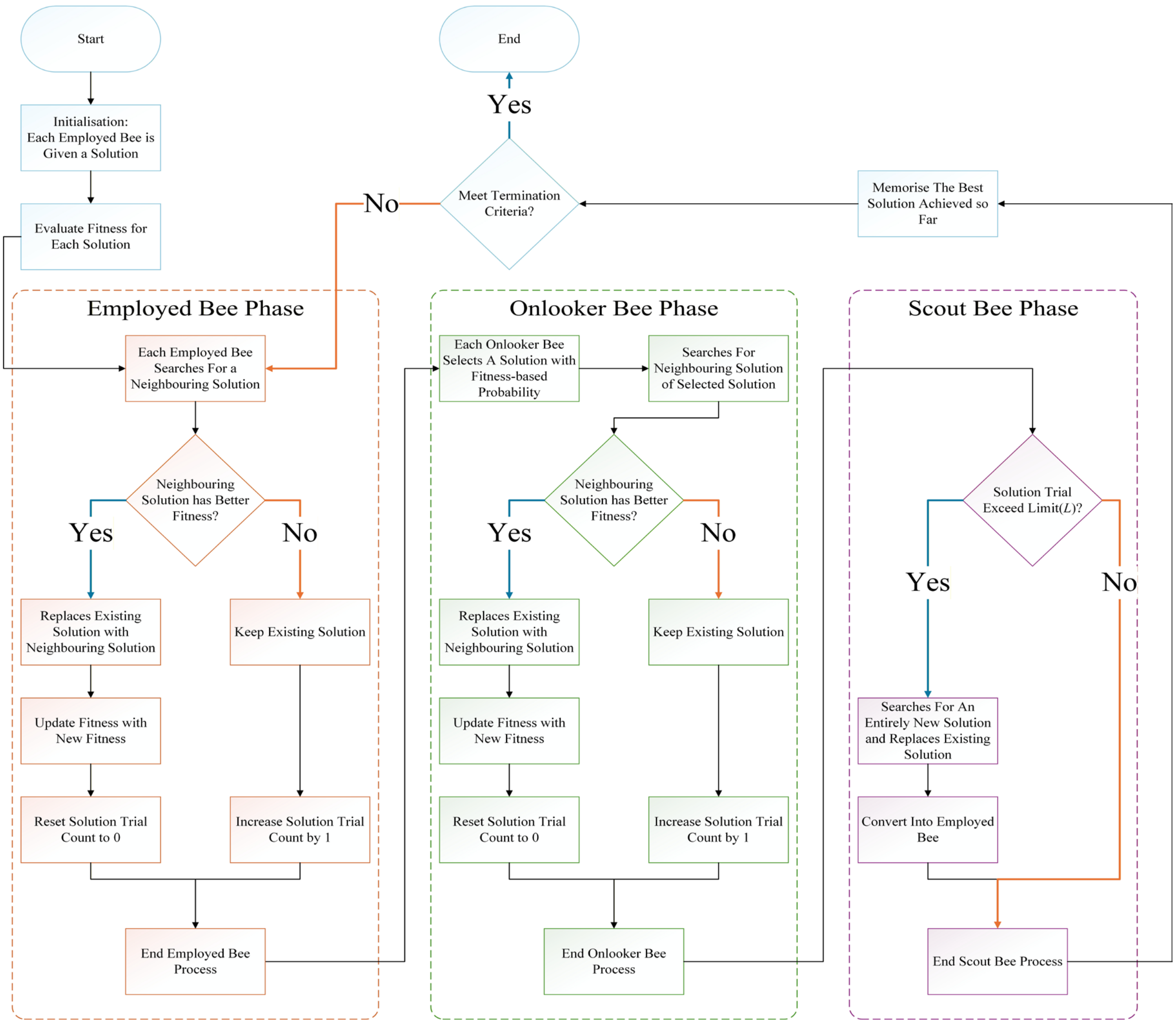

Figure 21 Flowchart of the ABC algorithm

3.5 Artificial Bee Colony-Extreme Learning Machine (ABC-ELM)

Since the performance of an ELM is highly sensitive to the random initialisation of its input weights and hidden biases, optimising these parameters is crucial. The ABC algorithm provides a way to improve the performance and optimise the stability of ELM by tuning its input weights and hidden biases. By optimising the hidden layer mapping space, the subsequent solution for output weights is determined based on a more robust feature representation rather than a potentially degenerate one caused by random initialisation.

This hybrid approach is referred to as Artificial Bee Colony-Extreme Learning Machine (ABC-ELM). In this study, the core architectural design of the ABC-ELM algorithm was adopted from Abobakr et al. [54] in 2018. ABC-ELM embeds the ELM into the ABC algorithm. The input weights and hidden biases are encoded as a candidate solution **S** for the ABC algorithm and the evaluation of the ELM prediction is encoded as the solution fitness to optimise the ELM.

The algorithmic formulation of ABC-ELM is outlined as follows:

A candidate solution **S** consists of the complete flattened set of input weights w and hidden biases b, which can be denoted as in Equation (36):

$$\mathbf{S} = [w_{11}, w_{12}, \dots, w_{1\tilde{N}}, w_{21}, w_{22}, \dots, w_{2\tilde{N}}, w_{n1}, w_{n2}, \dots, w_{n\tilde{N}}, b_1, b_2, \dots, b_{\tilde{N}}] \tag{36}$$

where $D = (n + 1) \times \tilde{N}$ and all w_{ai} and b_i are randomly initialised within the range $[-1,1]$.

The primary objective of optimising the ELM in this study for classification problems is to improve its ability to correctly identify the data patterns and classify them into their respective classes. Hence, the Matthews Correlation Coefficient (MCC) was utilised as the core performance metric. The MCC is calculated as follows in Equation (37):

$$\mathrm{MCC} = \frac{\mathrm{TP} \cdot \mathrm{TN} - \mathrm{FP} \cdot \mathrm{FN}}{\sqrt{(\mathrm{TP} + \mathrm{FP}) \cdot (\mathrm{TP} + \mathrm{FN}) \cdot (\mathrm{TN} + \mathrm{FP}) \cdot (\mathrm{TN} + \mathrm{FN})}} \tag{37}$$

Since the MCC values range between $[-1,1]$, the fitness function is mapped into the interval $[0,1]$. The fitness fit_h for a candidate solution $\mathbf{S}_h$ is defined in Equation (38):

$$\mathrm{fit}_{\boldsymbol{h}} = \frac{1 + \mathrm{MCC}_h}{2} \tag{38}$$

This ensures a perfectly linear mapping; the strictly non-negative fitness values prevent mathematical or logical errors. For example, it prevents negative values for the onlooker bees' selection probability p_h in Equation (46) and change percentage exceeding 100% in Equation (42). fit_h can be reverted to the original MCC_h value using the following Equation (39):

$$\mathrm{MCC}_{\boldsymbol{h}} = (2 \cdot \mathrm{fit}_{\boldsymbol{h}}) - 1 \tag{39}$$

Two distinct procedures are utilised to generate neighbouring solutions, denoted as Algorithm 1 and Algorithm 2. Algorithm 1 is inspired by the spatial dispersal concept of the Invasive Weed Optimisation (IWO) algorithm, which generates neighbouring solutions by applying a normally distributed random perturbation with a zero mean and an adaptive standard deviation across the search space. Algorithm 2 relies on the standard ABC mutation procedure, which produces neighbouring solutions by evaluating the differential vector between the current solution and another randomly selected solution within the population.

For a candidate solution **V** generated from the current solution **S,** all decision variables are bounded within the range $[-1,1]$, and its update is computed using either algorithm as defined below:

The formula for the adopted Algorithm 1 is expressed as Equation (40):

$$\mathbf{V}_{hk} = \mathbf{S}_{hk} + \mathrm{rand}(0, \sigma_{\mathrm{iter}}) \tag{40}$$

where $h = 1,2, \dots , SN, k = 1,2, \dots, D$. rand denotes a Gaussian noise generator that produces random values with zero mean and iteration-dependent standard deviation σ_{iter}, as defined in Equation (41).

$$\sigma_{\mathrm{iter}} = \left(\frac{iter_{\mathrm{max}} - iter}{iter_{\mathrm{max}}}\right)^{nmi} \cdot (\sigma_{\mathrm{initial}} - \sigma_{\mathrm{final}}) + \sigma_{\mathrm{final}} \tag{41}$$

The candidate solution $\mathbf{V}_h$ is generated by selectively perturbing a subset of elements in $\mathbf{S}_h$, rather than modifying the entire solution. The number of mutated dimensions is adaptively adjusted based on the fitness of the solution. The mutation percentage and the corresponding number of modified dimensions are computed as shown in Equations (42) and (43).

$$\mathrm{ChgPerct}_h = \frac{\mathrm{fit}_{\mathrm{best}} - \mathrm{fit}_h}{\mathrm{fit}_{\mathrm{best}}} (\mathrm{Chg}_{\mathrm{max}} - \mathrm{Chg}_{\mathrm{min}}) + \mathrm{Chg}_{\mathrm{min}} \tag{42}$$

$$\mathrm{Change}_h = \mathrm{round}\left(\frac{D \times \mathrm{ChgPerct}_h}{100}\right) \tag{43}$$

where fit_h is the fitness of $\mathbf{S}_h$ and $\mathrm{fit}_{\mathrm{best}}$ is the best fitness among whole population. $\mathrm{Chg}_{\mathrm{max}}$ and $\mathrm{Chg}_{\mathrm{min}}$ represent maximum and minimum percentages that can be changed in **S** respectively. To ensure validity, Change_h is constrained within $[1, D]$. The procedure is summarised in Algorithm 1.

Algorithm 1 Neighbourhood Generation (IWO-based Mutation)

Require: Current solution $\mathbf{S}_h$, iteration $iter$
Ensure: Neighbour solution $\mathbf{V}_h$

1: Compute σ_{iter} using Equation (41)
2: Compute ChgPerct_h using Equation (42)
3: Compute Change_h using Equation (43)
4: Limit the number of modified dimensions to be at least 1 and at most D
5: Set $\mathbf{V}_h \leftarrow \mathbf{S}_h$
6: Randomly select Change_h distinct dimensions
7: **for** each selected dimension k **do**
8:     Update $\mathbf{V}_{hk}$ using Equation (40)
9: **end for**
10: $\mathbf{V}_{hk}$ is bounded within $[-1, 1]$
11: **return** $\mathbf{V}_h$

The formula for the adopted Algorithm 2 is expressed as Equation (44):

$$\mathbf{V}_{hk} = \mathbf{S}_{hk} + \emptyset_{hk}(\mathbf{S}_{hk} - \mathbf{S}_{qk}) \tag{44}$$

where $h, q \in \{1,2, \dots, SN\}, h \neq q$ and $k = 1,2, \dots, D$. $\emptyset_{hk}$ is a random number between $[-1,1]$.

The element $\mathbf{V}_{hk}$ is updated from $\mathbf{S}_{hk}$ only if a randomly generated probability p_{hk}, drawn uniformly from the interval $[0,1]$, is greater than or equal to the adaptive threshold P_{copy} during the iteration $iter$; otherwise, the original value $\mathbf{S}_{hk}$ is retained.

The probability P_{copy} is calculated using Equation (45):

$$P_{\text{copy}} = \frac{P_{\text{final}} - P_{\text{initial}}}{iter_{\max}}(iter) + P_{\text{initial}} \tag{45}$$

where $iter_{\max}$ is the predefined maximum number of iterations. P_{final} and P_{initial} are the predefined final and initial probabilities, respectively. The neighbourhood generation process based on the standard ABC mutation strategy is summarised in Algorithm 2.

Algorithm 2 Neighbourhood Generation (ABC-based Mutation)

Require: Current solution $\mathbf{S}_h$, population $\{\mathbf{S}\}$, iteration $iter$
Ensure: Neighbour solution $\mathbf{V}_h$

```
Compute P_copy using Equation (45)
Randomly select a solution S_q from the population where q ≠ h
Set V_h ← S_h
for k = 1 to D do
    Generate random value p_hk between 0 and 1
    if p_hk ≥ P_copy then
        Update V_hk using Equation (44)
    end if
end for
V_hk is bounded within [−1, 1]
return V_h
```

During the onlooker bee phase, solutions are selected probabilistically based on the fitness information shared by the employed bees. The selection probability p_h for a specific solution $\mathbf{S}_h$ is based on its ratio to the best fitness, as calculated in Equation (46).

$$p_h = \left(\alpha_{\text{onlooker}} \cdot \frac{\text{fit}_h}{\text{fit}_{\text{best}}}\right) + (1 - \alpha_{\text{onlooker}}) \tag{46}$$

It is worth noting that this implementation refers to an open-source GitHub repository [55], which diverges from the canonical theoretical formulation originally proposed in the foundational ABC literature [53], where the probability of selection is proportional to the solution's contribution to the total population fitness as shown in Equation (47):

$$p_h = \frac{\text{fit}_h}{\sum_{q=1}^{SN} \text{fit}_q} \tag{47}$$

During the scout bee phase, solutions were abandoned when they exceeded the predefined limit (L). To avoid the solutions being considered as exhausted if the difference was caused by decimal precision, a numerical tolerance condition was provided, as defined in Equation (48).

$$|\text{fit}_h - \text{fit}_{\text{best}}| \leq \max(10^{-10} \cdot \max(|fit_h|, |fit_{best}|)\,, 10^{-12}) \tag{48}$$

Fitness Evaluation Protocol:

The fitness evaluation process for a candidate solution $\mathbf{S}_h$ (or neighbour solution $\mathbf{V}_h$) is defined as follows:

1. Construct an ELM network with $\widetilde{N}$ hidden nodes and activation function $g(x)$.
2. Decode the candidate solution $\mathbf{S}_h$ (or $\mathbf{V}_h$) and assign the values to ELM_h as the network's input weights and hidden biases.

3. Analytically compute the output weights using the training data.
4. Validate ELM_h to obtain the fitness score $\mathrm{fit}(\mathbf{S}_h)$ or $\mathrm{fit}(\mathbf{V}_h)$ using the predefined fitness function.

The complete ABC-ELM optimisation process, which is largely adopted from the referenced study with minor modifications and illustrated in Figure 22, is detailed as follows:

Step 1: Initialisation

Initialise the population with SN solutions $\mathbf{S}_{hk}, h = 1,2, \dots, SN, k = 1,2, \dots, D$

Apply the fitness evaluation protocol to determine $\mathrm{fit}(\mathbf{S}_h)$ for all solutions

Initialise the abandonment counter $\mathrm{trial}_h = 0, h = 1,2, \dots, SN$

Step 2: Iterative Optimisation Loop

Employed Bee Phase:

Produce a neighbouring candidate solution $\mathbf{V}_h$ from each solution $\mathbf{S}_h$ using the neighbourhood generation algorithm.

Apply the fitness evaluation protocol to $\mathbf{V}_h$

If $\mathrm{fit}(\mathbf{V}_h) > \mathrm{fit}(\mathbf{S}_h)$, replace $\mathbf{S}_h$ with $\mathbf{V}_h$ and set $\mathrm{trial}_h = 0$. Otherwise, retain $\mathbf{S}_h$ and increment trial_h by 1.

Calculate the probability p_h for each solution $\mathbf{S}_h$

Onlooker Bees Phase:

For each onlooker bee, select a solution $\mathbf{S}_h$ based on its probability p_h

Produce a neighbouring candidate solution $\mathbf{V}_h$ from the selected solution $\mathbf{S}_h$ using the neighbourhood generation algorithm.

Apply the fitness evaluation protocol for $\mathbf{V}_h$

If $\mathrm{fit}(\mathbf{V}_h) > \mathrm{fit}(\mathbf{S}_h)$, replace $\mathbf{S}_h$ with $\mathbf{V}_h$ and set $\mathrm{trial}_h = 0$. Otherwise, retain $\mathbf{S}_h$ and increment trial_h by 1.

Scout Bees Phase:

Identify if any solution $\mathbf{S}_h$ exceeds the predefined limit $(\mathrm{trial}_h > L)$.

Evaluate the proximity of the exhausted solution to the global best fitness. If $\mathbf{S}_h$ satisfies Equation (48), retain it and reset trial_h to zero.

Otherwise, replace $\mathbf{S}_h$ with a new randomly generated solution.

Apply the fitness evaluation protocol for to the new $\mathbf{S}_h$ and reset trial_h to zero.

Step 3: Termination

Repeat Step 2 until the termination criteria are met.

End the process and output the best solution found.

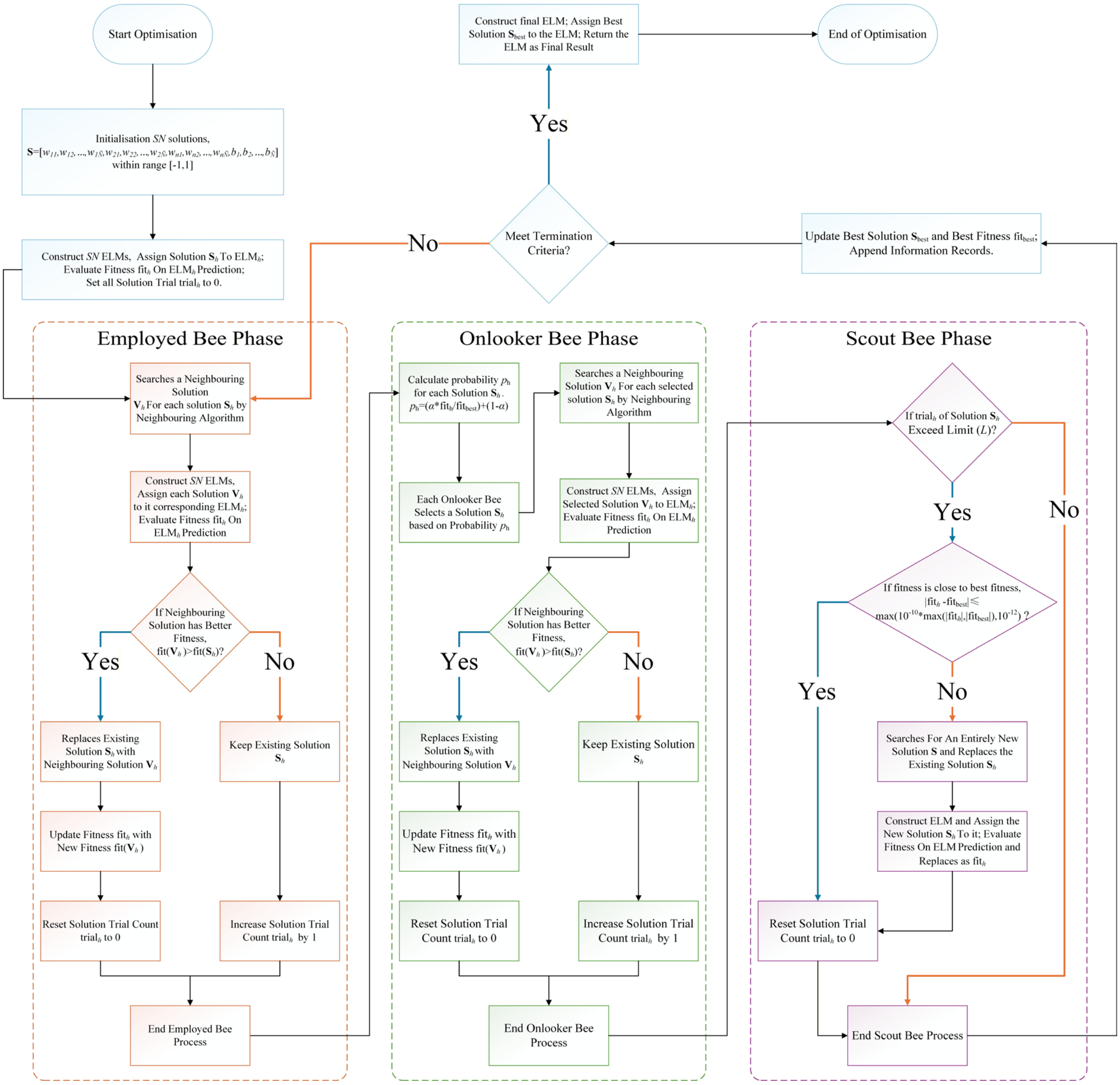

Figure 22 Flowchart of the ABC-ELM hybrid optimisation algorithm

3.6 ABC-ELM with Internal Cross-Validation (ABC-ELM-CV)

In the standard ABC-ELM architecture, the fitness evaluation relies on comparing a single scalar value. ABC-ELM with Internal Cross-Validation (ABC-ELM-CV) applies internal K-fold CV to mitigate the overfitting caused by a greedy search approach. However, integrating K-fold CV into the evaluation process yields a distribution of validation scores across the distinct folds. To preserve the algorithm's scalar comparison mechanism, these fold results must be aggregated into a single robust metric.

Therefore, the fitness of each candidate solution is evaluated using a risk-averse Lower Confidence Bound (LCB) derived from the standard deviation (STD) of the CV results, which penalises high-variance solutions that overfit to local optima. This penalised intermediate fitness score $\mathrm{fit}_h^{\mathrm{LCB}}$ of candidate solution $\mathbf{S}_h$ across all K folds is derived from Equation (28), resulting in Equation (49):

$$\begin{cases} \mu_h = \dfrac{1}{K}\sum_{f=1}^{K} \mathrm{fit}_f^{(h)} \\ \sigma_h = \sqrt{\dfrac{\sum_{f=1}^{K}\left(\mathrm{fit}_f^{(h)} - \mu_h\right)^2}{K-1}} \\ \mathrm{fit}_h^{\mathrm{LCB}} = \mu_h - (\alpha_{CV} \cdot \sigma_h) \end{cases} \tag{49}$$

Furthermore, to prevent non-positive fitness values that could destabilise the algorithm, a strict lower bound of 10^{-8} is enforced. The final fitness score $\mathrm{fit}_{\boldsymbol{h}}$ is given by Equation (50).

$$\mathrm{fit}_h = \max\left(10^{-8}, \mathrm{fit}_h^{\mathrm{LCB}}\right) \tag{50}$$

Apart from this modified fitness evaluation, all other algorithmic procedures remain identical to those of the ABC-ELM.

3.6.1 ABC-ELM-CV Variants

Standard ELM networks are characterised by very fast training and prediction speeds; therefore, the difference between constructing a single ELM and an ensemble model, which comprises a small number (K) of distinct ELM models is negligible. Hence, two simple ensemble methods inherited from ABC-ELM-CV were attempted to utilise the by-products: cross-fold ensemble and iteration ensemble. It was convenient to attempt this approach because the ABC-ELM-CV with internal K-fold CV already retained the normalisation parameters used for distinct folds and the best solutions across iterations within the model architecture. The models were referred to as ABC-ELM-CV via Cross-Fold Ensemble (ABC-ELM-CE) and ABC-ELM-CV via

Iterative Ensemble (ABC-ELM-IE). Both models followed the same procedure as ABC-ELM-CV, apart from constructing multiple ELM models for final prediction. The final ensemble model aggregated all the models' outputs via a majority voting mechanism.

3.6.1.1 ABC-ELM-CE

The core idea of ABC-ELM-CE was to utilise K distinct fold-specific scalers on their respective fold datasets to train multiple ELM models. This is similar to a bagging ensemble, although the subsets were fixed. Consequently, K ELM models, which share identical input weights and hidden biases (which is the best solution output from the ABC algorithm) but different output weights, were constructed.

3.6.1.2 ABC-ELM-IE

Unlike ABC-ELM-CE, ABC-ELM-IE applied the same data but distinct solutions. The ABC algorithm explores and exploits multiple solutions across its lifetime, generating a best solution for each iteration. Assuming that any best solution could potentially reach the early stopping criteria, all non-identical best solutions across iterations could become the final result. Based on this assumption, the solutions (excluding the initial non-iterated solution) were collected as candidates to construct multiple ELM models. The final solution was selected deterministically, and the remaining candidates were grouped by iteration. Candidates from each group were selected based on having the minimum highest cosine similarity with the existing solutions.

3.7 Performance Metrics

The metrics employed for model evaluation included accuracy, F_β-score, and the Matthews correlation coefficient (MCC). These classification scores are calculated based on the four fundamental components of a confusion matrix: True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN). In this context, "Positive" and "Negative" represent the predicted classes, while "True" and "False" denote whether the prediction matches the actual ground truth.

Accuracy measures the overall correctness of the model's predictions, calculated as shown in Equation (51):

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{51}$$

The F_β-score is a composite metric that focuses primarily on the true positive predictions. The coefficient β defines the relative importance of recall to precision. For example, the F_1-score considers precision to be as important as recall, whereas the F_2-score considers recall twice as important as precision. The F_β-score is expressed in Equation (52):

$$F_\beta = (1 + \beta^2) \cdot \frac{\text{Precision} \cdot \text{Recall}}{\beta^2 \cdot \text{Precision} + \text{Recall}} \tag{52}$$

where $\text{Precision} = \frac{\text{TP}}{\text{TP+FP}}$ and $\text{Recall} = \frac{\text{TP}}{\text{TP+FN}}$.The precision evaluates the exactness of the positive predictions, while recall evaluates the model's ability to identify all actual positive cases.

As introduced earlier in Equation (37), the Matthews correlation coefficient (MCC) is a comprehensive evaluation metric that incorporates all four components of the confusion matrix. Unlike accuracy, which suffers from information loss under class-imbalanced situations, and the F_β-score, which ignores the true negative cases, the MCC equally considers all four quadrants, ensuring a more robust evaluation of a model’s classification ability.

4 Experimental Setup

The experiments, parameter tuning, and testing were performed in parallel on four identically configured desktops: Dell OptiPlex 7490 AIO equipped with an Intel® Core™ i7-10700 CPU @ 2.90 GHz, 16 GB of RAM, and a 64-bit Windows 11 Enterprise operating system. The implementations and model evaluations were conducted using Python (3.13.5). Baseline models were implemented using scikit-learn (1.5.2), while statistical analyses performed with SciPy (1.15.3), statsmodels (0.14.4), and baycomp (1.0.3). The evaluated machine learning models included the Multilayer Perceptron (MLP), Random Forest (RF), Histogram-based Gradient Boosting Classifier (HGBC), Bagging Logistic Regression Classifier (Bagging LRC), Logistic Regression Classifier (LRC), and Linear Support Vector Classifier (Linear SVC).

4.1 Model Optimisation

The study was designed to evaluate two main aspects of the models' performance: stability and generalisation. During the tracing and parameter tuning phase, a total of 30 random seeds were fixed (seeds 131 to 160 in this study), and the predefined 5-fold training and validation sets were employed. For each architecture, a total of 150 records were collected for evaluation and decision-making. As the main optimisation of the ELM focused on mitigating the impact of random initialisation, the evaluation of each architecture was based on a risk-averse concept.

While performance across random seeds does not guarantee a Gaussian distribution, calculating non-parametric bounds across numerous seeds, folds, and hyperparameters during parameter tuning is computationally prohibitive. Therefore, the LCB was explicitly adopted as a computationally efficient surrogate metric, which serves purely as a directional heuristic to penalise high variance during the search phase.

Formally, the process computed the performance of a single seed r across the five folds, denoted as $\mathrm{LCB}_r^{(1)}$, and utilised this metric to evaluate the different architectures under the same seed initialisation, thereby yielding $\mathrm{LCB}_{Architecture}^{(2)}$. Subsequently, the optimal hyperparameters for the models, specifically the hidden node size and the λ value, were determined by maximising $\mathrm{LCB}_{Architecture}^{(2)}$.

During metaheuristic parameter tuning, multiple interacting factors must be considered. For a given model architecture, different hyperparameter configurations were evaluated based on the third-order score $\mathrm{LCB}_F^{(3)}$, which was derived from $\mathrm{LCB}_{Architecture}^{(2)}$. In this study, the internal algorithm of the ABC was the first factor to be evaluated and fixed, followed by the ratio between the population size (SN) and trial limit (L), and finally, the population size (SN) and the maximum number of iterations (MI). Notably, the risk-averse parameters in this study were bounded by $1 \leq s \leq 3$, with the penalty coefficient α_s defined in Equation (53):

$$\alpha_s = \begin{cases} 1 \;, & s = 1 \\ 2.58 \;, & s > 1 \end{cases} \tag{53}$$

This piecewise configuration was designed to ensure high confidence while preventing information loss caused by over-penalising.

4.2 ELM/RELM Parameter Tracing

For the ELM, the number of hidden nodes was optimised over the interval $[5, 120]$ with a step size of 5. Figure 23 illustrates the trend of $\mathrm{LCB}^{(2)}_{Architecture}$ with respect to the hidden node size, resulting in an optimal configuration of 40 nodes with a score of 0.4826.

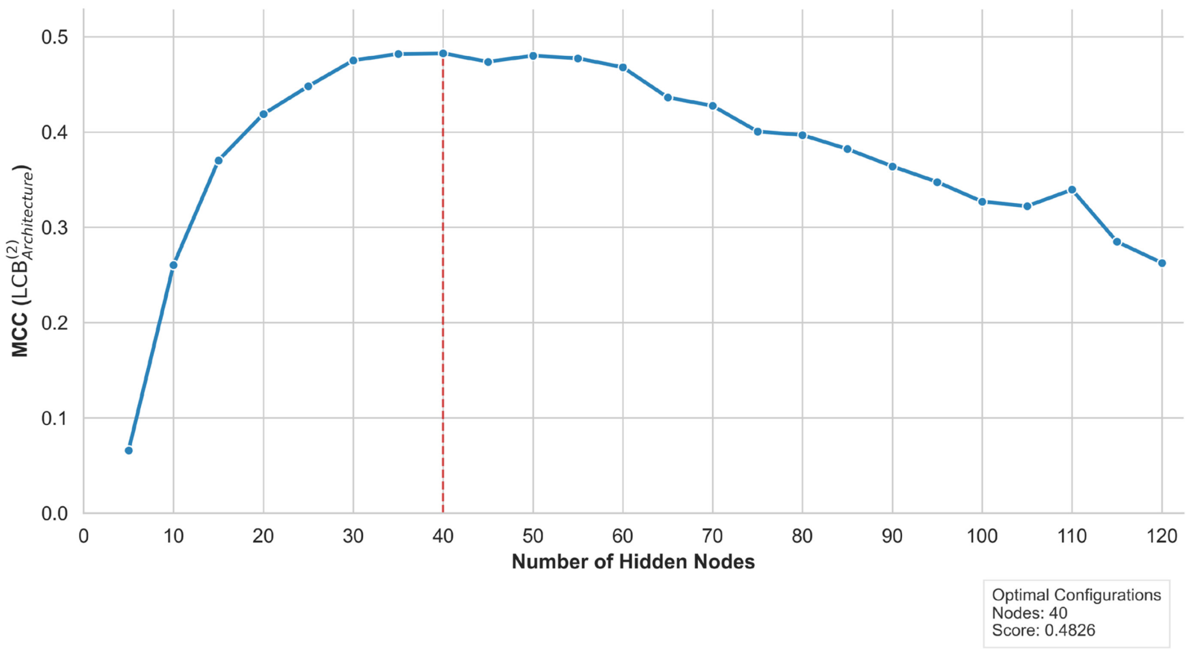

Figure 23 Performance trend of ELM hidden nodes

The RELM incorporates an additional regularisation parameter, λ. This optimisation was conducted via a geometric grid search over a narrower range of hidden nodes, crossed with a λ range of $[2^{-25}, 2^{10}]$ using a multiplicative step factor of $\sqrt{2}$. The performance trend concerning the RELM parameters is illustrated in Figure 24, which demonstrates that the

RELM achieved better performance than the optimal 40-node ELM when configured with 30 hidden nodes and $\lambda = \left(\sqrt{2}\right)^{-5}$, yielding an $\mathrm{LCB}^{(2)}_{Architecture}$ score of 0.5085.

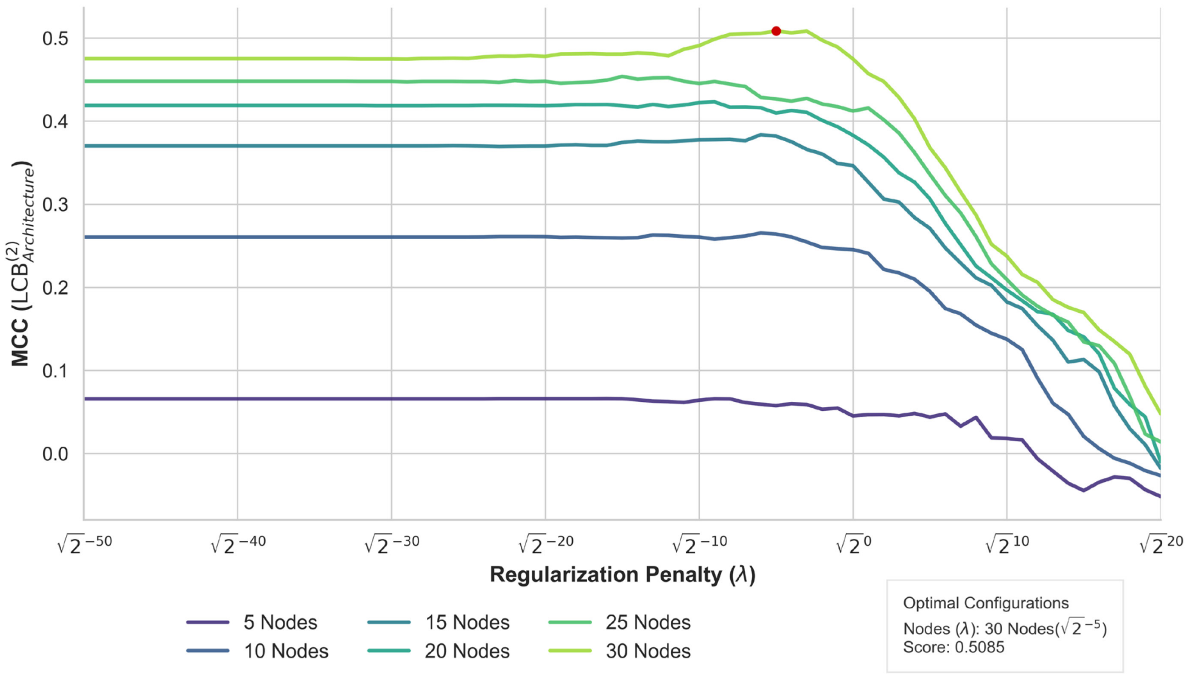

Figure 24 Performance trends of RELM parameters

Once the optimal hidden node count and λ value were established, all subsequent processes involving the ELM and RELM were based on these fixed configurations. The ensuing parameter tracing of the ABC algorithm was conducted utilising the RELM within the ABC-ELM-CV framework.

4.3 ABC Convergence Analysis

4.3.1 Neighbourhood Algorithms

First, the convergence behaviours of the ABC neighbourhood algorithms were evaluated by fixing the ratio between SN and L $\left(L = \frac{SN}{2}\right)$ across eight distinct configurations: $(SN, MI) \in \{(10,120), (10,1000), (20,250), (50,150), (80,120), (80,250), (100,120), (100,250)\}$. Table 6 details the mean and standard deviation across the eight $\mathrm{LCB}^{(2)}_{Architecture}$ scores, along with the final $\mathrm{LCB}^{(3)}_{Algorithm}$ for the algorithm combination. Furthermore, Figure 25 presents the collated boxplots of the $\mathrm{LCB}^{(2)}_{Architecture}$ scores across these configurations, comparing four different combinations of Algorithm 1 and Algorithm 2 for the employed and onlooker bees. Finally, Figure 26 depicts the convergence trace specifically for the model configured with the parameters $(SN = 80, MI = 120)$.

As illustrated in Figure 25, although the training results were comparable, the validation phase revealed distinct differences. Specifically, the combination utilising Algorithm 2 for employed bees and Algorithm 1 for onlooker bees exhibited the tightest interquartile range (IQR), demonstrating the highest convergence stability. Conversely, the reverse combination exhibited a slightly tighter IQR, but a wider dispersion of outliers compared to the pure Algorithm 1 combination, which displayed a relatively uniform distribution. Furthermore, this reverse combination yielded the lowest median score, indicating the poorest overall performance among the four configurations. Finally, the pure Algorithm 2 combination achieved the highest median while maintaining the second-best stability.

For each configuration in Figure 26, the pure Algorithm 1 combination converged to a stable state at the earliest stage, accompanied by a continuous increase in scout bee activations. In contrast, the pure Algorithm 2 combination exhibited continuous fluctuations until the final iterations, with scout bee activations peaking during the early stages and declining gradually. Unlike the pure combinations, the cross-combinations demonstrated stable convergence during the early stages but maintained minor fluctuations during the later stages, with scout bee activations peaking near the midpoint of the iteration cycle.

These observations align closely with the underlying mechanics of the neighbourhood algorithms. Algorithm 1 applies normal distribution-based mutations, resulting in smaller step sizes that facilitate localised, self-surrounding exploitation. In contrast, Algorithm 2 defines its neighbourhood by calculating the distance between the current and a selected neighbour solution, producing movement mutations of variable distances. This fundamental difference indicates that Algorithm 1 intrinsically favours exploitation, whereas Algorithm 2 prioritises exploration of the solution space. This distinction clearly explains the divergent behaviours observed across the different configurations. The pure combinations essentially amplify their respective characteristics, leading to premature convergence for Algorithm 1 and protracted, incomplete convergence for Algorithm 2.

Among the cross-combinations, assigning Algorithm 2 to employed bees and Algorithm 1 to onlooker bees yielded the highest convergence stability. In this setup, employed bees first broadly explore the solution space to identify diverse candidate regions. Guided by the resulting fitness scores, onlooker bees then apply Algorithm 1 to intensively exploit only the most promising solutions, thereby minimising performance variance. Conversely, the reverse approach disrupts this natural foraging logic. Employed bees prematurely exploit local areas before comparative evaluations occur. Subsequently, onlooker bees apply Algorithm 2 to explore away from these newly found optima, scattering the search effort and defeating the core purpose of the onlooker phase.

These empirical observations support the quantitative results: the configuration assigning Algorithm 2 to employed bees and Algorithm 1 to onlooker bees achieved the highest $\mathrm{LCB}^{(3)}_{Algorithm}$ score; although it yielded the lowest mean performance, this was effectively counterbalanced by incurring the smallest variance penalty. Consequently, this combination was selected for its robust convergence and low sensitivity to variations in the SN and MI parameters, even though the pure Algorithm 2 combination exhibited higher peak potential.

Table 6 Performance evaluation of ABC neighbourhood algorithm combinations

Employed Bee	**Onlooker Bee**	**Training** $\mathbf{LCB}^{(2)}_{Architecture}$	**Validation** $\mathbf{LCB}^{(2)}_{Architecture}$	$\mathbf{LCB}^{(3)}_{Algorithm}$
Algorithm 1	Algorithm 1	0.6463 ± 0.0246	0.5360 ± 0.0121	0.5047
Algorithm 1	Algorithm 2	0.6462 ± 0.0208	0.5333 ± 0.0117	0.5032
Algorithm 2	Algorithm 1	0.6467 ± 0.0206	0.5324 ± 0.0047	0.5202
Algorithm 2	Algorithm 2	0.6494 ± 0.0220	0.5360 ± 0.0088	0.5133

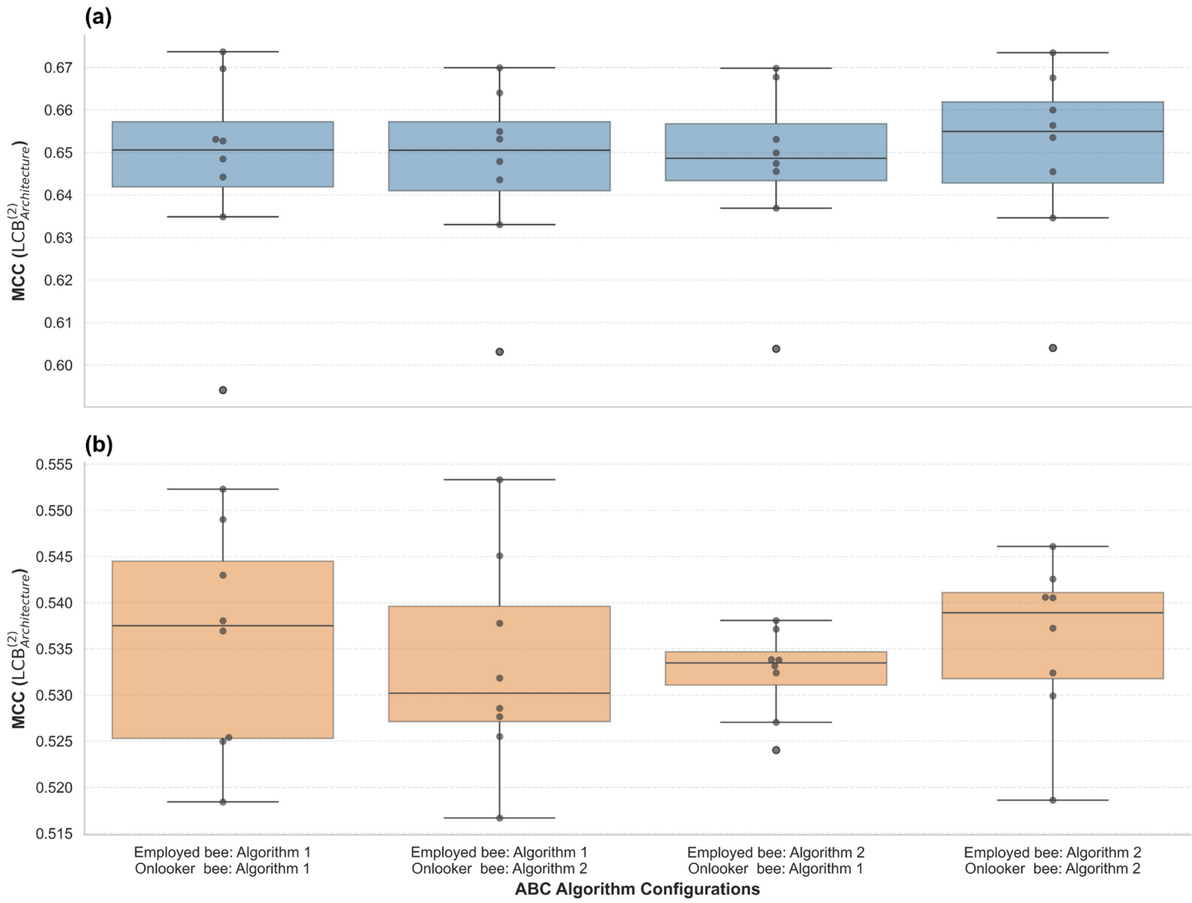

Figure 25 Distribution of architecture-level scores across ABC algorithm combinations (a) Training Convergence (b) Validation Convergence

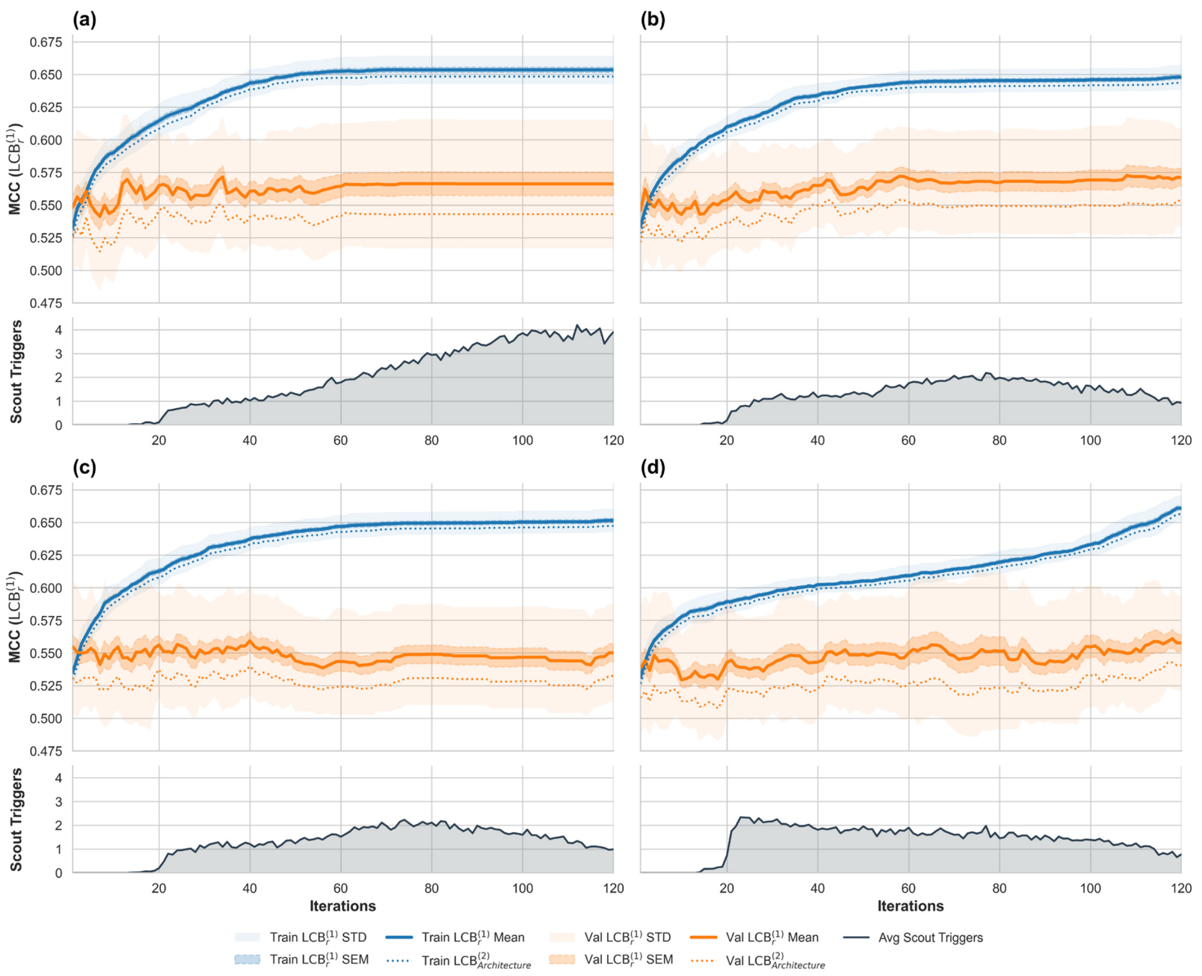

Figure 26 Convergence trends of ABC neighbourhood algorithm configurations (80SN, 40L, 120MI)
(a) Employed Bee: Algorithm 1; Onlooker Bee: Algorithm 1
(b) Employed Bee: Algorithm 1; Onlooker Bee: Algorithm 2
(c) Employed Bee: Algorithm 2; Onlooker Bee: Algorithm 1
(d) Employed Bee: Algorithm 2; Onlooker Bee: Algorithm 2

4.3.2 Trial Limit Ratio

Once the internal algorithms were fixed, the trial limit ratio was evaluated across the same eight distinct configurations. By design, a solution is evaluated by at least one employed bee per iteration, and potentially multiple times by onlooker bees, directly incrementing its trial limit counter. In an ABC algorithm with a large population, high-fitness solutions are evaluated more frequently by onlooker bees, thereby accelerating the exhaustion of their trial limits and prematurely triggering scout bees. Therefore, the trial limit L was defined as a proportional ratio relative to the population size SN .

Four different L/SN ratios were investigated. As detailed in Table 7 and Figure 27, a ratio of 1.5 distinctly outperformed the alternatives. It achieved the highest $\mathrm{LCB}^{(3)}_{L/SN}$ score, characterised by the highest mean performance and a marginally elevated variance; notably, its worst-case performance still surpassed the median results of the other ratios. A ratio of 0.5 imposed the strictest limit, preventing the bees from sufficiently exploiting a solution. However, it still achieved the second-best performance, which can be attributed to the rapid replacement of exhausted solutions. In contrast, a ratio of 2.0 provided an excessively lenient limit, causing the bees to waste computational effort exploiting poor solutions, thus yielding the worst performance. Although a ratio of 1.0 represented a balanced limit relative to the population, its performance was unremarkable, as the distinct benefits of rapid replacement and deep exploitation effectively neutralised each other. Consequently, the ratio of 1.5, acting as the optimal "sweet spot" that successfully balances exploration and exploitation, was selected for the L/SN parameter.

Table 7 Performance evaluation of ABC trial limit ratios

L/SN Ratio	Training $\mathbf{LCB}^{(2)}_{Architecture}$	Validation $\mathbf{LCB}^{(2)}_{Architecture}$	$\mathbf{LCB}^{(3)}_{L/SN}$
0.5	0.6467 ± 0.0206	0.5324 ± 0.0047	0.5281
1.0	0.6511 ± 0.0208	0.5303 ± 0.0055	0.5252
1.5	0.6532 ± 0.0185	0.5406 ± 0.0060	0.5356
2.0	0.6543 ± 0.0190	0.5257 ± 0.0063	0.5199

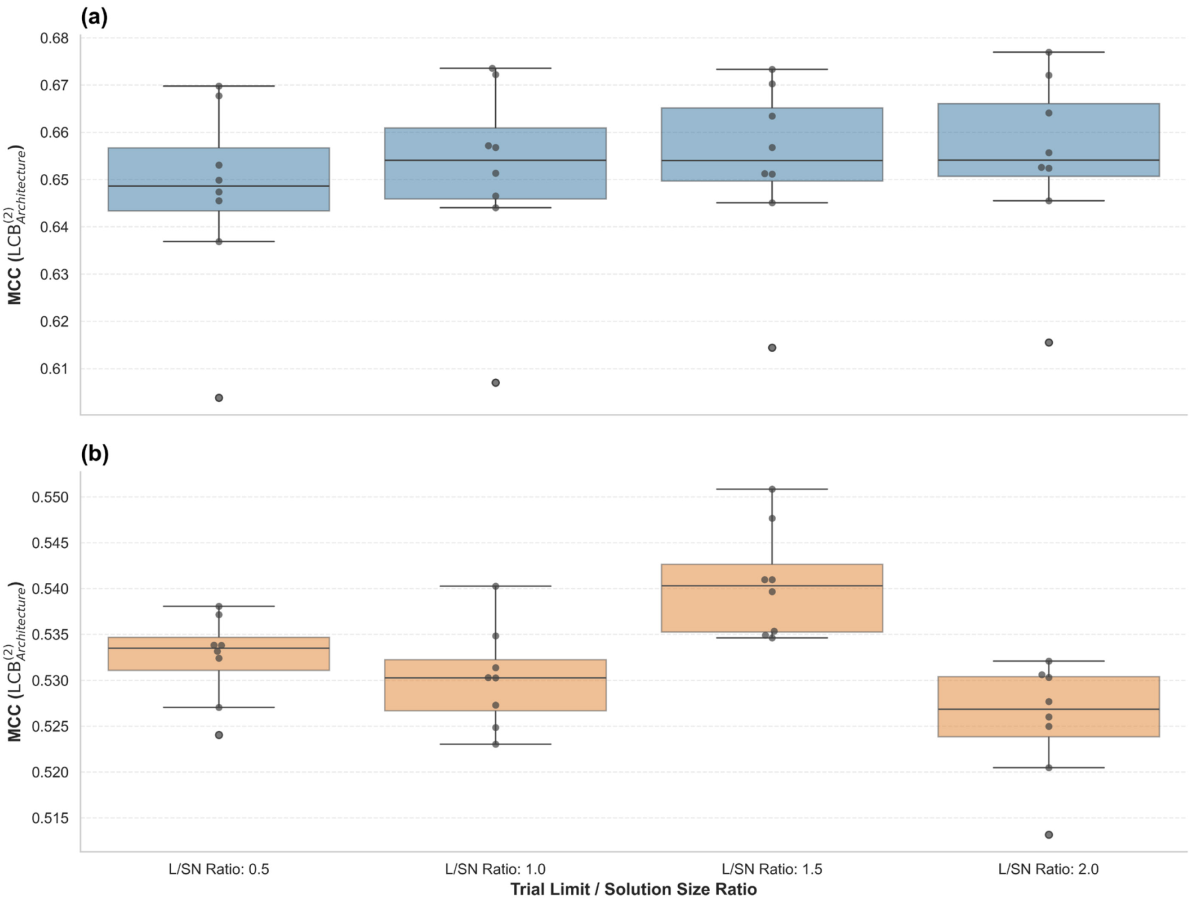

Figure 27 Distribution of architecture-level scores across ABC trial limit ratios (a) Training Convergence (b) Validation Convergence

4.3.3 Population Size and Iterations

Finally, having established the assignment of Algorithm 2 to employed bees and Algorithm 1 to onlooker bees, alongside a trial limit of $L = SN \times 1.5$, the population size (SN) and maximum number of iterations (MI) were evaluated. A grid search was conducted across 36 combinations, where $SN \in \{10,20,50,80,100,250\}$ and $MI \in \{50,100,120,250,500,1000\}$. Figure 28(a) presents the heatmap of the validation results based on the $\mathrm{LCB}^{(2)}_{Architecture}$ scores for each (SN, MI) configuration, illustrating the performance landscape across the parameter space.

Configurations with extreme lower bounds $(SN = 10, MI = 50)$ generally yielded inferior performance, attributable to restricted search capacity and insufficient iterations. Conversely, the upper extremes $(SN = 250, MI = 1000)$ also performed poorly, highlighting that excessively large populations or prolonged search durations elevate the risk of overfitting. Interestingly, the $(SN = 20, MI = 250)$ configuration achieved the highest absolute score of 0.5508. However, $SN = 20$ also produced the worst and second-worst performances when paired with $MI = 50$ and $MI = 120$, respectively. This severe fluctuation suggests that $SN = 20$ is a highly unstable population size, and its maximum score likely represents a spurious, isolated peak rather than a reliable optimum. Meanwhile, the region spanning $SN \in \{50,80,100\}$ over $MI \in \{100,120,250\}$ exhibited consistently high average performance, identifying it as a robust plateau region.

To isolate robust optimal regions from stochastic noise, a Gaussian filter with a standard deviation of $\sigma_{\text{spatial}} = 0.6$ was applied to smooth the raw heatmap values against their adjacent grid cells, providing a more reliable expectation of parameter stability. The resulting smoothed landscape is presented in Figure 28(b). Following this operation, the plateau region spanning $SN \in \{50,80,100\}$ and $MI \in \{100,120,250\}$ remained highly stable, whereas the isolated peak at $(SN = 20, MI = 250)$ receded to the third-highest position. Furthermore, Figure 29 illustrates the absolute difference between the raw and Gaussian-smoothed results; lower absolute differences indicate higher parameter stability and lower sensitivity to minor perturbations. Based on this comprehensive analysis of both peak potential and regional robustness, the configuration $(SN = 80, MI = 120)$ was selected as the final parameter set for the ABC algorithm.

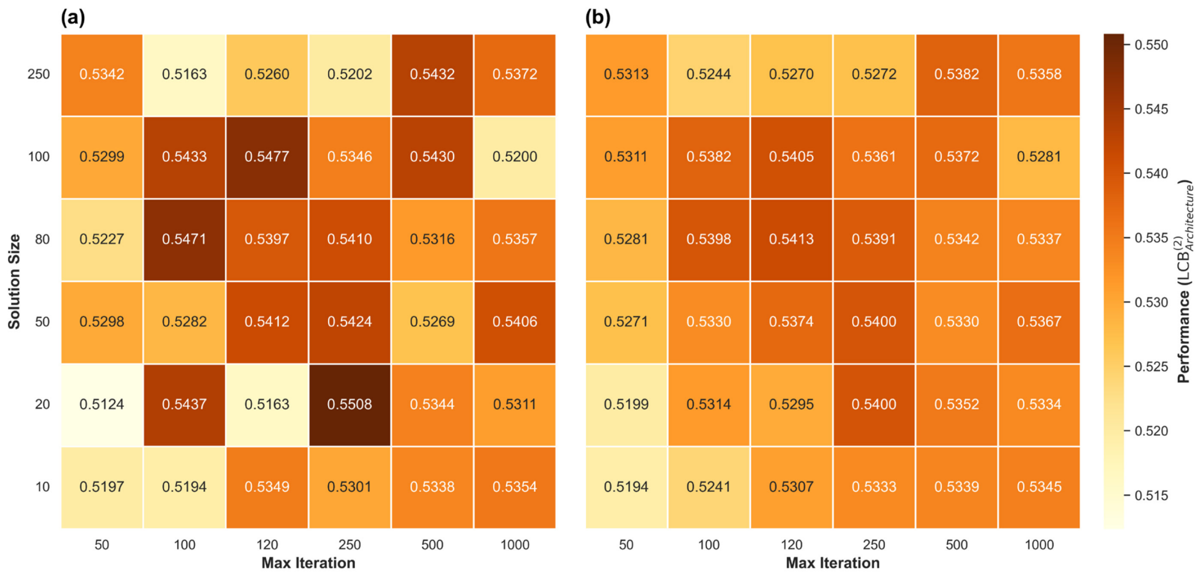

Figure 28 Performance landscape of ABC population size and iteration configurations (a) Empirical Observation (b) Expected Performance

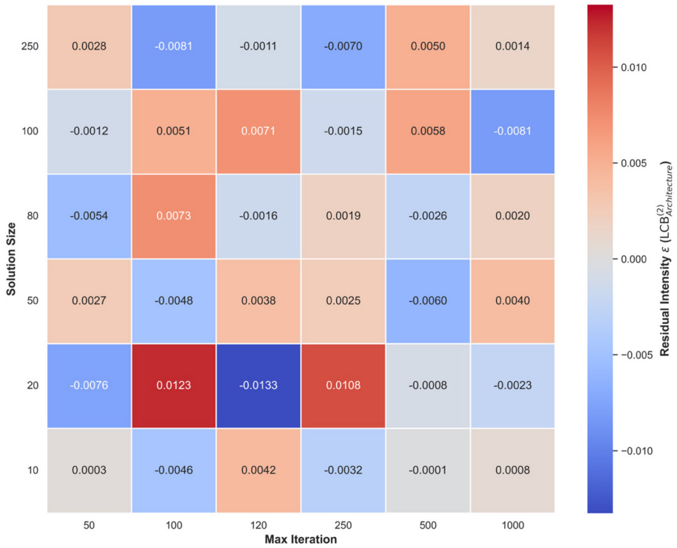

Figure 29 Residual analysis of ABC population size and iteration configurations

4.4 Model Performance Evaluation

During the final evaluation of models' performance across the five outer folds of the 80/20 data split, 50 random seeds were utilised to account for the high variance caused by the random initialisation of the ELM. Testing seeds were sampled from the range of 0 to 99999 excluding 131 to 160, using a master seed of 42, which avoids the potential bias in the testing results caused by seed overlap. To ensure a fair comparison, all models included in the testing phase applied the same 50 seeds, regardless of whether the specific model architecture was sensitive to random initialisation or other stochastic mechanisms. Furthermore, the hyperparameters for all Scikit-learn baseline models were determined via a grid search to maximise $\mathrm{LCB}^{(2)}_{Architecture}$ across the same 30 random seeds and the predefined 5-fold training and validation sets.

4.4.1 Statistical Analysis

To obtain a broader and more rigorous perspective of the model performance evaluation, the comparisons of models were not based solely on average scores; statistical tests such as the Wilcoxon signed-rank test and the paired t-test (SciPy) were included. The Holm-Bonferroni Correction (statsmodels) was applied to these tests to mitigate the false positive results. Bayesian analysis (baycomp) was additionally employed to quantify the exact probability of model superiority.

4.4.1.1 Wilcoxon Signed-Rank Test

The statistical significance of the model's outperformance across different random seeds was evaluated using the results $\mathrm{LCB}^{(1)}_r$ from the 5-fold CV for each individual seed r via the

Wilcoxon signed-rank test. This test was suitable for evaluating the significance of this outperformance because the large number of random seeds met the sample size requirement, and the variance across the seeds could not be assumed to follow a normal distribution, justifying a non-parametric approach rather than the computationally efficient surrogate metric adopted during parameter tuning. As the ELM initialisation was highly variable depending on the initial seeds, the Wilcoxon signed-rank test was primarily used to evaluate whether the optimisation process had effectively stabilised the model. Additionally, to obtain a fair perspective on the baseline comparison, the comparisons were conducted across other models, excluding the convex optimal models.

4.4.1.2 Paired *t*-test

In contrast, the generalisation ability of the models was evaluated using the averaged results of all seeds over each fold (to neutralise the initialisation variance). As the averaging process collapsed the records into five remaining aggregate results, the sample size became unsuited for Wilcoxon signed-rank test. Considering that the CV results were derived from overlapping data folds, the performance of the models should approximate a normal distribution. The paired *t*-test was utilised to evaluate the robustness of the models across varying data partitions.

4.4.1.3 Bayesian Evaluation

While the Wilcoxon signed-rank and paired *t*-test provided p values for statistical significance, they did not quantify the magnitude of the differences. A Bayesian evaluation of posterior probability was conducted to estimate the overall performance of the proposed model against several representative models, providing direct probabilistic insights into the percentage across three dimensions: outperforming, underperforming, or being practically equivalent.

4.4.1.4 Statistical Analysis Notation

Statistical significance was denoted using asterisks and +/- notation: * for $p < 0.05$, ** for $p < 0.01$, and *** for $p < 0.001$; a '+' indicates that the proposed model significantly outperforms the compared model, while a '-’ indicates the opposite. The result signs were placed above the comparatively weaker models, with orange symbols representing the Wilcoxon signed-rank test and blue symbols representing the paired *t*-test. During the Bayesian evaluation, the green and red colours represent the probability that the proposed model will outperform or underperform, respectively, while the grey colour represents the probability that both models fall within the Region of Practical Equivalence (ROPE). Furthermore, the percentage displayed on each colour represents the exact probability for each respective case.

5 Results and Discussion

The proposed ABC-RELM-CV model achieved average prediction scores of 0.7939 (± 0.0423) for accuracy, 0.7739 (±0.0474) for F1-score and 0.5947 (±0.0853) for MCC in the prediction test. The best-performing seed, evaluated among the tested seeds using $\mathrm{LCB}_r^{(1)}$, was seed 22742, which achieved an accuracy of 0.8018 (±0.0186), an F1-score of 0.7840 (±0.0279), and an MCC of 0.6095 (±0.0341).

These results represented a successful improvement over the standard RELM, which yielded an average accuracy of 0.7734 (±0.0338), an F1-score of 0.7532 (±0.0371), and an MCC of 0.5528 (±0.0680), recording its best performance at seed 77844 (accuracy: 0.81126 ±0.0387; F1-score: 0.7941 ±0.0444; MCC: 0.6282 ±0.0754). However, these overall results were slightly weaker than those of the ABC-RELM, which achieved an average accuracy of 0.7976 (±0.0391), an F1-score of 0.7765 (±0.0462), and an MCC of 0.6034 (±0.0777), recording its best performance at seed 78586 (accuracy: 0.8177 ±0.0260; F1-score: 0.7977 ±0.0350; MCC: 0.6447 ±0.0483).

As previously stated, the performance was evaluated across two dimensions: (a) stability across random seed initialisations, tracked using the MCC pessimistic floor $\mathrm{LCB}_r^{(1)}$ (a risk-averse score from 5-fold CV); and (b) generalisation ability, evaluated based on the averaged results across all 50 seeds. Furthermore, the representative models were included in a Bayesian evaluation to obtain a quantified perspective on the probability of outperformance. The region of practical equivalence was set at 0.035 by calculating the approximate change of MCC caused by a single classification difference across 64 tested samples, assuming balanced classes and an 80% accuracy per class.

5.1 Within-Family Baseline Comparison

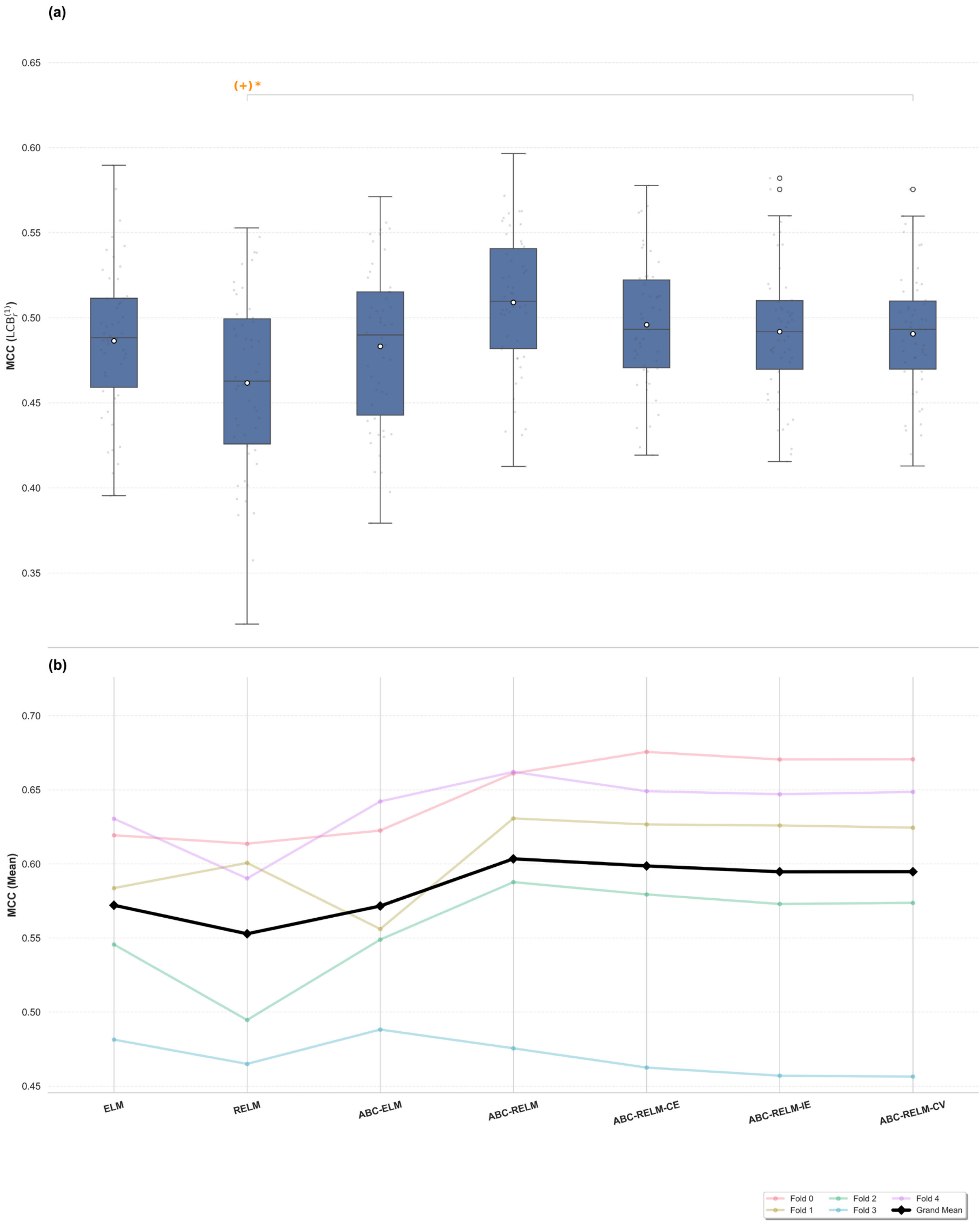

Figure 30 Performance analysis of ABC-RELM-CV and within-family baselines
(a) Cross-seed stability performance ($\mathrm{LCB}_r^{(1)}$*) (b) Cross-fold generalisation performance*

Table 8 Performance analysis of ABC-RELM-CV and within-family baselines

Model	Accuracy	F1-Score	MCC	*p* value	Direction
(a) Cross-Seed Stability Analysis ($\mathrm{LCB}_r^{(1)}$***):* Median *(Q_3 - Q_1)***					
ELM	0.7412 (0.0268)	0.7125 (0.0375)	0.4883 (0.0523)	1.0000	0
RELM	0.7274 (0.0362)	0.6990 (0.0463)	0.4628 (0.0735)	0.0168	+1
ABC-ELM	0.7415 (0.0348)	0.7108 (0.0427)	0.4898 (0.0725)	0.8731	0
ABC-RELM	0.7515 (0.0284)	0.7267 (0.0284)	0.5097 (0.0588)	0.1275	0
ABC-RELM-CE	0.7433 (0.0266)	0.7170 (0.0304)	0.4931 (0.0517)	0.5733	0
ABC-RELM-IE	0.7415 (0.0187)	0.7156 (0.0266)	0.4918 (0.0404)	1.0000	0
ABC-RELM-CV	0.7421 (0.0184)	0.7170 (0.0250)	0.4932 (0.0400)	-	-
***(b) Cross-Fold Generalisation Analysis:* Mean ± SD**					
ELM	0.7825 ± 0.0307	0.7616 ± 0.0367	0.5721 ± 0.0606	0.6282	0
RELM	0.7734 ± 0.0338	0.7532 ± 0.0371	0.5528 ± 0.0680	0.2650	0
ABC-ELM	0.7820 ± 0.0315	0.7600 ± 0.0376	0.5716 ± 0.0618	0.6282	0
ABC-RELM	0.7976 ± 0.0391	0.7765 ± 0.0462	0.6034 ± 0.0777	0.6282	0
ABC-RELM-CE	0.7957 ± 0.0416	0.7755 ± 0.0466	0.5986 ± 0.0839	0.1444	0
ABC-RELM-IE	0.7938 ± 0.0421	0.7738 ± 0.0470	0.5947 ± 0.0850	0.8855	0
ABC-RELM-CV	0.7939 ± 0.0423	0.7739 ± 0.0474	0.5947 ± 0.0853	-	-

Note: MCC: Matthews Correlation Coefficient; SD: standard deviation; Q_1, Q_3: quartiles. *p* value (based on MCC): Holm-Bonferroni adjusted (a: Wilcoxon; b: paired *t*-test). Direction: +1: ABC-RELM-CV wins; 0: no significant difference; -1: Baseline wins. Dash (–): reference/not applicable.

Table 8 and Figure 30 present the first-phase evaluation of the ABC-RELM-CV, comparing the proposed model against its within-family baselines. The ABC-RELM-CV outperformed the standard RELM with statistical significance. This suggested that the optimisation process was successful; under identical initialisation conditions, the proposed model consistently achieved superior relative performance. Furthermore, although not statistically significant across all pairwise comparisons, the ABC-RELM-CV achieved slightly higher median scores than the ELM, RELM, and ABC-ELM, while demonstrating the highest consistency, specifically the narrowest interquartile range (IQR), across different random seeds.

In contrast, the baseline ELM performed better than the RELM in terms of both median score and dispersion; this could be attributed to the parameter tuning of the RELM being systematically overfitted to the validation data. While the ABC-ELM obtained a slightly higher median than the ELM, its wider dispersion denoted that interchanging the inner model from the RELM to the ELM within the ABC framework without extra tuning was a technical error. Notably, the ABC-RELM achieved the highest overall median performance but exhibited a wider IQR compared to the ABC-RELM-CV lineage. This indicates that the architecture possesses higher potential but is more dependent on initialisations. These results imply that the achievement of performance optimisation stemmed mainly from ABC algorithm, whereas internal CV restricted the peak potential. Nevertheless, internal CV successfully narrowed the performance dispersion across varying initialisations.

Conversely, there was no statistically significant outperformance during the evaluation of the model's generalisation ability. The ABC-RELM still performed as the best model, demonstrating a slight advantage compared to the ABC-RELM-CV lineage, while the ELM, ABC-ELM, and RELM performed the worst. However, it is noteworthy that the ELM achieved the lowest variance across different folds, indicating that applying the fewest constraints to the models allowed them to achieve the most consistent performance across different data partitions. Additionally, the ABC-RELM-CV lineage performed best in fold 0 but worst in fold 3, exhibiting similar performances to the ABC-RELM on the other folds. Comparing these models with ELM, ABC-ELM and RELM, the performance trends of fold 0 and fold 1 were the inverse of those in fold 2 and fold 3, implying that the different data partitions possessed inconsistent characteristics.

5.2 Scikit-Learn Baseline Comparison

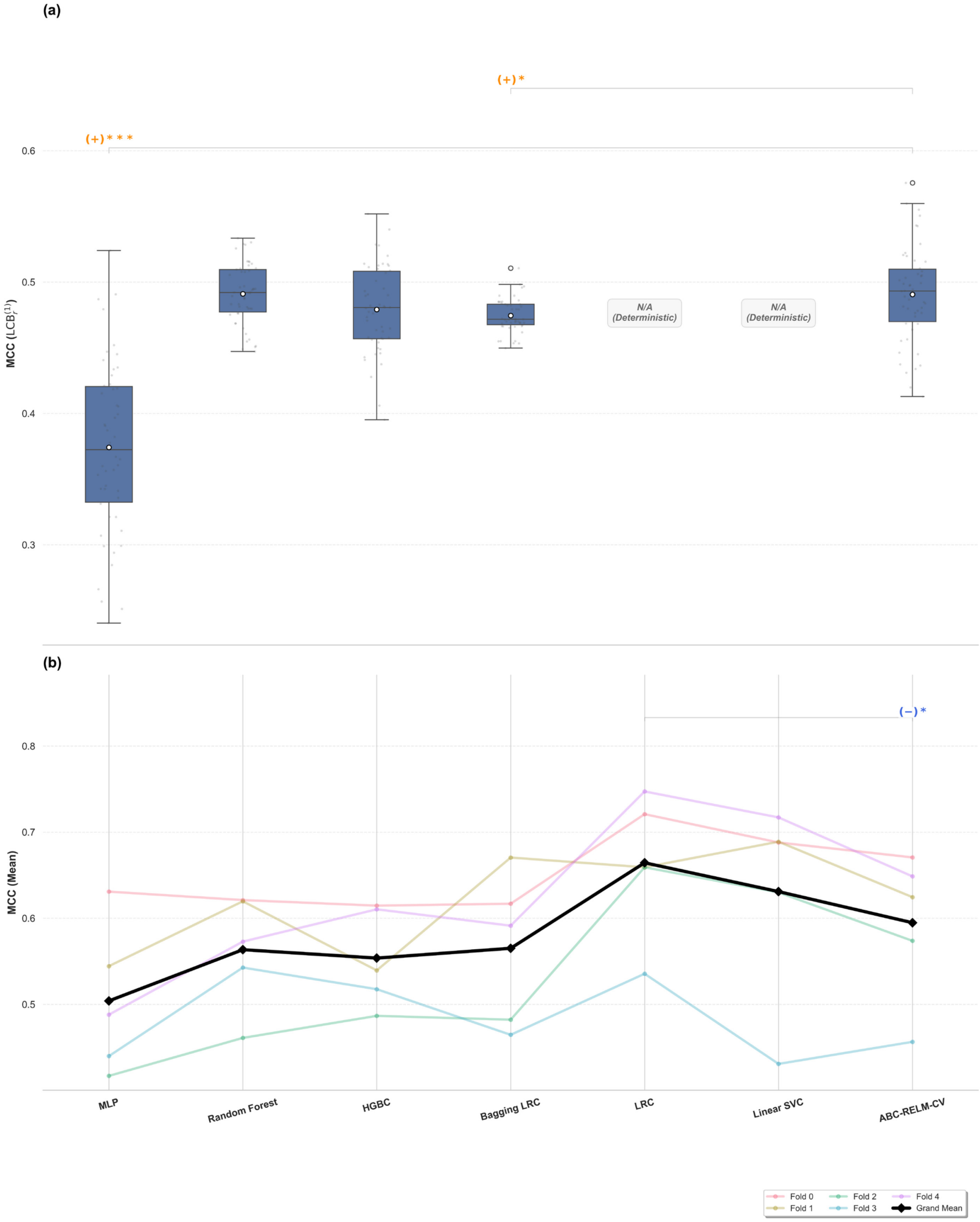

Figure 31 Performance analysis of ABC-RELM-CV and scikit-learn baselines
(a) Cross-seed stability performance ($\mathrm{LCB}_r^{(1)}$) (b) Cross-fold generalisation performance

Table 9 Performance analysis of ABC-RELM-CV and scikit-learn baselines

Model	**Accuracy**	**F1-Score**	**MCC**	***p* value**	**Direction**
(a) Cross-Seed Stability Analysis ($\mathbf{LCB}_r^{(1)}$***):* Median *(Q_3 - Q_1)***					
MLP	0.6761 (0.0482)	0.6811 (0.0664)	0.3723 (0.0881)	3.2951×10^{-12}	+1
Random Forest	0.7460 (0.0155)	0.7400 (0.0170)	0.4920 (0.0322)	0.8109	0
HGBC	0.7388 (0.0263)	0.7227 (0.0207)	0.4805 (0.0515)	0.1645	0
Bagging LRC	0.7347 (0.0075)	0.7173 (0.0095)	0.4716 (0.0156)	0.0434	+1
LRC	0.7872 (0.0000)	0.7626 (0.0000)	0.5826 (0.0000)	-	-
Linear SVC	0.7560 (0.0000)	0.7366 (0.0000)	0.5146 (0.0000)	-	-
ABC-RELM-CV	0.7421 (0.0184)	0.7170 (0.0250)	0.4932 (0.0400)	-	-
***(b) Cross-Fold Generalisation Analysis:* Mean ± SD**					
MLP	0.7456 ± 0.0428	0.7345 ± 0.0416	0.5040 ± 0.0862	0.1861	0
Random Forest	0.7809 ± 0.0326	0.7714 ± 0.0289	0.5634 ± 0.0661	0.6146	0
HGBC	0.7756 ± 0.0286	0.7635 ± 0.0287	0.5537 ± 0.0569	0.6146	0
Bagging LRC	0.7816 ± 0.0444	0.7691 ± 0.0479	0.5651 ± 0.0887	0.6146	0
LRC	0.8301 ± 0.0429	0.8161 ± 0.0535	0.6644 ± 0.0817	0.0247	-1
Linear SVC	0.8143 ± 0.0583	0.8018 ± 0.0652	0.6309 ± 0.1163	0.4547	0
ABC-RELM-CV	0.7939 ± 0.0423	0.7739 ± 0.0474	0.5947 ± 0.0853	-	-

Note: MCC: Matthews Correlation Coefficient; SD: standard deviation; Q_1, Q_3: quartiles. *p* value (based on MCC): Holm-Bonferroni adjusted (a: Wilcoxon; b: paired *t*-test). Direction: +1: ABC-RELM-CV wins; 0: no significant difference; -1: Baseline wins. Dash (–): reference/not applicable.

Table 9 and Figure 31 present the second-phase evaluation of the ABC-RELM-CV, comparing it against the scikit-learn baseline models. In terms of initialisation stability, the ABC-RELM-CV outperformed the MLP and Bagging LRC significantly; the MLP's performance exhibited the highest dispersion around the lowest median, while the Bagging LRC exhibited the lowest dispersion with the second-lowest median. Despite the differences in their performance patterns, the possible reason was similar: the dataset utilised was relatively small. This caused the MLP, which employed a highly complicated parameter and dimensional structure, to easily overfit to different local minima from different starting points. In contrast, the small dataset meant that the Bagging LRC could only utilise a smaller subset of data for training, resulting in performances that were stable across different splits but relatively low overall. Conversely, the ABC-RELM-CV obtained a slightly better median but was caught in the middle of the RF and HGBC in terms of dispersion. This denoted that even though the performances were relatively better, the ABC-RELM-CV was still sensitive to the initialisation.

During the evaluation of generalisation ability, the LRC outperformed the ABC-RELM-CV significantly, emerging as the best-performing model among all models with the highest mean MCC of 0.6644. Furthermore, even though the linear SVC did not achieve statistical significance, it still maintained a mean MCC that was 0.03 higher than that of the proposed model, securing second place among the models. Conversely, the MLP only achieved an MCC of 0.5040, becoming the worst-performance model of all. The ABC-RELM-CV performed slightly better than the RF and HGBC in this generalisation evaluation but exhibited a higher standard deviation. Excluding the LRC, which achieved similar performance, the RF and HGBC outperformed the other models in fold 3. However, their performances on the other folds were relatively worse than those of the Linear SVC and the ABC-RELM-CV. This suggested that the data partition for fold 3 likely exhibited a contrasting characteristic, which could cause high-dimensional function mapping to fail.

5.3 Model Evaluation

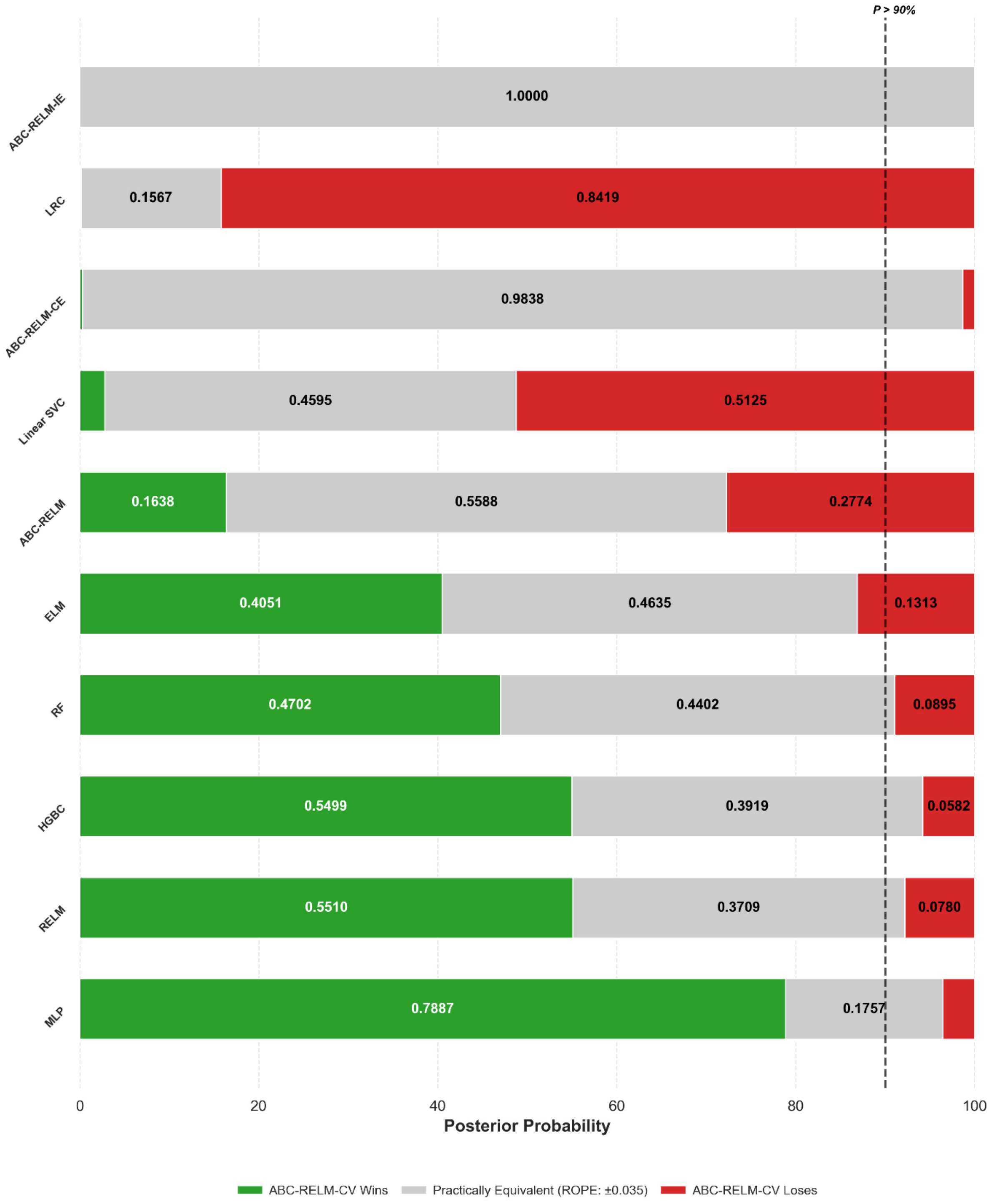

Figure 32 Bayesian evaluation of ABC-RELM-CV against representative models

Table 10 Performance evaluation of ABC-RELM-CV and representative models

Model	Accuracy	F1-Score	MCC	P_{Win}
ABC-RELM-CV	0.7939 ± 0.0423	0.7739 ± 0.0474	0.5947 ± 0.0853	-
ABC-RELM-IE	0.7938 ± 0.0421	0.7738 ± 0.0470	0.5947 ± 0.0850	0.0000
LRC	0.8301 ± 0.0429	0.8161 ± 0.0535	0.6644 ± 0.0817	0.0013
ABC-RELM-CE	0.7957 ± 0.0416	0.7755 ± 0.0466	0.5986 ± 0.0839	0.0029
Linear SVC	0.8143 ± 0.0583	0.8018 ± 0.0652	0.6309 ± 0.1163	0.0280
ABC-RELM	0.7976 ± 0.0391	0.7765 ± 0.0462	0.6034 ± 0.0777	0.1638
ELM	0.7825 ± 0.0307	0.7616 ± 0.0367	0.5721 ± 0.0606	0.4051
RF	0.7809 ± 0.0326	0.7714 ± 0.0289	0.5634 ± 0.0661	0.4702
HGBC	0.7756 ± 0.0286	0.7635 ± 0.0287	0.5537 ± 0.0569	0.5499
RELM	0.7734 ± 0.0338	0.7532 ± 0.0371	0.5528 ± 0.0680	0.5510
MLP	0.7456 ± 0.0428	0.7345 ± 0.0416	0.5040 ± 0.0862	0.7887

Note: MCC: Matthews Correlation Coefficient; *SD*: standard deviation. P_{Win}: Bayesian posterior probability of ABC-RELM-CV outperforming the baseline (threshold ROPE = 0.035 based on MCC). Dash (–): reference model.

The performance evaluation and corresponding Bayesian evaluation of the ABC-RELM-CV against representative models are detailed in Table 10 and Figure 32, respectively. To facilitate clearer observation, the models were classified into several groups: the original model (ABC-RELM), base models (ELM, RELM), variant models (ABC-RELM-CE, ABC-RELM-IE), the SLFN model (MLP), tree models (RF, HGBC), and linear models (LRC and Linear SVC).

There was only a 16.4% probability that the ABC-RELM-CV would outperform the ABC-RELM, yet a 27.7% chance that it would underperform. This was highly likely caused by two related reasons. The internal CV fitness architecture restricted the models from overfitting the entire data structure, which increased stability at the expense of peak potential. Furthermore, the data was partitioned again within the internal CV, resulting in training on smaller subsets of data. These limitations resulted in the ABC-RELM-CV abandoning rare characteristics by treating them as noise, producing a wider, flatter local optimum rather than a deeper, narrower global optimum. Consequently, the model was able to predict general data stably but failed to predict special cases.

On the other hand, even though the ABC-RELM-CV only slightly outperformed the base models in the first-phase evaluation, the Bayesian evaluation of posterior probability showed that the ABC-RELM-CV had a greater than 40% probability of outperforming them, with a less than 15% chance of underperforming. This confirmed the effectiveness of the ABC optimisation. Conversely, there was a greater than 98% probability that the ABC-RELM-CV and its ensemble variants would fall within the ROPE. This demonstrated that using the same solutions within the internal CV or different iteration solutions based on cosine similarity was inappropriate; the internal ensemble was unable to produce the diversity required by ensemble theory.

Furthermore, the ABC-RELM-CV obtained a greater than 45% probability of outperforming the tree models, with a less than 10% probability of underperforming, and a greater than 75% probability of outperforming the MLP. This demonstrated that the model was relatively superior, exhibiting higher adaptability to the smoothed, pre-processed dataset while maintaining balance through regularisation. However, when compared with the linear models, the ABC-RELM-CV showed a less than 5% probability of outperforming, with a greater than 80% probability of underperforming against the LRC and a nearly 50% probability of underperforming against the Linear SVC. The performance of the linear models dominated the overall results, suggesting that under the sample size constraints, the linear hyperplane acted as the principal complexity ceiling, while subtle non-linearities remained secondary.

In this case study, the convex optimal models predominantly outperformed the others, while the more complicated non-convex optimal models achieved worse results. The intermediate performance of the ABC-RELM-CV could be attributed to the fact that ELM acts as a convex optimisation model when the hidden layer output is fixed. These overall findings can be attributed to three main factors:

1. The intrinsic linear structure of the smoothed, pre-processed data.
2. The limited dataset (only 318 samples across 38 features) being insufficient to support the intricate boundaries of high-dimensional relationships.
3. The overfitting of complex models, where high sensitivity to specific noise obscured the underlying base patterns.

Overall, the ABC-RELM-CV had a low probability of underperforming compared to the other non-linear models. This suggests that it still achieved relatively good results, even though it was unable to outperform the linear models. However, the ABC-RELM remains preferable to the ABC-RELM-CV, unless further evaluations with a larger or different dataset suggest otherwise. This indicates that the effectiveness of internal CV for the ABC optimisation may be hindered by the limited dataset size, which could potentially result in a negative impact on performance. The CV mechanism does not always guarantee an improvement in generalisation ability, and more complex algorithms do not always yield better results.

6 Conclusions

In conclusion, the refinement of the dataset based on physiological consistency was executed to ensure data integrity. Building upon this, the ELM-based prediction of gallstone disease optimised by the ABC algorithm was successfully implemented. The final proposed ABC-RELM-CV model achieved an average accuracy of 0.7939, an F1-score of 0.7739, and an MCC of 0.5947. These results outperformed the baseline models of ELM and RELM, which achieved MCC values of 0.5721 and 0.5528, respectively. However, the model underperformed the linear models of LRC and SVC, which achieved superior MCC values of 0.6644 and 0.6309. While the integration of internal cross-validation was intended to enhance the model's stability, it did not yield a significant breakthrough in predictive performance compared to the standard ABC-RELM, which achieved a slightly higher 0.7976 accuracy, 0.7765 F1-score, and 0.6034 MCC. This suggests that for this specific dataset, the added complexity of internal cross-validation may have been limited by the sample size, thereby preventing a substantial improvement in generalisation. Nevertheless, by utilising metaheuristic optimisation, the ABC-optimised RELM models consistently outperformed the standard ELM while requiring 25% fewer hidden nodes (30 nodes compared to 40 in the base model), whereas the baseline RELM achieved relatively poorer generalisation performance, although it exhibited better results during the initial parameter tuning phase.

6.1 Limitations and Future Work

Although the optimisation of the ELM was successfully implemented, there were still several limitations. The number of hidden nodes and the regularisation parameters of the ELM and RELM applied in this study were determined using a grid search over manually set ranges and fixed for subsequent use. These hyperparameters were not optimised as the ABC algorithm's solution only included the input weights and hidden biases. Future attempts could explore incorporating these parameters into the ABC optimisation, but this would require a different neighbourhood algorithm capable of handling integer or logarithmic values.

Furthermore, although the ensemble attempts in this study were unsuccessful due to their simple designs, the ensembling of ELM models remains a possible research direction. In addition, the combined utilisation of metaheuristic algorithms, such as applying one for feature selection prior to another for result optimisation, could be explored. However, the different neighbourhood algorithms introduce numerous additional hyperparameters that may not be optimally tuned; furthermore, internal dynamic strategy switching could be employed to adaptively select the optimal algorithm composition during the search process. Appropriately tuning these parameters could potentially yield a more significant breakthrough than merely increasing the complexity of the algorithm.

Additionally, this study relied on a single gallstone dataset evaluating the full feature set without the implementation of a fully nested cross-validation framework. Furthermore, the data preprocessing workflow was developed using the author's methodological discretion, lacking validation from domain-specific medical experts. This meant that the handling of laboratory blood test data could not be customised according to the physiological indicators, which could lead to adverse effects on model performance. Consequently, these methodological constraints could introduce bias regarding the true gallstone prediction performance of the model architectures.

Considering that the dataset had a high feature-to-sample ratio, future work could aim to increase the sample size or apply dimensionality reduction and also include multiple different sources of gallstone datasets for model evaluation. Alternatively, exploring an ensemble of linear and non-linear models could potentially serve as a viable strategy to maximize information extraction from the UCI gallstone dataset. The preliminary research by Indirani et al. [34] and Akben and Yumrutas [42] reported that this ensemble approach outperformed single models to achieve an accuracy of over 85%. As a point of reference, most accuracy results from various studies achieved approximately 83% for linear models and nearly 80% for non-linear models on this dataset. Nonetheless, further rigorous validation remains necessary.

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Appendix A

Table A1 Table of Notations

Notation	Description	Index
X	Training dataset	x
N	Total sample size	j
n	Number of input features	a, b
A	Subset of n for evaluation	a
m	Number of output features	
$\widetilde{N}$	Hidden neurons size	i
D	Solution dimension	k
SN	Solution size	h, q
L	Trial limit	
MI	Max iteration	
K	Number of cross-validation fold	f
R	Random state seed	r
F	Blocking factor for stability assessment	
G	Total sample to evaluate through factor F	g
P, p	Probability	
W	Number of observations	z
α	Generic coefficient	
μ	Mean	
σ	Standard deviation	
ε	Residual / Error	
τ	Threshold	
s	Recursion order	

Appendix B

Table A2 Methodological Observations and Reporting Variations in Related Works

Reference	Special Attention
Esen et al., 2024 [32]	The dataset utilised appears to differ, as discussed previously. The results are derived from a 30% hold-out test set without CV.
Sarker et al., 2025 [33]	The study utilised 161 positive and 158 negative cases, with 1 denoting negative (Absent) and 0 denoting positive (Present). The discussion referenced previous studies showing: "CRP is associated with gallstone-related inflammation and disease severity, serving as a useful biomarker in gallbladder disorders [33]. Another medical research on 4484 participants has found that CRP levels are associated with a greater prevalence of gallstones [34]. Liu et al. have claimed that CRP is an independent risk factor for gallstone disease [35]. Another study on 6873 people has shown that higher dietary vitamin D consumption (D2+D3) is positively associated with gallstone incidence [36]. AAST is a liver enzyme; a higher level of AAST has a positive association with gallstone formation [37]. Another study on 55,531 has shown that liver enzymes may increase the risk of developing gallstones [38].", yet the study achieved the following result: "So this research has shown that the individuals who have lower CRP, higher Vitamin D, and AAST, have a higher chance of gallstones."
Cansu et al., 2025 [41]	The study does not specify how the labels were utilised nor the total number of positive or negative cases. The study omits details on how the data was processed, yet states that a total of 349 patients were included, reflecting the same variation in gender statistics as the original paper. This diverges from the actual dataset of 319 patients. Notably, the test set confusion matrix of 96 samples represents 27.5% of 349, or 30% of 319. The study declares that "The performance comparison presented in Table I reveals that the proposed method consistently outperforms the baseline results reported by Esen et al. [18] across all evaluation metrics and algorithms." However, the original study achieved a best result of 85.42%, whereas this study's reported best accuracy was 80.21%.

Akben and Yumrutas, 2026 [42]	The study utilised 161 positive and 158 negative cases, with 1 denoting negative (Absent) and 2 denoting positive (Present). Furthermore, the study declares "Clinical Findings: Gallbladder Wall Thickness, Stone Size, Pain Severity, Murphy's Sign" as part of the feature set, although these parameters are documented exclusively in the Kaggle website description.
Indirani et al., 2026 [34]	The study utilised 161 positive and 158 negative cases. However, the study indicates that "Additionally, Female patients exhibit a significantly elevated prevalence of gallstones, indicating a pronounced gender-associated risk.", but later describes "The data set contains 162 records in category 0 (likely Male or Female) and 157 records in category 1, suggesting a balanced representation of gender.", which caused its Figure 2 to lack clarity regarding which label denoted Male/Female and Gallstone Disease Present/Absent. The feature description remains identical to the original dataset, yet the study states that "Obesity percentage, total health expenditure, glucose levels, and AST/ALT ratios necessitate outlier removal and capping transformation." This incorporates two features not found in the source: "total health expenditure" and "AST/ALT ratios". Furthermore, the sampling methodology is not fully detailed. The study initially claims: "suggesting a balanced representation of gender. Based on the previous research referring to "class imbalance" and application of SMOTE was incorrect and has been removed - no oversampling (SMOTE) or class-weighting was used in the primary analyses", yet the text subsequently describes applying SMOTE to predictor features, stating "In terms of comorbidity, the majority of patients are classified in category 0 (217), with fewer being classified in categories 1, 2, and 3, suggesting a class imbalance that necessitates the application of SMOTE weights."
Rodriguez et al., 2026 [43]	The study utilised 158 positive and 161 negative cases, with 0 denoting negative (Absent) and 1 denoting positive (Present). The results are derived from a 30% hold-out test set without CV.
Shrestha et al., 2026 [44]	The study does not specify how the labels were utilised nor the total number of positive or negative cases. The study declares that: "Compared to our baseline model, RoF-BESA showed fewer misclassifications. Overall the classifier performed consistently well with only a small number of cases being misclassified in each fold." However, this statement contrasts with its own reported data, as the baseline model achieved an accuracy of 78%, while the proposed RoF-BESA model yielded 75.789%.

Appendix C

Table A3 Comparison of Pearson correlations before and after data correction

Feature	Pearson Correlation Coefficient (*r*)	
	Before Correction	**After Correction (Change)**
Vitamin D	-0.3549	-0.3565 (−0.0016)
Lean Mass (LM) (%)	-0.2257	-0.2247 (+0.0010)
Bone Mass (BM)	-0.2166	-0.2189 (−0.0023)
Hemoglobin (HGB)	-0.1969	-0.2054 (−0.0085)
Extracellular Water (ECW)	-0.1784	-0.1811 (−0.0027)
Extracellular Fluid/Total Body Water (ECF/TBW)	-0.1698	-0.1768 (−0.0070)
Aspartat Aminotransferaz (AST)	-0.1349	-0.1352 (−0.0003)
Creatinine	-0.1323	-0.1325 (−0.0002)
Total Body Water (TBW)	-0.1112	-0.1116 (−0.0004)
Height	-0.1079	-0.1097 (−0.0018)
Coronary Artery Disease (CAD)	-0.0970	-0.0965 (+0.0005)
Muscle Mass (MM)	-0.0933	-0.0854 (+0.0079)
Body Protein Content (Protein) (%)	-0.0864	-0.0853 (+0.0011)
Visceral Muscle Area (VMA) (Kg)	-0.0730	-0.0729 (+0.0001)
Low Density Lipoprotein (LDL)	-0.0565	-0.0579 (−0.0014)
Hypothyroidism	-0.0552	-0.0547 (+0.0005)
Triglyceride	-0.0508	-0.0503 (+0.0005)
Comorbidity	-0.0485	-0.0466 (+0.0019)
Alanin Aminotransferaz (ALT)	-0.0419	-0.0442 (−0.0023)
Intracellular Water (ICW)	-0.0353	-0.0590 (−0.0237)
Glomerular Filtration Rate (GFR)	-0.0283	-0.0324 (−0.0041)
Glucose	-0.0115	-0.0099 (+0.0016)
Total Cholesterol (TC)	0.0141	0.0149 (+0.0008)
Visceral Fat Rating (VFR)	0.0183	0.0206 (+0.0023)
Age	0.0363	0.0413 (+0.0050)
Weight	0.0487	0.0489 (+0.0002)
Obesity (%)	0.0539	-0.0089 (−0.0628)
Hepatic Fat Accumulation (HFA)	0.0903	0.0879 (−0.0024)
Diabetes Mellitus (DM)	0.1047	0.1061 (+0.0014)
Alkaline Phosphatase (ALP)	0.1099	0.1110 (+0.0011)

Body Mass Index (BMI)	0.1215	0.1228	(+0.0013)
Visceral Fat Area (VFA)	0.1404	0.1411	(+0.0007)
Gender	0.1535	0.1571	(+0.0036)
High Density Lipoprotein (HDL)	0.1586	0.1603	(+0.0017)
Hyperlipidemia	0.1619	0.1627	(+0.0008)
Total Fat Content (TFC)	0.1702	0.1719	(+0.0017)
Total Body Fat Ratio (TBFR) (%)	0.2255	0.2242	(−0.0013)
C-Reactive Protein (CRP)	0.2820	0.2837	(+0.0017)

Appendix D

The source code, together with the training/testing datasets and the experimental results (CSV files), is available at: https://github.com/youquanneu/COMP3200_Individual_Project.git.

In the implementation, Algorithm 1 and Algorithm 2 correspond to Algorithm 2 and Algorithm 3 in the provided code, respectively.

Appendix E

Agreed Project Brief

COMP3200-FYP (UoSM) 2025/26

Student Name	NEU YOU QUAN
Agreed Project Title	An Optimised Machine Learning Model for Gallstone Disease Prediction
Supervisor Name	Dr. Abobakr Al-Shamiri

Project Purpose	Gallstone disease (GD) is one of the most common disorders affecting the digestive system worldwide, with rising incidence due to a combination of genetic, biological, and lifestyle factors. Traditional diagnostic methods such as ultrasonography, CT scans, and MRI are effective but may be costly and less accurate in certain patient populations. This project aims to design and develop a machine learning-based model capable of predicting the risk of gallstone disease using clinical and demographic data. The study further seeks to employ metaheuristic algorithms to optimise model performance, thereby improving predictive accuracy and reliability.
Project Scope	This project will focus on the development of an optimised machine learning model for gallstone disease prediction using publicly available medical datasets. The project will include data preprocessing, feature selection, model training, evaluation, and optimisation using metaheuristic algorithms such as Artificial Bee Colony (ABC) algorithm. Excluded from the project are clinical trials, patient data collection from hospitals, and the design of new diagnostic hardware. The deliverables will include the trained models, performance evaluation metrics, comparative results before and after optimisation, and documentation of methodology.
Objectives (Tentative)	To analyse and identify representative features from existing medical datasets that improve the model's performance. To develop machine learning models capable of accurately predicting the risk of gallstone disease based on patient data. To investigate the integration of metaheuristic optimisation techniques with machine learning to enhance predictive accuracy.

Signature of the Student **Signature of the Supervisor**

Date: 23 Oct 2025

Figure A1 Agreed Project Brief

Appendix F

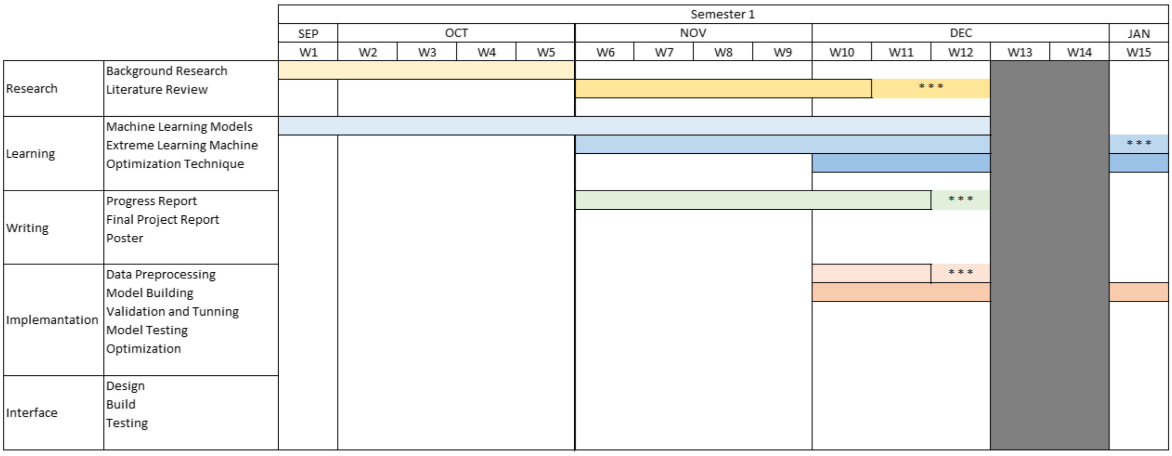

Figure A2 Gantt Chart: Semester 1

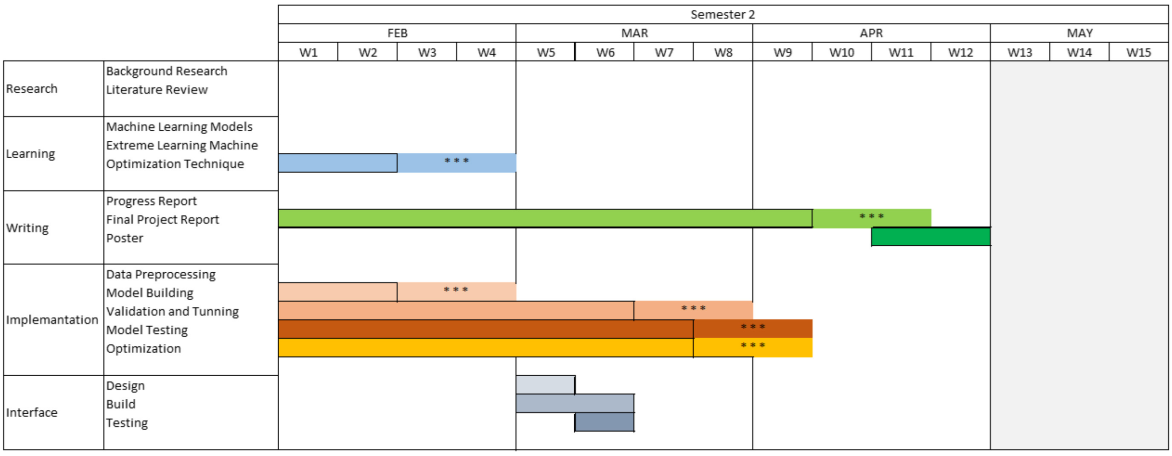

Figure A3 Gantt Chart: Semester 2